Savitribai Phule Pune University

(Formerly University of Pune)

Four Year Degree Program
B.Sc.(Computer Science)

With

Major: Computer Science

(Faculty of Science and Technology)



Syllabi for
F.Y.B.Sc. (Computer Science)

(For Colleges Affiliated to Savitribai Phule Pune University)

Choice Based Credit System (CBCS) Syllabus
Under National Education Policy (NEP)

To be implemented from Academic Year 2024-2025

Title of the Course: B.Sc.(Computer Science)

Preamble:

The B. Sc. (Computer Science) and B. Sc. (Computer Science) (Honors) and (Research) course is a systematically designed program with Computer Science as a major subject under the faculty of Science and Technology. The objective of the course is to prepare students to undertake careers involving problem solving using computer science and technologies, or to pursue advanced studies and research in computer science. The syllabus which comprises of Computer Science (Major) subject along with that of the three allied subjects (Mathematics, Electronics and Statistics) (Minor) covers the foundational aspects of computing sciences and also develops the requisite professional skills and problem solving abilities using computing sciences.

Introduction:

At the first year of under-graduation, the basic foundations of two important skills required for software development are laid. A course in problem solving and programming along with a course in database fundamentals forms the preliminary skill set for solving computational problems. The practical courses are designed to supplement the theoretical training in the year. Along with Computer Science (Major), VSC and SEC courses help in building a strong technical foundation. Another aspect of this course is IKS which tells about the rich heritage and advancement of India in the field of computation.

In the second year of under-graduation, computational problem solving skills are further strengthened by a course in Data structures, C++ and python programming. Software engineering concepts that are required for project design are also introduced. Essential concepts of computer networking are also introduced this year. The practical course included in both semesters complements the theory courses. Field projects/ OJT are introduced so that students can implement the concept they have learnt in first year.

In Second Year, the "Subject 1 : Computer Science" will be the Major Subject and the Minor subject will be chosen from "Subject 2 or Subject 3". Subject 2 and Subject 3 will not be available as Major Subjects in Second Year and Third Year

At the third year of under-graduation, all the subjects are designed to fulfill core Computer Science requirements as well as meet the needs of the software industry. Theory courses are adequately supplemented by hands-on practical courses. Major elective courses are taking care of recent advancement in the field of computer science. Minor and Skill Enhancement courses enable the students to acquire additional skills.

At the fourth year (honors) and (research) of under-graduation, all the subjects are designed to fulfill core Computer Science requirements as well as meet the needs of the software industry. Practical courses and field projects enable students to get hands-on training. Various learning tracks are open through Major elective courses. Research methodology course will create interest among the students to carry research in the field of computer science.

Objectives:

- To develop problem solving abilities using a computer.
- To build the necessary skill set and analytical abilities for developing computer based solutions for real life problems.
- To train students in professional skills related to the Software Industry.
- To prepare the necessary knowledge base for research and development in Computer Science.
- To help students build-up a successful career in Computer Science and to produce entrepreneurs who can innovate and develop software products.

Eligibility

a) H.S.C.(10 + 2) Science stream with Mathematics.
b) Three years diploma course after S.S.C.(10th std.) of Board of Technical Education conducted by Government of Maharashtra or its equivalent.

Programme Out comes:

PO No	Outcomes
PO1	Develop creative skills, critical thinking, analytical skills and research to address the real world problems using computational skills
PO2	Understand and apply mathematical foundation, computing and domain knowledge and develop computing models for defined problems
PO3	Understand software project management and computing principles with computing knowledge to manage projects in multidisciplinary environments
PO4	Illustrate the concepts of systems fundamentals, including architectures and organization, operating systems, networking and communication
PO5	Understand and apply the concepts of Digital Electronics, Computer Architecture, IoT etc.
PO6	Recognize the need for and develop the ability to engage in continuous learning as a Computing professional
PO7	Apply modern computing tools, skills and techniques necessary for innovative software solutions
PO8	Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations
PO9	Gain Self Discipline and commit Professional Ethics in global economic environment
PO10	Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment
PO11	Identify opportunities, entrepreneurship vision and use innovative ideas to create value and wealth for the betterment of the individual and society

Savitribai Phule Pune University

Structure of UG Program as per NEP-2020

Name of Program :- B.Sc. (Computer Science)

Major Course:- Computer Science

Level:- 4.5 (First Year) **Sem:-I**

Course Type	Course Code	Course Title	Credits		Teaching Scheme Hr/Week		Evaluation Scheme and Max Marks		
			TH	PR	TH	PR	CE	EE	Total
Subject 1	CS-101-T	Problem Solving using 'C' Programming	2		2		15	35	50
	CS-102-P	Lab Course based on CS-101-T		2		4	15	35	50
Subject 2	MTC-101-T	Matrix Algebra	2		2		15	35	50
	MTC-102-P	Mathematics Practical I		2		4	15	35	50
Subject 3	ELC-101-T	Principles of Analog Electronics	2		2		15	35	50
	ELC-102-P	Electronics Practical Course I		2		4	15	35	50
IKS(2)	IKS-100-T	Generic IKS	2		2		15	35	50
GE/OE* (2)	OE-101-CS -T/ OE-102-CS -T/ OE-103-CS-T / OE-104-CS-T	Office Automation I / Introduction to Computers and Basics of Internet / Introduction to Google Apps I / Fundamentals of Computers I	2		2		15	35	50
SEC (2)	SEC-101-CS	Statistical Methods for Computer Science I		2		4	15	35	50
AEC(2)	AEC-101-ENG	English	2		2		15	35	50
VEC(2)	VEC-101-ENV	EVS-I	2		2		15	35	50
Total			14	08	14	16			550

** The subjects offered to other faculty students under OE vertical are OE-101-CS -P/ OE-102-CS -T/OE-103-CS-P / OE-104-CS-T. The students of B.Sc. (Computer Science) will opt the subjects offered by other faculty given in University Basket.*

Level:- 4.5 (First Year) **Sem:-II**

Course Type	Course Code	Course Title	Credits		Teaching Scheme Hr/Week		Evaluation Scheme and Max Marks		
			TH	PR	TH	PR	CE	EE	Total
Subject 1	CS-151-T	Advanced C Programming	2		2		15	35	50
	CS-152-P	Lab Course Based on CS-151-T		2		4	15	35	50
Subject 2	MTC-151-T	Graph Theory	2		2		15	35	50
	MTC-152-P	Mathematics Practical II		2		4	15	35	50
Subject 3	ELC-151-T	Principles of Digital Electronics	2		2		15	35	50

	ELC-152-P	Electronics Practical Course II		2		4	15	35	50
GE/OE* (2)	OE-151-CS-T / OE-152-CS-T / OE-153-CS-T OE-154-CS-T	Office Automation II / Computer Fundamentals / Introduction to Google Apps II/ Fundamentals of Computers II		2		4	15	35	50
SEC(2)	SEC-151-CS-P	Statistical Methods for Computer Science II		2		4	15	35	50
AEC(2)	AEC-151-ENG	English	2		2		15	35	50
VEC(2)	VEC-151-ENV	EVS-II	2		2		15	35	50
CC(2)	CC-151-T	From University Basket	2		2		15	35	50
Total			12	10	12	20			550

** The subjects offered to other faculty students under OE vertical are OE-151-CS -P/ OE-152-CS -T/OE-153-CS-P / OE-154-CS-T. The students of B.Sc. (Computer Science) will opt the subjects offered by other faculty given in University Basket.*

Exit option: *Award of UG Certificate in Major with 44 credits and an additional 4 credits core as per university guidelines OR Continue with Major and Minor*
Continue option: Student will select one subject among the (subject 2 and subject 3) as minor and subject 1 will be major subject

In Second Year, the "Subject 1 : Computer Science" will be Major Subject and the Minor subject will be chosen from "Subject 2 or Subject 3". Subject 2 and Subject 3 will not be available as Major Subjects in Second Year and Third Year

Level:- 5.0 (Second Year) Sem:-III

Course Type	**Course Code**	**Course Title**	**Credits**		**Teaching Scheme Hr/Week**		**Evaluation Scheme and Max Marks**		
			TH	PR	TH	PR	CE	EE	Total
Major Core (4+2)	CS-201-MJ-T	Data Structure -I	2		2		15	35	50
	CS-202-MJ-T	Database Management System I	2		2		15	35	50
	CS-203-MJ-P	Lab Course based on CS-201-MJ-T & CS-202-MJ-T		2		4	15	35	50
VSC(2)	CS-221-VSC-T	Software Engineering	2		2		15	35	50
IKS	IKS-200-T	Computations in Ancient India	2		2		15	35	50
FP/OJT/ CEP(2)	CS-231-FP	Mini Project		2		4	15	35	50
Minor (2+2)	CS-241-MN-T	Mathematics or Electronics	2		2		15	35	50
	CS-242-MN-P	Mathematics or Electronics		2		4	15	35	50
GE/OE(2)	OE-201-CS-T OE -202-CS-P OE-203-CS-T	E commerce / Web Design / Digital Marketing	2		2		15	35	50
AEC(2)	AEC-201-T	From University Basket	2		2		15	35	50

CC(2)	CC-201-T	From University Basket	2		2		15	35	50
Total			16	06	16	12			550

Level:- 5.0 (Second Year) Sem:-IV

Course Type	**Course Code**	**Course Title**	**Credits**		**Teaching Scheme Hr/Week**		**Evaluation Scheme and Max Marks**		
			TH	PR	TH	PR	CE	EE	Total
Major Core (4+2)	CS-251-MJ-T	Data Structure - II	2		2		15	35	50
	CS-252-MJ-T	Database Management System II	2		2		15	35	50
	CS-253-MJ-P	Lab Course based on CS-251-MJ-T & CS-252-MJ-T		2		4	15	35	50
VSC(2)	CS-221-VSC-P	Advanced Python Programming		2		4	15	35	50
FP/OJT/ CEP(2)	CS-281-FP	Mini Project		2		4	15	35	50
Minor (2+2)	CS-291-MN-T	Mathematics or Electronics	2		2		15	35	50
	CS-292-MN-P	Mathematics or Electronics		2		4	15	35	50
GE/OE(2)	OE-251-CS-T / OE-252-CS-P / OE-253-CS-T	E commerce / Web Design / Digital Marketing		2		4	15	35	50
SEC(2)	SEC-251-CS-P / SEC-252-CS-P	Computer Networks / Statistical Analysis using R Software		2		4	15	35	50
AEC(2)	AEC251	From University Basket	2		2		15	35	50
CC(2)	CC-251-T	From University Basket	2		2		15	35	50
Total			10	12	10	24			550

Exit option: Award of UG Diploma in Major and Minor with 88 credits and an additional 4 credits core as per university guidelines OR Continue with Major and Minor

Level:- 5.5 (Third Year) Sem:-V

Course Type	**Course Code**	**Course Title**	**Credits**		**Teaching Scheme Hr/Week**		**Evaluation Scheme and Max Marks**		
			TH	PR	TH	PR	CE	EE	Total
Major Core (8+4)	CS-301-MJ-T	Core Java	2		2		15	35	50
	CS-302-MJ-T	Operating Systems	2		2		15	35	50
	CS-303-MJ-T	Web Technology-I	2		2		15	35	50
	CS-304-MJ-T	Theory of Computer Science	2		2		15	35	50
	CS-305-MJ-P	Lab Course based on CS-302-MJ-T		2		4	15	35	50

	CS-306-MJ-P	Lab Course based on CS-301-MJ-T & CS-303-MJ-T		2		4	15	35	50
Major Elective (2+2)	CS-307-MJ-T	Data Science	2		2		15	35	50
	CS-308-MJ-P	Lab Course based on CS-307-MJ-T		2		4	15	35	50
	OR								
	CS-309-MJ-T	Database Technologies	2		2		15	35	50
	CS-3010-MJ-P	Lab Course on CS-309-MJ-T		2		4	15	35	50
	OR								
	CS-3011-MJ-T	Embedded Systems	2		2		15	35	50
	CS-3012-MJ-P	Lab Course on CS-3011-MJ-T		2		4	15	35	50
VSC(2)	CS-321-VSC-P	Advanced Python Programming		2		4	15	35	50
FP/OJT/ CEP(2)	CS-331-FP	Project		2		4	15	35	50
Minor (2)	CS-341-MN-T	Mathematics or Electronics	2		2		15	35	50
Total			12	10	12	20			550

Level:- 5.5 (Third Year) **Sem:-VI**

Course Type	**Course Code**	**Course Title**	**Credits**		**Teaching Scheme Hr/Week**		**Evaluation Scheme and Max Marks**		
			TH	PR	TH	PR	CE	EE	Total
Major Core (8+4)	CS-351-MJ-T	Advanced Java	2		2		15	35	50
	CS-352-MJ-T	Design Framework	2		2		15	35	50
	CS-353-MJ-T	Web Technology-II	2		2		15	35	50
	CS-354-MJ-T	Compiler Construction	2		2		15	35	50
	CS-355-MJ-P	Lab Course based on CS-352-MJ-T		2		4	15	35	50
	CS-356-MJ-P	Lab Course based on CS-351-MJ-T & CS-353-MJ-T		2		4	15	35	50
Major Elective (2+2)	CS-357-MJ-T	Android Programming	2		2		15	35	50
	CS-358-MJ-P	Lab Course based on CS-357-MJ-T		2		4	15	35	50
	OR								
	CS-359-MJ-T	Software Testing Tools	2		2		15	35	50
	CS-3510-MJ-P	Lab Course based on CS-359-MJ-T		2		4	15	35	50
	OE								
	CS-3511-MJ-T	Internet of Things							
	CS-3512-MJ-P	Lab Course based on CS-3511-MJ-T							
VSC(2)	CS-321-VSC-P	Agile Processes		2		4	15	35	50
FP/OJT/ CEP(4)	CS-381-OJT	OJT		4		8	30	70	100
Total			10	12	10	24			550

Level:- 6.0 (Fourth Year) Sem:-VII (Research)

Course Type	Course Code	Course Title	Credits		Teaching Scheme Hr/Week		Evaluation Scheme and Max Marks		
			TH	PR	TH	PR	CE	EE	Total
Major Core (6+4)	CS-401-MJ-T	Advanced Operating System	2		2		15	35	50
	CS-402-MJ-T	Artificial Intelligence	2		2		15	35	50
	CS-403-MJ-T	Principles of Programming Language	2		2		15	35	50
	CS-404-MJ-P	Lab Course based on CS-401-MJ-T		2		4	15	35	50
	CS-405-MJ-P	Lab Course based on CS-402-MJ-T		2		4	15	35	50
Major Elective (2+2)	CS-406-MJ-T	Advance Databases and Web Technologies	2		2		15	35	50
	CS-407-MJ-P	Lab Course on CS-406-MJ-T		2		4	15	35	50
	OR								
	CS-408-MJ-T	Cloud Computing	2		2		15	35	50
	CS-409-MJ-P	Lab Course on CS-408-MJ-T		2		4	15	35	50
	OR								
	CS-410-MJ-T	C# .NET Programming	2		2		15	35	50
	CS-411-MJ-P	Lab Course on CS-410-MJ-T		2		4	15	35	50
FP/OJT/ CEP/RP(4)	CS-431-RP	Research Project		4		8	30	70	100
	CS-451-MN	Research Methodology	4		4		30	70	100
Total			12	10	12	20			550

Level:- 6.0 (Fourth Year) Sem:-VIII (Research)

Course Type	Course Code	Course Title	Credits		Teaching Scheme Hr/Week		Evaluation Scheme and Max Marks		
			TH	PR	TH	PR	CE	EE	Total
Major Core (6+4)	CS-451-MJ-T	Design and Analysis of Algorithms	2		2		15	35	50
	CS-452-MJ-T	Mobile App Development Technologies	2		2		15	35	50
	CS-453-MJ-T	Software Project Management	2		2		15	35	50

	CS-454-MJ-P	Lab Course based on CS-451-MJ-T		2		4	15	35	50
	CS-455-MJ-P	Lab Course based on CS-452-MJ-T		2		4	15	35	50
Major Elective (2+2)	CS-456-MJ-T	Full Stack Development I	2		2		15	35	50
	CS-457-MJ-P	Lab Course based on CS-456-MJ-T		2		4	15	35	50
	OR								
	CS-458-MJ-T	Web Services	2		2		15	35	50
	CS-459MJ-P	Lab Course based on CS-458-MJ-T		2		4	15	35	50
	OR								
	CS-460-MJ-T	ASP DOT Net Programming	2		2		15	35	50
	CS-461-MJ-P	Lab Course based on CS-460-MJ-T		2		4	15	35	50
FP/OJT/ CEP(8)	CS-481-FP	Research Project		8		16	60	140	200
Total			08	14	08	28			550

Level:- 6.0 (Fourth Year) Sem:-VII (Honors)

Course Type	**Course Code**	**Course Title**	**Credits**		**Teaching Scheme Hr/Week**		**Evaluation Scheme and Max Marks**		
			TH	PR	TH	PR	CE	EE	Total
Major Core (10+4)	CS-401-MJ-T	Advanced Operating System	2		2		15	35	50
	CS-402-MJ-T	Artificial Intelligence	2		2		15	35	50
	CS403MJ-T	Principles of Programming Language	2		2		15	35	50
	CS-404-MJ-P	Lab Course based on CS401MJ		2		4	15	35	50
	CS-405-MJ-P	Lab Course based on CS402MJ		2		4	15	35	50
	CS-406-MJ-T	Advanced Networking	2		2		15	35	50
	CS-407-MJ-T	Digital Marketing	2		2		15	35	50
Major Elective (2+2)	CS-408-MJ-T	Advance Databases and Web Technologies	2		2		15	35	50
	CS-409-MJ-P	Lab Course on CS-408-MJ-T		2		4	15	35	50
	OR								
	CS-410-MJ-T	Cloud Computing	2		2		15	35	50
	CS-411-MJP-T	Lab Course on CS-410-MJ-T		2		4	15	35	50
	OR								
	CS-412-MJ-T	C# .NET Programming	2		2		15	35	50
	CS-413-MJ-P	Lab Course on CS-412-MJ-T		2		4	15	35	50
	CS-441-MN-T	Research Methodology	4		4		30	70	100
Total			16	06	16	12			

Level:- 6.0 (Fourth Year) Sem:-VIII (Honors)

Course Type	Course Code	Course Title	Credits		Teaching Scheme Hr/Week		Evaluation Scheme and Max Marks		
			TH	PR	TH	PR	CE	EE	Total
Major Core (10+4)	CS-451-MJ-T	Design and Analysis of Algorithms	2		2		15	35	50
	CS-452-MJ-T	Mobile App Development Technologies	2		2		15	35	50
	CS-453-MJ-T	Software Project Management	2		2		15	35	50
	CS-454-MJ-P	Lab Course based on CS-451-MJ-T		2		4	15	35	50
	CS-455-MJ-P	Lab Course based on CS-452-MJ-T		2		4	15	35	50
	CS-456-MJ-T	Crypto Currency Technologies	2		2		15	35	50
	CS-457-MJ-T	Cyber Security	2		2		15	35	50
Major Elective (2+2)	CS-458-MJ-T	Full Stack Development I	2		2		15	35	50
	CS-459-MJ-P	Lab Course based on CS-458-MJ-T		2		4	15	35	50
	OR								
	CS-460-MJ-T	Web Services	2		2		15	35	50
	CS-461-MJ-P	Lab Course based on CS-460-MJ-T		2		4	15	35	50
	OR								
	CS-462-MJ-T	ASP DOT Net Programming	2		2		15	35	50
	CS-463-MJ-P	Lab Course based on CS-462-MJ-T		2		4	15	35	50
FP/OJT/ CEP(4)	CS-481-OJT	OJT		4		8	30	70	100
Total			12	10	12	20			

Semester I

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – I
Course Type: Subject 1 Code : CS-101-T
Course Title :Problem Solving Using C Programming

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Previous knowledge of any programming concepts is assumed.
- Knowledge of mathematical operators.
- Students think out of the box i.e. imagination power.

Course Objectives

- To introduce the foundations of computing, programming and problem- solving using computers.
- To develop the ability to analyze a problem and devise an algorithm to solve it.
- To formulate algorithms, pseudocodes and flowcharts for arithmetic and logical problems.
- To understand structured programming approaches.
- To implement algorithms in the ‘C’ language.
- To test, debug and execute programs.

Course Outcomes

On completion of the course, student will be able to :

- Explore algorithmic approaches to problem solving.
- Control the sequence of the program and give logical outputs.
- Understand and manage Input /Output operations in ‘C’ program
- Develop modular programs using control structures and arrays in ‘C’.

Course Contents

Chapter 1	**Problem Solving Aspects**	**5 Hrs**

1.1. Introduction to problem solving using computers.
1.2. Problem solving steps.
1.3 Algorithms-definition, characteristics,examples, advantages and limitations.
1.4 Flowcharts - definition, notations, examples , advantages and limitations, Comparison with algorithms.
1.5 Programming Languages as tools, programming paradigms, types of languages

1.6 Compilation process (compilers, interpreters), linking and loading, syntax and semantic errors, testing a program
1.7 Good Programming Practices (naming conventions, documentation, indentation).

Chapter 2	**'C' Fundamentals**	**6Hrs**

2.1 History of 'C' language.
2.2 Application areas.
2.3 Structure of a 'C' program.
2.4 'C' Program development life cycle.
2.5 Function as building blocks.
2.6 'C' tokens
2.7 Character set, Keywords , Identifiers
2.8 Variables, Constants (character, integer, float, string, escape sequences, enumeration constant).
2.9 Data Types (Built-in and user defined data types).
2.10 Operators, Expressions, types of operators, Operator precedence and Order of evaluation.
2.11 Character input and output.
2.12 String input and output.
2.13 Formatted input and output.

Chapter 3	**Control Structures**	**6 Hrs**

3.1 Decision making structures:- if ,if-else, switch and conditional operator.
3.2 Loop control structures:- while ,do while, for.
3.3 Use of break and continue.
3.4 Nested structures.
3.5 Unconditional branching (goto statement).

Chapter 4	**Functions**	**6Hrs**

4.1 Concept of function, Advantages of Modular design.
4.2 Standard library functions.
4.3 User defined functions:- declaration , definition, function call, parameter passing (by value , by reference), return statement.
4.4 Recursive functions.
4.5 Scope of variables and Storage classes.

Chapter 5	**Arrays**	**7Hrs**

5.1 Concept of array.
5.2 Types of Arrays – One, Two and Multidimensional array.
5.3 Array Operations - declaration, initialization, accessing array elements.
5.4 Memory representation of two-dimensional array (row major and column major)
5.5 Passing arrays to function.
5.6 Array applications - Finding maximum and minimum, Counting occurrences, Linear search, Sorting an array (Simple exchange sort, bubble sort), Merging two sorted arrays, Matrix operations (trace of matrix, addition, transpose, multiplication, symmetric, upper/ lower triangular matrix)

Reference Books:

R1. How to Solve it by Computer, R.G. Dromey, Pearson Education.
R2. Problem Solving and Programming Concept, Maureen Sprankle,7th Edition, Pearson Publication.
R3. C: the Complete Reference, Schildt Herbert, 4th edition, McGraw Hill
R4. A Structured Programming Approach Using C, Behrouz A. Forouzan, Richard F. Gilberg, Cengage Learning India
R5. The 'C' programming language, Brian Kernighan, Dennis Ritchie, PHI
R6. Programming in C ,A Practical Approach, Ajay Mittal , Pearson
R7. Programming with C, B. Gottfried, 3rd edition, Schaum's outline Series, Tata McGraw Hill.
R8. Programming in ANSI C, E. Balagurusamy, 7th Edition, McGraw Hill.

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – I
Course Type: Subject 1 Code : CS-102-P
Course Title :Lab Course based on CS-101-T

Teaching Scheme 3 4 Hrs /Week	No. of Credits 2	Examination Scheme IE:15 Marks UE: 35 Marks

Prerequisites

- Previous knowledge of any programming concepts is assumed.
- Knowledge of mathematical operator.
- Student think the out of box ie imagination power.

Course Objectives.

- Explore and develop the algorithmic approaches to problem solving.
- Understand and implement modular programs using control structures and arrays in 'C'.
- Implement programming logic and also test, debug and execute programs.
- Implement Control the sequence of the program and give logical outputs.

Course Outcomes:-

On completion of this course, students will be able to:

- Explore and develop the algorithmic approaches to problem solving.
- Understand and implement modular programs using control structures and arrays in 'C'.
- Implement programming logic and also test, debug and execute programs.
- Implement Control the sequence of the program and give logical outputs.

Guidelines:

LabBook: The lab book is to be used as a hands-on resource, reference and record of assignment submission and completion by the student. The lab book contains the set of assignments which the student must complete as a part of this course.

Submission:

Problem Solving Assignments:

The problem solving assignments are to be submitted by the student in the form of a journalcontainingindividualassignmentsheets.EachassignmentincludestheAssignmentTitle, Problem statement, Date of submission, Assessment date, Assessment grade and instructors sign.

Programming Assignments:

Programs should be done individually by the student in the respective login. The codes should be uploaded on either the local server, Moodle, Github or any open source LMS. Print-outs of the programs and output may be taken but not mandatory for assessment.

Assessment:

Continuous assessment of laboratory work is to be done based on overall performance and lab assignments performance of student. Each lab assignment assessment will be assigned grade/marks based on parameters with appropriate weightage. Suggested parameters for overall assessment as well as each lab assignment assessment include- timely completion, performance, innovation, efficient codes and good programming practices.

Operating Environment:

For 'C' Programming:

Operating system: Linux

Editor: Any linux based editor like vi, edit etc.

Compiler: cc or gcc

LAB Course Contents

A) C Programming

Assignment 1	**Problem Solving Aspects**
• Pseudo-code to programs. • Compilation process (compilers , interpreters), linking and loading, syntax and semantic errors, testing a program • Practices (naming conventions, documentation, indentation).	
Assignment 2	**'C' Fundamentals**
• 'C' tokens and Character set, Keywords , Identifiers • character, integer, float, string, escape sequences, enumeration constant. • Built-in and user defined data types and Operators, Expressions, types of operators, Operator precedence and Order of evaluation.	
Assignment 3	**Control Structures : Conditional Structures**

• Use of if ,if-else, and. • Use of Switch case • Use of conditional operator	
Assignment 4	**Control Structures : Loop Control Structures**
• Use of While loop • Use of Do While loop • Use of for lo • Use of break and continue. • Nested structures and goto statement.	
Assignment 5	**Control Structures : Break continue and Nested Loop**
• Use of break and continue. • Nested structures and goto statement.	
Assignment 6	**Functions**
• User defined functions:- declaration , definition, function call, parameter passing (by value), return statement. •	
Assignment 7	**Recursive Functions**
• Use of Recursive functions.	
Assignment 8	**Scope of variables**
• Use of Scope of variables • Use of Storage classes.	
Assignment 9	**One Dimensional Arrays**
• One Dimensional Arrays (1D) Operations - declaration, initialization, accessing array elements. •	
Assignment 10	**One Dimensional Arrays : passing array to function**
• Assignment on Passing 1D arrays to function	
Assignment 11	**One Dimensional Arrays : Array Operations**
• Finding maximum and minimum, Counting occurrences, Linear search,	
Assignment 12	**One Dimensional Arrays : Sorting and Searching**
• Sorting an array (Simple exchange sort, bubble sort (ie arrange the data in ascending and descending order))	
Assignment 13	**Two Dimensional Arrays : Basic Operations**
• Two and Multidimensional array(2D) Operations - declaration, initialization, accessing array.	
Assignment 14	**Two Dimensional Arrays : Passing 2D arrays to functions**
• Passing 2D arrays to function. • Merging two sorted arrays,	
Assignment 15	**Two Dimensional Arrays : matrix operations**
• Matrix operations :	

- Transpose
- Addition,
- Subtraction
- Multiplication
- Symmetric,

Diagonal/upper/ lower triangular matrix

Reference Books:

R1. How to Solve it by Computer, R.G. Dromey, Pearson Education.
R2. Problem Solving and Programming Concept, Maureen Sprankle,7th Edition, Pearson Publication.
R3. C: the Complete Reference, Schildt Herbert, 4th edition, McGraw Hill
R4. A Structured Programming Approach Using C, Behrouz A. Forouzan, Richard F. Gilberg, Cengage Learning India
R5. The 'C' programming language, Brian Kernighan, Dennis Ritchie, PHI
R6. Programming in C ,A Practical Approach, Ajay Mittal , Pearson
R7. Programming with C, B. Gottfried, 3rd edition, Schaum's outline Series, Tata McGraw Hill.
R8. Programming in ANSI C, E. Balagurusamy, 7th Edition, McGraw Hill.

Note: Lab Book be prepared for this course by BOS.

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – I
Course Type: Subject 2 Code : MTC-101-T
Course Title :Matrix Algebra

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs / week	2	IE : 15 marks UE: 35 marks

Aims :

- To give the students a sufficient knowledge fundamental principles ,methods and a clear perception of innumerous power of mathematical ideas and tools and know how to use them by modeling , solving and interpreting.
- Reflecting the broad nature of the subject and developing mathematical tools for continuing further study in various fields of science.
- Enhancing students' overall development and to equip them with mathematical modeling abilities, problem solving skills,
- Creative talent and power of communication necessary for various kinds of employment.
- Enabling students to develop a positive attitude towards mathematics as an interesting and valuable subject of study
- To test, debug and execute programs.

Course Objectives

- A student should be able to recall basic facts about mathematics and should be able to display knowledge of conventions such as notations , terminology and recognize basic geometrical figures and graphical displays, state important facts resulting from their studies.

- A student should get a relational understanding of mathematical concepts and concerned structures, and should be able to follow the patterns involved, mathematical reasoning .
- A student should get adequate exposure to global and local concerns that explore them many aspects of Mathematical

Course Outcomes

Upon successful completion of this course, the student will be able to:

- Work with graphs and identify certain parameters and properties of the given graphs.
- Perform certain algorithms, justify why these algorithms work, and give some estimates of the running times of these algorithms.
- Solve basic exercises of the type: given a graph with properties X, prove that the graph also has property Y.
- Develop an appreciation for the literature on the subject and be able to read and present results from the literature.
- Write cohesive and comprehensive solutions to exercises and be able to defend their arguments.

Course Contents

Chapter 1	**Matrices**	**10 Hrs**

1.1 Matrix Operations
1.2 Elementary Matrices, Elementary Row operations
1.3 Row reduction and echelon forms
1.4 LU factorization of a matrix

Chapter 2	**Determinants**	**5 Hrs**

2.1 Introduction to determinants
2.2 Properties of determinants
2.3 Determinant by Row reduction
2.4 Determinant by Cofactor expansion along any row or column

Chapter 3	**Invertible matrices**	**5 Hrs**

3.1 3.1 The inverse of a matrix
3.2 Characterization of invertible matrices
3.3 To find inverse of a matrix by Row reduction

To find inverse of a matrix by Adjoint Method

Chapter 4	**Linear Equations**	**10 Hrs**

3.1 System of Linear equations
3.2 The matrix equation Ax=b
3.3 Gauss Elimination Method
3.4 Cramer's rule
3.5 Application of LU decomposition to solve system Ax=b

Books:

Text Book : Linear Algebra and its Applications (5th Edition)

David C Lay, Steven R. Lay, Judi J. MacDonald

Pearson Publication,2016

ISBN 978-0-321-98238-4

Unit 1: Sections 1.1, 1.2, 1.3.

Unit 2 : Sections 3.1, 3.2.

Unit 3 : Sections 2.1, 2.2, 2.3.

Unit 4: Sections 1.4, 1.5,1.6 , 3.3.

Reference Books :

1. Elementary Linear Algebra with supplemental Applications
   Author : Howard Anton and others
   Wiley Student Edition
2. Matrix and Linear Algebra (aided with MATLAB)
   Author :Kanti Bhushan Datta
   Eastern Economic Edition

Savitribai Phule Pune University **F.Y.B.Sc. (Computer Science) - Sem – I** **Course Type: Subject 2 Code : MTC-101-P** **Course Title :Mathematics Practical I**		
Teaching Scheme 3 4 Hrs /Week	No. of Credits 2	Examination Scheme IE:15 Marks UE: 35 Marks
Assignment 1	**Introduction to Python**	
• Installation of Python • Values and Types : int , float, str etc • Variables : assignment statements , printing variable values , types of variables • Boolean operators, Logical operators • Mathematical functions from math ,cmath,modules.		
Assignment 2	**Python Strings**	
• Accessing values in strings • Updating strings • String special operators • Concatenation • Repetition		
Assignment 3	**Python List and Python Tuple**	

• Accessing Values • Updating • Delete elements • Basic operations • Indexing, Slicing • Built-in Functions
Assignment 4 **Python Set**
• To create a set • To change a set in Python • To remove elements from a set • Python Set Operations • Built-in Functions with Set
Assignment 5 **Python Dictionary**
• To create a Dictionary • To change a Dictionary in Python • To remove elements from a Dictionary • Python Dictionary Operations • Built-in Functions with Dictionary.
Assignment 6 **Decision making Statements**
• IF statement • IF...ELIF...ELSE Statements: • Nested IF statements: • while loop • for loop
Assignment 7 **Use SymPy for basic Operations On Matrices**
• Addition , Subtraction , Multiplication , power etc • To Accessing elements, Row, Column of Matrix. • To create some standard Matrices.
Assignment 8 **Use SymPy for Operations on Matrices**
• To insert an element in any row or column • To insert matrix into matrix • To delete any row or column • Elementary row opreations
Assignment 9 **Use Sympyto obtain**
• The determinants of Matrix. • The rank of Matrix • The transpose Of Matrix • The reduced row echelon form of Matrix
Assignment 10 **Use SymPy to obtain**
• The inverse of a matrix • The inverse of a matrix by Row reduction • The minor and co factors of matrix • The inverse of a matrix by Adjoint Method
Assignment 11 **Use SymPy to obtain**
• Lower triangular matrix • Upper triangular matrix

● LU decomposition of matrix	
Assignment 12	**Use SymPy to solve System of Linear equations**
● Cramer's Rule ● Gauss Elimination Method ● Gauss Jordan Method ● LU decomposition Method .	

Savitribai Phule Pune University **F.Y.B.Sc. (Computer Science) - Sem – I** **Course Type: Subject 3 Code : ELC-101-T** **Course Title : Principles of Analog Electronics**		
Teaching Scheme 02 Hrs/ week	No. of Credits 2	Examination Scheme IE : 15 marks UE: 35 marks

Course Objectives

- To study various types of semiconductor devices
- To study elementary electronic circuits and systems
- To study Instrumentation System
- To study various blocks of instrumentation System
- To study smart instrumentation system

Course Outcomes

- Understand the concept of semiconductor diodes.
- Understand the different applications of FET, BJT and MOSFET.
- Understand working principle of different sensors.
- Use Op-amp for different application.

Course Contents

Chapter 1	**Semiconductor Diodes**	**05 Hrs**
Semiconductor, P and N type semiconductors, Formation of PN junction diode, it's working. Zener diode, LED, Photo diode (Symbol, working principal, list of applications only)		
Chapter 2	**Bipolar Junction Transistor (BJT)**	**05 Hrs**
Bipolar Junction Transistor (BJT) symbol, types, construction, working principle, Transistor. Amplifier configurations - CB, CC (only concept), CE configuration: input and output characteristics, Definition of α, β and ϒ , Concept of Biasing (numerical problems not expected),		
Chapter 3	**Oscillators**	**05 Hrs**
Barkhauson Criteria, Low frequency Wein-bridge oscillator, High frequency crystal oscillator		
Chapter 4	**Data converters**	**05 Hrs**

Need of Digital to Analog converters, parameters, weighted resistive network, R-2R ladder network, need of Analog to Digital converters, parameters, Flash ADC

Chapter 5	Introduction to Instrumentation System	05 Hrs

Block diagram of Instrumentation system, Definition of sensor and transducer Classification of sensors: Active and passive sensors. Specifications of sensors: Accuracy, range, linearity, sensitivity, resolution, reproducibility. Temperature sensor (Thermistor, LM-35), Passive Infrared sensor (PIR),

Actuators: DC Motor, stepper motor

Chapter 6	OPAMP as signal Conditioner	05 Hrs

Concept, block diagram of Op amp, basic parameters (ideal and practical): input and output impedance, bandwidth, differential and common mode gain, CMRR, slew rate, IC741/ LM324, Concept of virtual ground.

Reference Books:

1. Electronic Devices and Circuits I – T. L. Floyd- PHI Fifth Edition
2. Principles of Analog Electronics - A.P. Malvino
3. Sedha R.S., A Text Book Of Applied Electronics, S. Chand & Company Ltd
4. Sensors and Transducers : D. Patranabis, PHI publication, 2nd Edition
5. Sensors and Transducers : Prof A.D. Shaligram
6. Op Amp and Linear Integrated Circuits: Ramakant Gaykwad

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – I
Course Type: Subject 3 Code : ELC-102-P
Course Title : Electronics Practical Course I

Teaching Scheme	No. of Credits	Examination Scheme
04 Hrs/ week	2	IE : 15 marks UE: 35 marks

Course Objectives

- To study different semiconductor diodes.
- To understand applications of IC 555 as a multivibrator.
- To study different applications of op-amp.
- To understand applications of sensors

Course Outcomes

- Use different semiconductor diodes for various applications.

- Understand the different applications of FET, BJT and MOSFET.
- Use of different sensors for parameter measurement

Course Contents

Group A (Any 13)

1.	Study of forward and reverse bias characteristics of PN junction diode.
2.	To study the forward characteristics of LED for different colours
3.	Study of Zener diode as a voltage regulator
4.	Study of Optocoupler (mechanism and characteristics, Working principle of Light emitting diode, photo diode)
5.	Study of Transistor as a switch.
6.	Study of Transistor as a switch.
7.	Study of IC 555 as astable multivibrator used as square wave generator / clock
8.	Study of Digital to Analog Converter using R-2R ladder network
9.	Study of optical sensor (LDR)
10.	Study of temperature sensor (LM35)
11.	Study of PIR sensor
12.	Study of Study of Op amp as inverting/non-inverting amplifier
13.	Op Amp as a Unity gain follower
14.	Study of Op-amp as adder/subtractor
15.	Study of Flash ADC.
16.	Study of Wein-bridge oscillator.
17.	Study of crystal oscillator

Group B: Activity (Any 1: Equivalent to 2 Practicals)

1. Bipolar Junction Transistor (BJT) symbol, types, construction, working principle, Transistor. Amplifier configurations - CB, CC (only concept), CE configuration: input and output characteristics, Definition of α, β and Υ , Concept of Biasing (numerical problems not expected), Identification of components (Passive and Active) and study of multimeter -

   a. Minimum 10 different types of components are expected.
   b. Identification based on visual inspection / data sheets.
   c. Measure the various parameters using multimeter.

2. Technical survey of 5 electronic appliances used in different fields (Home, Hospital, Agriculture, Chemical industry, Automobile industry)

(Note: basics of the devices will be explained in theory and practical will be based on applications of different types and configurations of the devices learnt in theory. In this way they will learn in class as well as in lab and more concepts can be covered in given number of credits.)

Suggested Readings/Material:

1. Electronic Devices and Circuits I – T. L. Floyd- PHI Fifth Edition
2. Principles of Analog Electronics - A.P.Malvino
3. Sedha R.S., A Text Book Of Applied Electronics, S.Chand& CompanyLtd
4. Sensors and Transducers : D. Patranabis, PHI publication, 2nd Edition
5. Sensors and Transducers : Prof A.D.Shaligram
6. Op Amp and Linear Integrated Circuits: Ramakant Gaykwad

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – I
Course Type: GE/OE Code : OE-101-CS-T
Course Title :Office Automation I

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Previous knowledge of Computer concepts is assumed.
- Knowledge of Computer as operational tool is required.

Course Objectives

- To introduce the foundations of office automation especially word processing.
- To develop the ability to prepare the well formatted word documents.
- To prepare the documents using word processing tools such as tables, figures, shapes etc.
- To prepare the word documents using advanced automated features.

Course Outcomes

On completion of the course, student will be able to :

- Prepare the professional word documents
- Explore various tools in the word processing software.
- Develop documents using word processing advanced tools.

Course Contents

Chapter 1	**Working with Documents**.	**2 Hrs**

1.1. Opening & Saving files,
1.2. Editing text documents, Inserting, Deleting,
1.3. Cut, Copy, Paste, Undo, Redo,
1.4. Find, Search, Replace,
1.5. Formatting page &setting Margins,
1.6. Converting files to different formats,
1.7. Importing & Exporting documents, Sending files to others,

1.8. Using Tool bars, Ruler, Using Icons, using help		
Chapter 2	**Formatting Documents**	**2 Hrs**
2.1 Setting Font styles 2.2 Font selection- style, size, colour, etc. 2.3 Type face - Bold, Italic, Underline, 2.4 , Case settings, Highlighting, Special symbols. 2.5 Setting Paragraph style. 2.6 Alignments, Indents, Line Space, Margins, 2.7 , Bullets & Numbering		
Chapter 3	**Setting Page Style**	**4 Hrs**
3.1 Formatting Page 3.2 Page tab : Margins, Layout settings, Paper tray 3.3 Border & Shading 3.4 Columns 3.5 Header & Footer 3.6 Setting Footnotes & End notes 3.7 Shortcut Keys; Inserting manual page break, Column break and line break 3.8 Creating sections & frames 3.9 Anchoring & Wrapping 3.10 Printing Documents		
Chapter 4	**Setting Document Styles**	**2 Hrs**
4.1 Table of Contents 4.2 Index 4.3 Page Numbering 4.4 date & Time, Author, etc. 4.5 Creating Master Documents 4.6 Web page		
Chapter 5	**Creating Tables**	**7 Hrs**
5.1 Table settings and Drawing - Inserting ClipArts, Pictures/Files etc., 5.2 Borders, Alignments, 5.3 Insertion, deletion, 5.4 , Merging, Splitting, 5.5 Sorting, 5.6 Formula		
Chapter 6	**Special Features**	**6 Hrs**
6.1 Inserting Formula, equation, symbols 6.2 Inserting Cliparts, pictures, objects, word art 6.3 Drawing: shapes, smart art, etc 6.4 Charts 6.5 Hyperlinks, bookmarks, cross-references, Digital Signature		
Chapter 7	**Tools**	**7 Hrs**

7.1 Word Completion, Spell Checks,
7.2 Mail merge
7.3 Templates,
7.4 Creating contents for books, Creating Letter/Faxes, Creating Web pages 7.5 Hyperlinks, bookmarks, cross-references
7.5 Using Wizards
7.6 Tracking Changes, Security,.

Reference Books :

1. Illustrated Microsoft Office 365 & Word 2019 Comprehensive by Jennifer Duffy
2. Microsoft Word 365 2019 by Joan Lambert
3. Microsoft Word 2013 Bible by Lisa A Bucki

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – I
Course Type: GE/OE Code : OE-102-CS-T
Course Title : Introduction to Computers and Basics of Internet

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Basic knowledge of Computer concepts is assumed.
- Knowledge of Computer as operational tool is required.

Course Objectives

- To introduce the fundamental concepts of computers
- To introduce the basic concepts of Internet
- To develop the ability to analyses and use the computer peripherals effectively
- To develop the ability to analyses and use the internet effectively

Course Outcomes

On completion of the course, student will be able to :

- Use the computer peripherals effectively
- Use the internet for the day to day life
- Explore various applications available over the internet.

Course Contents

Chapter 1	**Fundamentals of Computers**	**8 Hrs**

1.1 Overview of a Computer-Definition, functionalities of Computer
1.2 Generations and Classification of Computers
1.3 Functional Components of a Computer
1.4 Applications Of Computers
1.5 Software and Hardware-Definition, types of software

1.6 Introduction to various Operating systems-Windows,Linux,Android,IOS		
Chapter 2	**Introduction to various Computer applications**	**6 Hrs**
2.1 Various Explorers 2.2 Editors such as Notepad, wordpad 2.3 Calculator, calendar, etc 2.4 , Paint. 2.5 Various browsers 2.6 Internet settings		
Chapter 3	**Basics of Internet**	**6 Hrs**
3.1 Definition and History of Internet 3.2 Uses and Applications of Internet 3.3 Definition of Web 3.4 Website Address and URL 3.5 Different types of Internet Connections: ● Dial up Connection ● Broad Band (ISDN, DSL, Cable) ● Wireless (Wi-Fi, WiMax, Satellite, Mobile) naming convention 3.6 Modes of Connecting Internet (Hotspot, Wi-Fi, USB Tethering)		
Chapter 3	**Browsers and Email**	**10 Hrs**
3.1 Search Engines 3.2 Web Browsers ● Popular Web Browsers (Microsoft Edge, Google Chrome, Mozilla Firefox,Safari, etc.) ● Popular Search Engines.(Google, Bing, Startpage ,DuckDuckGo etc..) 3.3 Portals 3.4 Social Networking sites, blogs 3.5 Using Browsers : ● Viewing webpage ● Downloading and uploading the website 3.6 E-mail: ● Configuring an E-mail Account ● Composing and Sending Mail ● Receiving, Replying to and Forwarding Mail ● Attachments to email		
Reference Books: 1. Computer Fundamentals by P.K. Sinha &Priti Sinha, 3rd edition, BPB pub. 2.Fundamental of Computers – By V. Rajaraman B.P.B. Publications 3. The Internet Book by Douglas E Comer		
E-Books and Online Learning Material 1. https://www.geeksforgeeks.org/computer-fundamentals-tutorial/ 2. https://www.javatpoint.com/computer-fundamaentals		

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – I
Course Type: GE/OE Code : OE-103-CS-T
Course Title : Introduction to Google Apps I

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Basic knowledge of Computer concepts is assumed.
- Knowledge of Computer as operational tool is required.
- Knowledge of Internet is required

Course Objectives

- To introduce the foundations of various Google tools.
- To develop the ability to analyses and use the tools effectively

Course Outcomes

On completion of the course, student will be able to :

- Use the google tools for the day to day life
- Explore various applications available in the google tools.
- Develop the skills to implement the skills available in the google tools.

Course Contents

Chapter 1	**Gmail**	**2 Hrs**

1.1 Configuring an E-mail Account
1.2 Composing and Sending Mail
1.3Receiving, Replying to and Forwarding Mail
1.4 Attachments to email

Chapter 2	**Google Drive**	**3 Hrs**

2.1 Opening the Drive
2.2 Creating folders, google docs, google sheets, google slides
2.3 Managing Files and folders
2.4 Sharing files and folders and managing permissions
2.5 Downloading the files and folders
2.6 Uploading files and folders
2.7 Printing files

Chapter 3	**Google Docs, Sheets and Slides**	**8 Hrs**

3.1 Creating Google docs, sheets and slides
3.2 Formatting the documents
3.3 Managing the document permissions
3.4 Uploading/downloading the documents
3.5 Special features in the docs, sheets and slides

Chapter 4	**Google Forms**	**7 Hrs**

4.1 Creating a google form

4.2 Adding various styles of the questions 4.3 settings of the google form 4.4 Creating the links of the google form and sharing the link 4.5 Creating and managing the permissions 4.6 Managing the data collected through google form		
Chapter 5	**Other Google tools**	**10 Hrs**
5.1 Google Calendar 5.2 Google Meet 5.3 Google Chat 5.4 Google Contacts 5.5 Google Photos 5.6 Google Maps		
Reference Books: 1. Complete Beginners guide to Google Apps Script by Daniel Lawrie. 2.Google Apps made easy by James Bernstein 3. My Google Apps by Sherry Kinkoph Gunter		

Savitribai Phule Pune University **F.Y.B.Sc. (Computer Science) - Sem – I** **Course Type: GE/OE Code : OE-104-CS-T** **Course Title : Fundamentals of Computers I**		
Teaching Scheme 02 Hrs/ week	No. of Credits 2	Examination Scheme IE : 15 marks UE: 35 marks
Prerequisites ● Basic knowledge of Computer concepts is assumed.		
Course Objectives ● To converse with basic terminology of computer ● To understand basics of Computer and working with Operating System ● To develop working skills with productivity enhancing tools ● To perform documentation and accounting operations		
Course Outcomes On completion of the course, student will be able to : ● Understand the concept of input and output devices of Computers ● Learn the functional units and classify types of computers ● Understand concept of software and working of operating system ● Learn basic Word processing, Spreadsheet and Presentation Graphics Software skills ● Study to use the Information Technology safely, legally, and responsibly ● Describe various uses of offices automation tools in accounting Operations		
Course Contents		
Chapter 1	**Introduction to Computers**	15 Hrs
1.1 Basics of Computers - Definition, Block Diagram, Computer Hierarchy, (Classification),		

Characteristics of Computer, Computer Memory Input and Output Devices.

1.2 Introduction to Software - Software Types - System Software, Application Software, Types of Operating Systems, Functions of Operating Systems.

1.3 Working with Windows Operating System:- Structure of Windows, Windows Explorer, File and Folder Operations, The Search, The Recycle Bin, Adding or Removing New Programs using, Control Panel, Applications in windows (Paint, Notepad, WordPad, and Calculator)

Data Processing: Files and Records, File Organization (Sequential, Direct/Random, Index)

Chapter 2	**Office Automation Tools**	15 Hrs

2.1 Definition of InformationTechnology (IT) Benefits of Information Technology (IT) Applications of Information Technology (IT)

2.2 Office Automation Tools:

1.2.1 MS-Word: Introduction, Starting MS-Word, MS-Word Screen and its Components, Elementary Working with MS-Word

1.2.2 MS-Excel: Introduction, Starting MS-Excel, Basics of Spread sheet, MS-Excel Screen and its Components, Elementary Working with MS-Excel

MS-PowerPoint: Introduction, Starting MS-PowerPoint, Basics of PowerPoint, MS-PowerPoint Screen and Its Components, Elementary Working with MS PowerPoint.

Reference Books:

1. Computer Fundamentals by: Anita Goel, Pearson Education India ISBN: 9788131742136
2. Connecting with Computer Science, by Greg Anderson, David Ferro, Robert Hilton, Course Technology, Cengage Learning,ISBN:9781439080351
3. Fundamentals of Computer : For undergraduate courses in commerce and management, ITL Education Solutions Limited, Pearson Education, ISBN:9788131733349
4. Introduction to Computer Science, 2/e, ITL Education Solutions Limited, Pearson Education, ISBN:9788131760307
5. Frontiers of Electronic Commerce, Ravi Kalakota, Andrew B. Whinston, Pearson Education,ISBN:9788177583922
6. Internet: The Complete Reference, Margaret Levine Young, Tata McGraw Hill Education Private Limited, ISBN: 9780070486997
7. On the Way to the Web: The Secret History of the Internet and Its Founders, A. Banks,Apress Publication, ISBN: 9781430208693
8. Computers and Commerce: A Study of Technology and Management at Eckert-MauchlyComputer Company, Engineering Research Associates, and Remingto, Arthur L. Norberg, MIT Press (MA),ISBN:9780262140904
9. Essential of E-commerce technology by V.Rajaraman, Prentice Hall India Learning Private Limited ISBN 9788120339378
10. Fundamentals of Computers by E. Balagurusamy, McGraw Hill
11. Computer Fundamentals by Priti Sinha, Pradeep K. Sinha, BPB Publications

Continuous Internal Evaluation – Max. Marks 15 Marks (Min. Passing Marks: 06)(Min. Passing Percentage: 40% of Max. Marks)	
The colleges need to adopt any Two Methods out of the following Methods for Continuous Internal Evaluation: 1) Offline Written Examination 2) Power Point Presentations 3) Assignments / Tutorials 4) Oral Examination 5) Open Book Test 6) Offline MCQ Test 7) Group Discussion 8) Analysis of Case Studies	
Semester End Examination: Max. Marks 35 and Duration of Examination is 2 Hours (Min. Passing Marks: 14)(Min. Passing Percentage: 40% of Max. Marks)	
Instructions: *1. Attempt all questions*	
Q. 1. Fill in the Blanks on all Units	05 Marks
Q. 2. Theory Question on Unit-1 OR Unit-2	08 Marks
Q. 3. Numerical Problem on Unit-1 OR Unit-2	14 Marks
Q. 4.Write Short Notes on all Units (Any 2 out of 4)	08 Marks

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – I
Course Type: SEC Code : SEC-101-CS-P
Course Title : Statistical Methods for Computer Science I

Teaching Scheme	No. of Credits	Examination Scheme
04 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Basic knowledge of Computer concepts is assumed.
- Basic Concepts of statistics is assumed.

Course Outcomes

On completion of the course, student will be able to :

- Present the complex data in tabular format.
- Use various diagrammatic and graphical techniques to represent statistical data and interpret the data.
- Compute various measures of central tendency, dispersion, skewness, and kurtosis using MS-Excel and interpret the results
- Establish relation between variables and estimate response for given bivariate data using software and interpret the results

List of experiments

Sr. No.	Title of the Experiment	No. of Experiments

1	Tabulation and construction of frequency distribution. (Use of at least two data sets more than 50 observations- each for constructing frequency distribution)	1
2	Diagrammatic representation of statistical data using EXCEL and data interpretation. (problems like in Paper I of SET and NET examination and other competitive examinations). (simple bar diagram, subdivided bar diagram, multiple bar diagram, percentage bar diagram, pie diagram, spike plot for Likert scale) Data interpretation from diagrams.	1
3	Graphical representation of statistical data: Histogram and frequency curve. Determination of mode graphically,Ogive curves and Pareto chart. Determination of median graphically. Data interpretation from graphs.	1
4	Summary statistics – I: Computation of measures of central tendency for ungrouped data (AM, Median and Mode)using MS-Excel by regular formula method and using direct command. (Discuss use of an appropriate measure).	1
5	Summary statistics – II: Computation of measures of central tendency for grouped data (AM, Median and Mode) using MS-Excel by regular formula method and bydirect command. (Discuss use of an appropriate measure).	1
6	Summary statistics – III: Computation of measures of dispersion for ungrouped data (Range, quartiles, variance, standard deviation, coefficient of variation) using MS-Excel by regular formula method and by direct command. Use of an appropriate measure and interpretation of results	1
7	Summary statistics – IV: Computation of measures of dispersion for grouped data (Range, quartiles, variance, standard deviation, coefficient of variation using MS-Excel by regular formula method and by direct command. Use of an appropriate measure and interpretation of results.	1
8	Computations of raw and central moments (not using the relation formula), measures of skewness and kurtosis (calculations in MS Excel by regular formula method and by direct command, using Karl Pearson's formula and moments). Use of an appropriate measure and interpretation of results.	1
9	Measures of Attributes (without MS Excel)	1
10	Scatter diagram and computation of covariance and Karl Pearson's correlation coefficient (calculations in MS-Excel by regular formula method and by direct command). Use of an appropriate measure and interpretation of results.	1

11	Fitting of line of regression $Y = a + bX$, calculations in MS-Excel by regular formula method and by direct command (use scatter plot for explaining the linear relationship).	1
12	Data Collection, its condensation and representation using MS-Excel.	1

Note:

1. Every practical is equivalent to four theory lectures per batch per week.
2. One hour is reserved for theory explanation of corresponding practical.
3. For project, a group of maximum 8 students be made. All the students in a group are given equal marks for project. Different data sets from primary or secondary sources may be collected.

Semester II

SavitribaiPhule Pune University
F.Y.B.Sc. (Computer Science) - Sem – II
Course Type: Major Course Code : CS-151-T
Course Title :Advanced C Programming

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Problem Solving tools like algorithms, flowcharts and pseudocodes.
- Basic knowledge of 'C' language.

Course Objectives

- To study advanced concepts of programming using the 'C' language.
- To understand code organization with complex data types and structures.
- To work with files.

Course Outcomes

On completion of the course, student will be able to :

- Develop modular programs using control structures, function ,pointers, arrays, strings and structures
- Design and develop solutions to real world problems using C.
- Understand and repeat the sequence of instructions and points for a memory location.
- Identification, analyzation, development, verify and document the requirements for a computing environment.

Course Contents

Chapter 1	**Pointers**	**8Hrs**

1.1. Introduction to Pointers.
1.2. Declaration, definition, initialization, dereferencing.
1.3. Pointer arithmetic.
1.4. Relationship between Arrays & Pointers- Pointer to array, Array of pointers.
1.5. Multiple indirection (pointer to pointer).
1.6. Functions and pointers- Passing pointer to function, Returning pointer from function, Function pointer.
1.7. Dynamic memory management- Allocation(malloc(),calloc()), Resizing(realloc()), Releasing(free()).,
1.8. Memory leak, dangling pointers.
1.9. Types of pointers.

Chapter 2	**Strings**	**6Hrs**

2.1 String Literals, string variables, declaration, definition, initialization.
2.2 Syntax and use of predefined string functions
2.3 Array of strings.
2.4. Strings and Pointers
2.5. Command line arguments.

Chapter 3	Structures And Unions	8Hrs
3.1. Concept of structure, definition and initialization, use of typedef. 3.2. Accessing structure members. 3.3. Nested Structures 3.4. Arrays of Structures 3.5. Structures and functions- Passing each member of structure as a separate argument, Passing structure by value / address. 3.6. Pointers and structures. 3.7. Concept of Union, declaration, definition, accessing union members. 3.8. Difference between structures and union.		
Chapter 4	**File Handling**	**6Hrs**
4.1. Introduction to streams. 4.2. Types of files. 4.3. Operations on text files. 4.4. Standard library input/output functions. 4.5. Random access to files.		
Chapter 5	**Preprocessor**	**2Hrs**
5.1. Role of Preprocessor 5.2. Format of preprocessor directive 5.3. File inclusion directives (#include) 5.4. Macro substitution directive, argumented and nested macro 5.5. Macros versus functions		
Reference Books:		
R1. C: the Complete Reference, Schildt Herbert, 4th edition, McGraw Hill R2. A Structured Programming Approach Using C, Behrouz A. Forouzan, Richard F. Gilberg, Cengage Learning India R3. The 'C' programming language, Brian Kernighan, Dennis Ritchie, PHI R4. Programming in C ,A Practical Approach, Ajay Mittal , Pearson R5. Programming with C, B. Gottfried, 3rd edition, Schaum's outline Series, Tata McGraw Hill. R6. Programming in ANSI C, E. Balagurusamy, 7th Edition, McGraw Hill.		

SavitribaiPhule Pune University **F.Y.B.Sc. (Computer Science) - Sem – II** **Course Type: Major Course Code : CS-151-P** **Course Title :Lab Course based on CS-151-P**		
Teaching Scheme 04 Hrs/ week	No. of Credits 2	Examination Scheme IE : 15 marks UE: 35 marks
Prerequisites ● Problem Solving of mathematical operator and function and array. ● Basic knowledge of 'C' language.		

Course Objectives

- To study advanced concepts of programming using the 'C' language.
- To understand code organization with complex data types and programming structures.
- To work with files and its types.

Course Outcomes

On completion of the course, student will be able to :

- Develop modular programs using function, pointers, arrays, strings and structures
- Design and develop solutions to real world problems using Advanced C programming.

Guidelines:

LabBook: The lab book is to be used as a hands-on resource, reference and record of assignment submission and completion by the student. The lab book contains the set of assignments which the student must complete as a part of this course.

Submission:

Advanced 'C' Programming Assignments:

The problem solving assignments are to be submitted by the student in the form of a journal containing individual assignment sheets. Each assignment includes the Assignment Title, Problem statement, Date of submission, Assessment date, Assessment grade and instructors sign.

Programming Assignments:

Programs should be done individually by the student in the respective login. The codes should be uploaded on either the local server, Moodle, Github or any open source LMS. Print-outs of the programs and output may be taken but not mandatory for assessment.

Assessment:

Continuous assessment of laboratory work is to be done based on overall performance and lab assignments performance of student. Each lab assignment assessment will be assigned grade/marks based on parameters with appropriate weightage. Suggested parameters for overall assessment as well as each lab assignment assessment include- timely completion, performance, innovation, efficient codes and good programming practices.

Operating Environment:

For Advanced 'C' Programming:

Operating system: Linux

Editor: Any linux based editor like vi, gedit etc.

Compiler: cc or gcc

LAB Course Contents

Advance C Programming

Assignment 1	**Pointers : Operations on pointers**

- Pointers - Declaration,

● definition, initialization, d ● dereferencing ● Pointer arithmetic.	
Assignment 2	**Pointers : Pointers and arrays**
● Pointer to array, ● Array of pointers ● pointer to pointer	
Assignment 3	**Pointers :pointers and functions**
● Passing pointer to function, ● Returning pointer from function, ● Function pointer	
Assignment 4	**Pointers :Dynamic Memory allocation**
Dynamic memory management (Allocation) ● malloc(), ● calloc(), ● Resizing(realloc()),	
Assignment 5	**Pointers :dangling pointers and free**
● Releasing (free ())., ● dangling pointers	
Assignment 6	**Strings : basic operations**
● String Literals, string variables, declaration, definition, initialization and Syntax and use of predefined string functions	
Assignment 7	**Strings : array of strings & pointers**
● Array of strings and Pointers	
Assignment 8	**Structures : Basics**
● Structure, definition and initialization, use of typedef. ● Accessing structure members and Nested Structures	
Assignment 9	Arrays of Structures and functions
● Arrays of Structures and functions- Passing each member of structure as a separate argument, ● Passing structure by value / address.	
Assignment 10	Pointers and Structures
Use of Pointers and Structures	
Assignment 11	**Unions**
Concept of Union, declaration, definition, accessing union members	
Assignment 12	**Command line arguments : basics**
● To access command-line arguments	

● Functions - atoi(), atol() and atof()
Assignment 13 | **Command line arguments : use of files**
● Arithmetic operation on arguments ● Accessing string and file using command line arguments
Assignment 14 | **File Handling**
● Streams and Types of files. ● Operations on text files. ● Standard library input/output functions and Random access to files.
Assignment 15 | **Preprocessor**
● Preprocessor and Format of preprocessor directive ● File inclusion directives (#include) ● Macro substitution directive, argumented and nested macro and macros versus functions
Reference Books:
R1. C: the Complete Reference, Schildt Herbert, 4th edition, McGraw Hill R2. A Structured Programming Approach Using C, Behrouz A. Forouzan, Richard F. Gilberg, Cengage Learning India R3. The 'C' programming language, Brian Kernighan, Dennis Ritchie, PHI R4. Programming in C ,A Practical Approach, Ajay Mittal , Pearson R5. Programming with C, B. Gottfried, 3rd edition, Schaum's outline Series, Tata McGraw Hill. R6. Programming in ANSI C, E. Balagurusamy, 7th Edition, McGraw Hill.

Note: Lab Book be prepared for this course by BOS.

Savitribai Phule Pune University **F.Y.B.Sc. (Computer Science) - Sem – II** **Course Type: Subject 2 Code : MTC-151-T** **Course Title :Graph Theory**		
Teaching Scheme 02 Hrs/ week	No. of Credits 2	Examination Scheme IE : 15 marks UE: 35 marks
Aims : ● To give the students a sufficient knowledge fundamental principles ,methods and a clear perception of innumerous power of mathematical ideas and tools and know how to use them by modeling , solving and interpreting. ● Reflecting the broad nature of the subject and developing mathematical tools for continuing further study in various fields of science. ● Enhancing students' overall development and to equip them with mathematical modeling abilities, problem solving skills, ● Creative talent and power of communication necessary for various kinds of employment. ● Enabling students to develop a positive attitude towards mathematics as an interesting and valuable subject of study		

- To test, debug and execute programs.

Course Objectives

- A student should be able to recall basic facts about mathematics and should be able to display knowledge of conventions such as notations , terminology and recognize basic geometrical figures and graphical displays, state important facts resulting from their studies.
- A student should get a relational understanding of mathematical concepts and concerned structures, and should be able to follow the patterns involved, mathematical reasoning .
- A student should get adequate exposure to global and local concerns that explore them many aspects of Mathematical

Course Outcomes

Upon successful completion of this course, the student will be able to:

- Work with graphs and identify certain parameters and properties of the given graphs.
- Perform certain algorithms, justify why these algorithms work, and give some estimates of the running times of these algorithms.
- Solve basic exercises of the type: given a graph with properties X, prove that the graph also has property Y.
- Develop an appreciation for the literature on the subject and be able to read and present results from the literature.
- Write cohesive and comprehensive solutions to exercises and be able to defend their arguments.

Course Contents

Chapter 1	**An Introduction to graph**	**8 Hrs**

1.1.Definitions, Basic terminologies and properties of graph.

1.2. Special types of graphs, some applications of special types of graph.

1.3.Matrix representation and elementary results, Isomorphism of graphs.

Chapter 2	**Connected graph**	**7 Hrs**

2.1 Walk, trail, path, cycle, more definitions and elementary properties of connectedness.

2.2 Cut edge (Bridge), Cut vertex, cut set, vertex connectivity, edge connectivity, definitions and properties.

2.3 Shortest path problem, Dijkstra's algorithm.

Chapter 3	**Euler and Hamilton path**	**7 Hrs**

3.1. The Konigsberg bridge problem, Euler trail, path, circuit and tour, elementary properties and examples, Fleury's algorithm

3.2. Hamilton path, circuit, definitions, elementary properties and examples

Chapter 4	**Trees**	**8 Hrs**

4.1. Definitions, basic terminologies, properties and applications of trees.

4.2. Weighted graph, definition and properties of spanning tree, shortest spanning tree, Kruskal's algorithm, Prim's algorithm.

4.3.Binary tree, definitions and properties, tree traversal: preorder , inorder, postorder, infix, prefix, postfix notations and examples.

Books:

Text Book:

Kenneth Rosen, Discrete Mathematics and its applications. Seventh Edition (Tata McGraw Hill).

Reference Books:

1. John Clark and Derek Holton, A first look at Graph theory, (Allied Publishers)
2. Narsingh Deo, Graph Theory with applications to computer science and engineering.(Prentice Hall)
3. C.L. Liu, Elements of Discrete Mathematics, (Tata McGraw Hill)
4. Douglas B. West, Introduction to Graph Theory, second edition.(Pearson Education)

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – II
Course Type: Subject 2 Code : MTC-152-P
Course Title :Mathematics Practical II

Teaching Scheme	No. of Credits	Examination Scheme
4 Hrs /Week	2	IE:15 Marks UE: 35 Marks

Assignment 1	**Using networkx from python do the following**

1.Generate graph G with vertex(node) set {1,2,3,4,5} and the edge set {(1,5), (1,3),(1,2),(2,3),(2,4), (3,4), (4,5)} . Draw graph G.

2. Generate graph G1 with vertex set { ‘a’,’b’,’c’,’d’} and the edge set {x=(‘a’, ’d’), y=(’b’,’c’), z=(’b’,’’d’), w =(‘a’,’c’)}. Draw graph G1 showing labeled vertices and edges.

3. Generate graph G2 with vertex set {1,2,3,4,5} and edge set {(4,5),(5,3), (2, 2),(2,3),(2,4), (3,4), (1,5)} . Draw graph G2 with vertices in red colour and edges in green.

4. Find the number of vertices, number of edges and degrees of all vertices in above graphs.

5. Verify Hand shaking lemma for above graphs.

Assignment 2	**Using networkx from python do the following**

1. Draw a regular graph on 4 vertices with degree 2.
2. Draw a regular graph on 5 vertices with degree 3.
3. Draw the star graphs on 4, 7 and 8 vertices
4. Draw the Petersen graph . Determine whether G is 2_regular. Is it 3_regular graph?
5. Find adjacency matrix and incidence matrix of each of above graphs. Find the number of vertices, number of edges and degrees of all vertices in above graphs

Assignment 3	**Using networkx from python do the following**

1. Draw the null graphs different number of vertices for example N_7, N_{17} ,N_{12} etc.
2. Draw the complete graphs for example K_5, $K_{30,}$ K_{45} etc.
3. Draw the cycle graphs such as C_8, $C_{12,}$ $C_{20,}$ C_{35} etc.
4. Draw the wheel graphs for such as W_5, $W_{10,}$ $W_{21,}$ $W_{30.}$
5. Draw the complete bipartite graphs $K_{4,3}$, $K_{1,8}$, $K_{5,9}$ etc.

Assignment 4	**Using networkx from python do the following**

1. Draw a directed graph D1 with vertex set V= {1, 2, 3, 4,5} and directed edge set E = {(1,4) , (2,3), (1,2), (5,3), (5,1), (4,1), (3,2),(5,2), (5,4)}. Draw underlying graph of D1, Find in degrees and out degrees of all vertices in D1.
2. Draw a directed graph D2 with vertex set V= {1, 2, 3, 4 } and directed edge set

i. {(2,4) , (2,3), (1,3), (4,1), (3,2),(1,2)}.Draw underlying graph of D2, Find indegrees and out degrees of all vertices in D2.

3. Draw any symmetric directed graph on given number of vertices.
4. Draw any asymmetric directed graph on given number of vertices.
5. Draw any complete symmetric directed graph on given number of vertices.
6. Draw any complete asymmetric directed graph on given number of vertices.

Assignment 5	**Using networkx from python do the following**

1. Create a simple graph G. Draw graph G with nodes and edges in colors of your choice.
2. Create and draw complement of above G. Determine whether the complement is simple graph.
3. Determine whether G is bipartite.
4. Find the number of components in the graph G.
5. Determine whether G is connected. Determine whether the complement of G is connected.

Assignment 6	**Using networkx from python do the following**

1. Draw K = Complete graph K_5, H= complement of N_5 . Determine whether K is isomorphic to H.
2. Generate and draw any 2 graphs with names G1 and G2. Determine whether G1 is isomorphic to G2.
3. Draw union of graphs G1 and G2.
4. Draw intersection of graphs G1 and G2.
5. Draw product of graphs G1 and G2.

Assignment 7	**Using networkx from python do the following**

1. Draw any graph G.
2. In the graph G add some vertices and add some edges.
3. From the graph G delete some vertices and delete some edges.
4. Determine whether G is connected graph.

5. Find the vertex connectivity and edge connectivity of the graph G.	
Assignment 8	**Using networkx from python do the following**
1. Draw any connected graph G. 2. Find all bridges, all cut vertices (articulation points) and cut set in G. 3. Find the vertex connectivity and edge connectivity of G. 4. Find the eccentricity of every vertex in G. 5. Find center, radius and diameter of graph G.	
Assignment 9	**Using networkx from python do the following**
1. Draw any connected graph G. 2. Find all paths in G and all trails in G. 3. Draw paths of some given lengths from G. 4. Find all cycles in graph G. Is it Hamiltonian graph? 5. Determine whether G is Eulerian graph, whether it is Semi Eulerian graph.	
Assignment 10	**Using networkx from python do the following**
1. Draw any connected graph G. Determine whether G is a tree. 2. Draw spanning tree T in G. 3. Find the number of vertices in spanning tree of G. 4. Find the number of edges in spanning tree of G. 5. Determine whether spanning tree T of G is a binary tree.	
Assignment 11	**Using networkx from python do the following**
1. Draw any graph T containing n number of vertices and n – 1 edges. 2. Determine whether T is a tree. 3. Determine whether T is a binary tree. 4. Determine whether T is a bipartite graph . 5. Find center, radius and diameter of graph T.	
Assignment 12	**Using networkx from python do the following**
1. Draw balanced binary trees of heights 2 ,4, and 5 etc. 2. Draw ternary trees of heights 1 and 3 etc. 3. Draw any n-ary tree of height h for given n and h . 4. Find the no of vertices and edges in given trees, Verify the relation between them. 5. Find center, radius and diameter of above trees.	

Savitribai Phule Pune University **F.Y.B.Sc. (Computer Science) - Sem – II** **Course Type: Subject 3 Code : ELC-151-T** **Course Title : Principles of Digital Electronics**		
Teaching Scheme 02 Hrs/ week	No. of Credits 2	Examination Scheme IE : 15 marks UE: 35 marks
Course Objectives		

- To learn different number system and their inter conversion.
- To understand logic gates and their applications.
- To study rules and laws of Boolean Algebra.
- To understand design of combinational circuit and their different types.

Course Outcomes

- 1. To learn different number system and their inter conversion.
- 2. To understand logic gates and their applications.
- 3. To study rules and laws of Boolean Algebra.
- 4. To underst and design of combinational circuit and their different types..

Course Contents

Chapter 1	**Number Systems and Digital Codes**	**07 Hrs**

Introduction to decimal, binary, octal and hexadecimal number system and their inter-conversions, the concept of 1's and 2's complements, binary addition, binary subtraction using 1's and 2's complements. BCD code, Excess-3 code, Gray code and ASCII code.

Chapter 2	**Logic Gates**	**03 Hrs**

Logic gates – basic and derived (symbol, Boolean equation and truth table), concept of universal gates.

Chapter 3	**Logic Families**	**04 Hrs**

Introduction of CMOS and TTL logic families. Parameters of logic families: voltage levels, propagation delay, noise margin, fan in, fan out, power dissipation Comparison between CMOS and TTL logic families.

Chapter 4	**Boolean Algebra**	**05 Hrs**

Laws of Boolean Algebra, De-Morgan's theorems, simplification of logic equations using Boolean algebra, minterms, maxterms, Boolean expression in SOP and POS form, conversion of SOP/POS expression to its standard SOP/POS form.

Chapter 5	**Introduction to Karnaugh Map**	**05 Hrs**

Introduction to Karnaugh map,problems based on SOP (up to 4 variables), digital designing using K-map for 3-bit gray tobinary and binary to gray conversion. Ex-OR gate as a 4-bit Parity Checker and Generator.

Chapter 6	**Combinational Circuits**	**05 Hrs**

Introduction to Arithmetic Circuits, half adder, full adder, half subtractor, full subtractor, four-bit parallel adder, universal adder / subtractor, digital comparator, introduction to ALU. Introduction, Multiplexer (2:1, 4:1), demultiplexer (1:2, 1:4) and their applications. Encoders: decimal to BCD/binary, 3x4 matrix keyboard encoder and priority encoder. Decoders: BCD to decimal and BCD to seven segment decoder.

Reference Books:

1. Digital Design - M. Morris Mano, PHI, New Delhi.
2. Digital Systems Principles and Applications - Ronald J. Tocci.
3. Digital electronics - G. K. Kharate, Oxford University Press.
4. Fundamentals of Digital Circuits - Anand Kumar.

5. Digital Principles and Applications - Malvino and Leach, TMG Hill Edition.

Savitribai Phule Pune University **F.Y.B.Sc. (Computer Science) - Sem – II** **Course Type: Subject 3 Code : ELC-152-P** **Course Title : Electronics Practical Course II**		
Teaching Scheme 04 Hrs/ week	No. of Credits 2	Examination Scheme IE : 15 marks UE: 35 marks

Course Objectives

- To understand logic gates ICs and their applications in Digital Design.
- To design different digital circuits using logic gates.
- To study different combinational circuits.

Course Outcomes

- Understand the design and build of digital circuits using logic gates.
- Use breadboard / tag-board for building small electronic circuits.
- Design digital circuits for different applications.
- Validate observed outputs with expected theoretical outputs.

Course Contents

Group A (Any 13)

1.	Verification of logic gates by using digital ICs.
2.	Realization of basic gates using discrete components
3.	Realization of basic gates using universal logic gates.
4.	Verification of De Morgan’s theorems.
5.	Study of half adder and full adder using logic gates.
6.	Study of half subtractor and full subtractor using logic gates.
7.	4-bit binary parallel adder and subtractor using IC7483.
8.	3-bit binary to Gray conversion using logic gates.
9.	3-bit Gray to Binary conversion using logic gates.
10.	Study of EX-OR gate as a 4-bit parity generator.
11.	Study of EX-OR gate as a 4-bit parity checker.
12.	Study of 1-bit digital comparator.
13.	Study of ALU using IC 74181.
14.	Study of multiplexer and demultiplexer.
15.	Study of Decimal to BCD/Binary encoder.

16.	Study of Priority Encoder IC 74148
17.	Study of BCD to seven segment decoder using IC 7447

Group B: Activity (Any 1: Equivalent to 2 Practicals)

1. Perform any 2 experiments from Group A using circuit simulation software LTSPICE / Circuit Mod / Proteus etc. (Give preference to not performed experiments)

2. Perform survey of following topics –

   a. Study of laboratory safety and precautionary measures.

   b. Study of e-waste management or any relevant topic of Electronics.

Suggested Readings/Material:

1. Digital Design - M. Morris Mano, PHI, New Delhi.
2. Digital Systems Principles and Applications - Ronald J. Tocci.
3. Digital electronics - G. K. Kharate, Oxford University Press.
4. Fundamentals of Digital Circuits - Anand Kumar.
5. Digital Principles and Applications - Malvino and Leach, TMG Hill Edition.

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – II
Course Type: GE/OE Code : OE-151-CS-T
Course Title :Office Automation II

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Previous knowledge of Computer concepts is assumed.
- Knowledge of Computer as operational tool is required.

Course Objectives

- To introduce the foundations of office automation especially Presentation Skills.
- To develop the ability to prepare the well formatted powerpoint presentations.
- To prepare the presentations using powerpoint presentation tools such as tables, figures, shapes, images, audio, video etc.
- To prepare the presentations using advanced automated features such as animation, slide shows, etc.

Course Outcomes

On completion of the course, student will be able to :

- Prepare the professional presentations
- Explore various tools in the powerpoint presentation software.
- Develop documents using powerpoint advanced tools.

Course Contents

Chapter 1	Introduction to Presentation.	5 Hrs
1.1. Opening new presentation, 1.2. Different presentation templates, 1.3. Setting backgrounds, 1.4. Selecting presentation layouts		
Chapter 2	**Creating and Formatting Presentation**	**8 Hrs**
2.1 Presentation style, 2.2 Adding text to the Presentation. 2.3 Adding style 2.4 Colour, gradient fills 2.5 Arranging objects 2.6 Adding Header & Footer 2.7 Slide Background, Slide layout		
Chapter 3	**Adding Graphics and effects to Presentation**	**12 Hrs**
3.1 Inserting pictures, movies, tables etc into presentation, 3.2. Drawing Pictures using Draw 3.3. Setting Animation 3.4 Transition Effect		
Chapter 4	**Printing and showing a presentation**	**5 Hrs**
4.1 Printing Handouts 4.2 Generating Standalone Presentation viewer 4.3 Presenting the presentation using various styles		

Reference Books :

1. Microsoft Powerpoint by James Holler
2. Powerpoint for dummies, office 2021 Edition Doug Lowe
3. Learn Microsoft Office 2019 by Linda Foulkes

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – II
Course Type: GE/OE Code : OE-152-CS-T
Course Title : Computer Fundamentals

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Basic knowledge of Computer concepts is assumed.
- Knowledge of Computer as operational tool is required.

Course Objectives

- To introduce the fundamental concepts of computers.
- To study the basics of Computer System
- To introduce the computer peripherals and other devices

- To learn how to configure computer devices
- To Learn Basic Commands of Operating system and application software

Course Outcomes

On completion of the course, student will be able to :

- Use the computers for the day to day life
- Learn the fundamental concepts of computer science
- Explore various applications available in the computers.
- Explain the needs of hardware and software required for a computation task

Course Contents

Chapter 1	**Introduction to Computers**	**7 Hrs**

1.1 Introduction, Characteristics of Computers,
1.2 Block diagram of computer
1.3 Computer Generations
1.4 Types of computers and features- Mini Computers, Micro Computers, Mainframe Computers, Super Computers, Laptops and Tablets
1.5 Types of Programming Languages- Machine Languages, Assembly Languages, High Level Languages

Chapter 2	**Introduction to Computer Peripherals**	**8 Hrs**

2.1 Primary And Secondary storage devices
2.2 Primary storage devices – RAM, ROM, PROM, EPROM
2.3 Secondary Storage Devices - CD, HD, Pen drive
2.4 I/O Devices- Scanners, Digitizers, Plotters, LCD, Plasma Display
2.5 Pointing Devices –Mouse, Joystick, Touch Screen
2.6 Number Systems, Octal, Hexadecimal system Conversion, Simple
2.7 Addition, Subtraction, Multiplication, Division

Chapter 3	**Number System**	**6 Hrs**

3.1 Introduction to Binary Number System
3.2 Introduction to Octal Number System
3.3 Introduction to Hexadecimal Number System
3.4 Addition, Subtraction, Multiplication, Division

Chapter 4	**Operating Systems**	**4 Hrs**

4.1 Definition of Operating System
4.2 Functions of Operating System
4.3 Role of Operating System
4.4 Types of Operating System

Chapter 5	**Introduction to Computer Networking**	**5 Hrs**

5.1 Network definition Common terminologies: LAN, WAN, Node, Host, Workstation, bandwidth, Interoperability
5.2 Network Administrator,
5.3 Network Security
5.4 Network Components: Severs, Clients

5.5 Communication Media 5.6 Types of network: Peer to Peer, Clients Server
Reference Books: 1. Computer Fundamentals by P.K. Sinha &Priti Sinha, 3rd edition, BPB pub. 2.Fundamental of Computers – By V. Rajaraman B.P.B. Publications 3. Computer Networks – By Tennenbum Tata MacGrow Hill Publication
E-Books and Online Learning Material 3. https://www.geeksforgeeks.org/computer-fundamentals-tutorial/ 4. https://www.javatpoint.com/computer-fundamaentals

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – II
Course Type: GE/OE Code : OE-153-CS-T
Course Title : Introduction to Google Apps II

Teaching Scheme	No. of Credits	Examination Scheme
02 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Basic knowledge of Computer concepts is assumed.
- Knowledge of Computer as operational tool is required.
- Knowledge of Internet is required

Course Objectives

- To introduce the specialized Google tools.
- To develop the ability to analyses and use the tools effectively and skillfully

Course Outcomes

On completion of the course, student will be able to :

- Use the google tools for the day to day life
- Explore various applications available in the google tools.
- Develop the skills to implement the skills available in the google tools.

Course Contents

Chapter 1	**Creating You Tube Channel**	**10 Hrs**
1.1 Creating a you tube channel 1.2 Managing the channel, permissions, playlists, etc 1.3 Uploading the videos on the channel 1.4 Live Streaming		
Chapter 2	**Creating a website**	**10 Hrs**
2.1 Creating a website 2.2 Managing home page 2.3 Creating Menus on the website 2.4 Adding pages to the site		

2.5 Setting up themes 2.6 Adding tools such as tables, placeholders, hyperlinking, buttons, maps, etc 2.7 Publishing a site		
Chapter 3	**Google Classroom**	**10 Hrs**
3.1 Creating Google classroom 3.2 Creating a class 3.3 Streaming a class 3.4 Adding students to class 3.5 Adding class work : assignments, quiz, question, material, etc 3.5 Giving grades to the assignments 3.6 joining a class (from student side)		
Reference Books: 1. Complete Beginners guide to Google Apps Script by Daniel Lawrie. 2.Google Apps made easy by James Bernstein 3. My Google Apps by Sherry Kinkoph Gunter		

Savitribai Phule Pune University **F.Y.B.Sc. (Computer Science) - Sem – II** **Course Type: GE/OE Code : OE-154-CS-T** **Course Title : Fundamentals of Computers II**		
Teaching Scheme 02 Hrs/ week	No. of Credits 2	Examination Scheme IE : 15 marks UE: 35 marks
Prerequisites ● Basic knowledge of Computer concepts is assumed.		
Course Objectives ● To build an understanding of the fundamental concepts of computer networking. ● To familiarize the student with the basic terminology of the computer networking area. ● To understand computer network technology with various devices independently. ● To identify the different types of network topologies and protocols.		
Course Outcomes On completion of the course, student will be able to : ● Understand the basic concepts of Networking and Cyber Security. ● Describe Cyber Security Laws and concepts of Digital Signature ● Identify the different types of Network devices and their functions within a Network. ● Elaborate the Internet Services and related terms of Internet. ● Evaluate information security threats.		
Course Contents		
Chapter 1	**Introduction to Computers Network and E-Commerce:**	**15 Hrs**
1.1 Introduction Importance of Networking Computer Network (LAN, WAN, MAN) Network Components (Hub, Switch, Bridge, Gateway, Router, Modem). 1.2 Network Topology, Wireless Network Internet and Internet application Introduction, Internet evolution, Working of Internet, Use of Internet, Overview of World Wide Web (Web Server		

and Client).
1.3 IEEE802.11 -Wi-Fi: Types of Wi-Fi, Uses of Wi-Fi
Near by Share : Applications of Nearby share

Chapter 2	Introduction to Internet & Cyber Security	15 Hrs

2.1 Concept of Internet, Internet Service Providers(ISP), Services Provided by the Internet: E-mail, Search Engine
2.2 Information security overview – Background and current scenario Types of Attacks, Goals of security:

2.3 Overview of security threats, Weak / Strong passwords and password cracking, Insecure Network connections, Digital Signature.
2.4 Cyber Security: Cybersecurity definition, Cybercrime: Classification of Cybercrime: a)Email spoofing b) Spamming c) Identity Theft d) Online Fraud e) ATM Skimming f) Credit Card Fraud
2.5 Overview of Indian Information Technology Act 2002

Reference Books:

1. Computer Fundamentals by: Anita Goel, Pearson Education India ISBN: 9788131742136
2. Connecting with Computer Science, by Greg Anderson, David Ferro, Robert Hilton, Course Technology, Cengage Learning,ISBN:9781439080351
3. Fundamentals of Computer : For undergraduate courses in commerce and management, ITL Education Solutions Limited, Pearson Education, ISBN:9788131733349
4. Introduction to Computer Science, 2/e, ITL Education Solutions Limited, Pearson Education, ISBN:9788131760307
5. Frontiers of Electronic Commerce, Ravi Kalakota, Andrew B. Whinston, Pearson Education,ISBN:9788177583922
6. Internet: The Complete Reference, Margaret Levine Young, Tata McGraw Hill Education Private Limited, ISBN: 9780070486997
7. On the Way to the Web: The Secret History of the Internet and Its Founders, A. Banks, Apress Publication, ISBN: 9781430208693
8. Computers and Commerce: A Study of Technology and Management at Eckert-Mauchly Computer Company, Engineering Research Associates, and Remingto, Arthur L. Norberg, MIT Press (MA),ISBN:9780262140904
9. Essential of E-commerce technology by V.Rajaraman, Prentice Hall India Learning Private Limited ISBN 9788120339378
10. Fundamentals of Computers by E. Balagurusamy, McGraw Hill
11. Computer Fundamentals by Priti Sinha, Pradeep K. Sinha, BPB Publications
12. Computer Networks - Andrew Tanenbaum (III Edition)
13. Complete Guide to Networking - Peter Norton
14. Data Communications & Networking - Behrouz Ferouzan (III Edition)
15. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives – Nina Godbole, SunitBelapure, Wiley: April 2011 India Publications Released.
16. Cyber Security Essentials- James Graham Richard Howard Ryan Olson

Indian Information Technology Act 2002

Continuous Internal Evaluation – Max. Marks 15 Marks (Min. Passing Marks: 06)(Min. Passing Percentage: 40% of Max. Marks)	
The colleges need to adopt any Two Methods out of the following Methods for Continuous Internal Evaluation: 9) Offline Written Examination 10) PowerPointPresentations 11) Assignments / Tutorials 12) OralExamination 13) Open Book Test 14) Offline MCQ Test 15) Group Discussion 16) Analysis of Case Studies	
Semester End Examination: Max. Marks 35 and Duration of Examination is 2 Hours (Min. Passing Marks: 14)(Min. Passing Percentage: 40% of Max. Marks)	
Instructions: *1. Attempt all questions*	
Q. 1. Fill in the Blanks on all Units	05 Marks
Q. 2. Theory Question on Unit-1 OR Unit-2	08 Marks
Q. 3. Numerical Problem on Unit-1 OR Unit-2	14 Marks
Q. 4.Write Short Notes on all Units (Any 2 out of 4)	08 Marks

Savitribai Phule Pune University
F.Y.B.Sc. (Computer Science) - Sem – II
Course Type: SEC Code : SEC-151-CS-P
Course Title : Statistical Methods for Computer Science II

Teaching Scheme	No. of Credits	Examination Scheme
04 Hrs/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Basic knowledge of Computer concepts is assumed.
- Basic Concepts of statistics is assumed.

Course Outcomes

On completion of the course, student will be able to :

- Fit second-degree curve, and exponential curves.
- Estimate trends by using time series data.
- Understand concept of probability.
- Estimate probabilities of standard probability distributions
- Perform tests based on normal, Chi-Square, t and F distributions.

List of experiments

Sr. No.	Title of the Experiment	No. of Experiments

1	Fitting of second-degree curve $Y = a + bX + cX^2$ using MS-Excel. (Use of scatter plot for explaining the nonlinear relationship between two variables)	1
2	Fitting of exponential curve of type $Y = ab^x$, $Y = aX^b$ using MS-Excel. (Use of scatter plot for explaining the nonlinear relationship between two variables)	1
3	Time Series- Estimation of trend by using the method of moving averages by using regular formula method and by using MS-Excel.	1
4	Time Series- Estimation of trend by using exponential smoothing by regular formula and by using MS-Excel.	1
5	Problems on probability theory (P(A∪B), P(A'∪B'), P(A'∩B), P(A'∩B') Use Venn diagram whenever possible.	1
6	Computation of probability values for Normal distribution and Chi-square distribution using MS Excel command.	1
7	Computation of probability values for t distribution and F distribution using MS Excel command.	1
8	Test for means and proportion (Z-test) i) $H_0: \mu = \mu_0$ vs $H_1: \mu \neq / > / < \mu_0$, σ^2 known ii) $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 \neq / > / < \mu_2$, σ^2 known iii) $H_0: P = P_0$ vs $H_1: P \neq / > / < P_0$ $H_0: P_1 = P_2$ vs $H_1: P_1 \neq / > / < P_2$	1
9	Test based on students t i) $H_0: \mu = \mu_0$ vs $H_1: \mu \neq / > / < \mu_0$, σ^2 unknown ii) $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 \neq / > / < \mu_2$, σ^2 unknown iii) Paired t-test	1
10	Test based on χ^2 i) Goodness of fit ii) Independence of attributes (2 x 2). iii) Independence of attributes (2 x 3 or 3 x 2 or 3 x 3)	1
11	Tests based on F-distribution i) $H_0: \sigma_1^2 = \sigma_2^2$ vs $H_1: \sigma_1^2 \neq / > / < \sigma_2^2$, means known ii) $H_0: \sigma_1^2 = \sigma_2^2$ vs $H_1: \sigma_1^2 \neq / > / < \sigma_2^2$, means unknown	1
12	Project (Part-II) - Analysis of data collected in semester-I	1

Note: 1. Every practical is equivalent to four theory lectures per batch per week 2. One hour is reserved for theory explanation of corresponding practical. 3. For project, a group of maximum 8 students be made. All the students in a group are given equal marks for project. Different data sets from primary or secondary sources may be collected.

BOS: Banking, Finance and Insurance
FY Semester-II
Subject: Retail Insurance, Paper-II

Subject Code	Type of Course	Name of the Subject	Credits	Lectures per Week in Clock Hour
OE160COM	GE / OE	Retail Insurance, Paper-II	2	2

Course Objectives:

1. To acquaint students with the concepts of insurance.
2. To cultivate awareness regarding the basic knowledge of the insurance sector.
3. To discern the disparity between Life and General Insurance.

Unit	Title and Contents	No. of Lectures in Clock Hours
1	**Introduction of Retail Insurance:** 1.1 Concept and Definition 1.2 Characteristics of Insurance 1.3 Need of Insurance 1.4 Importance and Scope of Insurance 1.5 Principles of Insurance 1.6 Economic and Commercial Significance of Insurance 1.7 Difference between Life and General Insurance	10
2	**Life Insurance Product:** 2.1 Life Insurance – Meaning and Origin 2.2 Features of Life Insurance 2.3 Advantages of Life Insurance 2.4 Importance of Life Insurance Policies 2.5 Types of Life Insurance Policies 2.6 Procedure of taking Life Insurance Policies 2.7 Nomination and Assignment of Life Insurance Policies	10
3	**General Insurance Product:** 3.1 General Insurance – Meaning and Origin 3.2 Importance of General Insurance 3.3 Types of General Insurance Policies 3.4 Fire Insurance 3.5 Marine Insurance 3.6 Motor Vehicle Insurance 3.7 Theft & Burglary Insurance 3.8 Procedure for taking Fire Insurance Policy 3.9 Procedure for taking Marine Insurance Policy	10

Course Outcomes: After completion of the course, students will be able to:

1. Understand the concepts of insurance.
2. Remember the basic knowledge of the insurance sector.
3. Analyze the disparity between Life and General Insurance.

Teaching Methodology

The Teacher can use the following Methods as Teaching Methodology:

1. Class Room Lectures
2. Guest Lectures of Professionals, Industry Experts etc.
3. Teaching with the help of ICT tools
4. Visits to various Professionals Units, Companies and Business / Industry Units
5. Group Discussion / Debates
6. Assignments, Tutorials, Presentations, Role Play etc.
7. YouTube Lectures developed by MHRD, UGC, Government of Maharashtra, University etc.
8. Analysis of Case Studies

Internship for Students if any: Not applicable

List of Recommended Books and Study Materials

1. Insurance Principles and Practice" by M.N. Mishra 2019, Publisher: McGraw Hill Education India
2. "Insurance Law and Practice" by Dr. S.K. Kataria 2020, Publisher: Bharat Law House
3. "General Insurance Underwriting Principles & Practice" by Dr. Rakesh Agarwal 2018, Publisher: Taxmann Publications
4. "Life Insurance: Principles and Practice" by Dr. C.L. Tyagi 2017, Publisher: S. Chand & Company Ltd.
5. "Insurance Handbook" by Dr. H.L. Kumar 2021, Publisher: New Age International Publishers
6. Principles and Practice of Life Insurance" by S.B. Mathur and R.K. Gupta 2016, Publisher: Taxmann Publications
7. "General Insurance: Principles and Practice" by Dr. N. D. Vohra 2018, Publisher: S. Chand & Company Ltd.
8. "Insurance: Theory and Practice" by S. S. Hundal and V.K. Srivastava 2019, Publisher: Himalaya Publishing House).
9. "Risk Management and Insurance Planning" by Dr. Jyotsna Sethi 2020, Publisher: Excel Books
10. "Insurance Planning and Management" by Dr. V. Ravi Kumar 2017, Publisher: McGraw-Hill Education
11. Insurance Regulatory and Development Authority of India (IRDAI): https://www.irdai.gov.in/
12. Insurance Information Bureau of India (IIB): https://www.iib.gov.in/
13. Insurance Institute of India (III): https://www.insuranceinstituteofindia.com/
14. National Insurance Academy (NIA): https://www.niapune.org.in/
15. Insurance Companies Websites
16. https://www.licindia.in/
17. https://www.icicilombard.com/
18. https://www.hdfcergo.com/

Scheme of Examination

Scheme of Examination: 1. Internal Assessment: 30% and 2. External Assessment: 70%

Scheme of Examination	Exam Format	Min. Passing Marks
Continuous Internal Evaluation (CIE) (15 Marks)	The colleges need to adopt any Two Methods out of the following Methods for Continuous Internal Evaluation: 1. Offline Written Examination 2. Power Point Presentations 3. Assignments / Tutorials 4. Oral Examination 5. Open Book Test 6. Offline MCQ Test 7. Group Discussion 8. Analysis of Case Studies	Min. 06 Marks (40% of Passing)

SEE / External Exam (35 Marks) (Total 2 Hours Duration)	***Instructions:*** *1. Question No. 1 is Compulsory.* *2. Attempt any Three Questions from Question No. 2 to 5.* Q. 1: Fill in the Blanks on all Units = 05 Marks Q. 2: Theory Question on Unit-1 = 10 Marks Q. 3: Theory Question on Unit-2 = 10 Marks Q. 4: Theory Question on Unit-3 = 10 Marks Q. 5: Short Notes on all Units (Any 2 out of 4) = 10 Marks	Min. 14 Marks (40% of Passing)
Total 50 Marks **Separate Passing for Internal Assessment (CIE) and External Exam (SEE)**		

Restructured Syllabus (2024 Pattern as per NEP-2020)

Faculty: COMMERCE & MANAGEMENT (Commerce)

Generic Elective / Open Elective for Other Faculty

BOS: Banking, Finance and Insurance

Subject: Indian Banking System

Academic Year: 2024-2025

Programme Name	Subject Code	Type of Course	Name of the Subject	Credits	Lectures per Week in Clock Hour
Humanities, Science & Technology, and Inter-Disciplinary		GE / OE	Indian Banking System	2	2

Course Objectives:

1. To inform the students about the structure and various types of Banks working in India.
2. To make the students understand the role and functions of Commercial Banks in India.
3. To provide details about the Cooperative Banking in India.

Unit	Title and Contents	No. of Lectures in Clock Hours
1	**Structure of Indian Banking System:** 1.1 Evolution of Banking in India 1.2 Reserve Bank of India: History and Functions 1.3 Structure of Indian Banking System 1.4 Difference between Scheduled and Non-Scheduled Banks	8
2	**Commercial Banking in India:** 2.1 Public Sector Banks in India: SBI and Nationalized Banks in India 2.2 Private Sector Banks in India: Old and New Age Private Sector Banks, Foreign Banks working in India 2.3 Regional Rural Banks 2.4 Differentiated Banks in India: Payment Banks, Small Finance Banks and Digital Banking Units	10
3	**Cooperative Banking System in India:** 3.1 Objectives, Functions, Progress and Problems of: 3.1.1 Primary Agricultural Cooperative Credit Societies 3.1.2 District Central Cooperative Banks 3.1.3 State Cooperative Banks 3.2 Land Development Bank and Branches of LDB 3.3 Urban Cooperative Banks	8
4	**Review of Indian Banking System:** 4.1 Mergers and Acquisitions in Banking Sector 4.2 Role of Banking System in the Economic and Development	4

Course Outcomes: After completion of the course, students will be able to:

1. Understand the structure of Indian Banking System.
2. Comprehend the working of commercial banks in India.
3. Understand role and functions Co-operative Banks in India.
4. Analyse relevant case studies.

Teaching Methodology

The Teacher can use the following Methods as Teaching Methodology:

1. Class Room Lectures
2. Guest Lectures of Professionals, Industry Experts etc.
3. Teaching with the help of ICT tools
4. Visits to various Professionals Units, Companies and Business / Industry Units
5. Group Discussion / Debates
6. Assignments, Tutorials, Presentations, Role Play etc.
7. YouTube Lectures developed by MHRD, UGC, Government of Maharashtra, University etc.
8. Analysis of Case Studies

Internship for Students if any: Not applicable

List of Recommended Books and Study Materials

1. P. L. Varshneya, Banking: Law and Practice
2. N. S. Toor, Handbook of Banking Information
3. Reserve Bank of India Publications, Functions and Working of the RBI
4. Vasant Desai, Central Banking and Economic Development
5. S. N. Maheshwari, Banking
6. H. R. Machiraju, Indian Banking System
7. Annual Reports of Banks and RBI
8. RBI Bulletin
9. Report on Currency & Finance: Reserve Bank of India Publication.
10. Report on Trend and Progress of Banking in India, Financial Stability Report: RBI Publications
11. S.B. Gupta, Monetary Economics
12. Reserve Bank of India Publication, Report on Trends and Progress of Banking in India
13. www.rbi.org
14. www.nabard.org
15. www.mahagramin.in
16. www.vkgb.co.in
17. www.sahakarayukta.maharashtra.gov.in

Scheme of Examination

Scheme of Examination: 1. Internal Assessment: 30% and 2. External Assessment: 70%

Scheme of Examination	Exam Format	Min. Passing Marks
Continuous Internal Evaluation (CIE) (15 Marks)	The colleges need to adopt any Two Methods out of the following Methods for Continuous Internal Evaluation: 1. Offline Written Examination 2. Power Point Presentations 3. Assignments / Tutorials 4. Oral Examination 5. Open Book Test 6. Offline MCQ Test 7. Group Discussion 8. Analysis of Case Studies	Min. 06 Marks (40% of Passing)
SEE / External Exam (35 Marks) (Total 2 Hours Duration)	***Instructions:*** *1. Question No. 1 is Compulsory.* *2. Attempt any Three Questions from Question No. 2 to 5.* Q. 1: Fill in the Blanks on all Units = 05 Marks Q. 2: Theory Question on Unit-1 = 10 Marks Q. 3: Theory Question on Unit-2 = 10 Marks	Min. 14 Marks (40% of Passing)

	Q. 4: Theory Question on Unit-3 = 10 Marks Q. 5: Short Notes on all Units (Any 2 out of 4) = 10 Marks	
Total 50 Marks **Separate Passing for Internal Assessment (CIE) and External Exam (SEE)**		

DATA REPORTING SKILLS

(Effective for the academic year 2024-2025)

SEMESTER-II

Subject	ADD ON
Number of Lecture Hours/Week	02
Total Number of Lecture Hours	30

Course Learning Objectives: This course will enable students to

- Identify the key features of MS Excel and MS PowerPoint required for effective data analysis and presentation.
- Utilize tools such as Pivot Tables, Conditional Formatting, and What-If Analysis to solve business challenges.
- Develop financial statements like Cash Flow, Funds Flow, and Ratio Analysis using advanced Excel techniques.
- Access the effectiveness of presentations by incorporating design principles, animations, and transitions in PowerPoint.

Module-1	**Lecture Hours**
Introduction to the MS Excel interface, basic functionalities, workbook creation, and basic data entry and formatting	05
Module-2	
Data analysis using functions like SUMIF, COUNTIF, VLOOKUP, and others. Students learn to prepare financial statements and analyse data.	06
Module-3	
Conditional formatting and sort/filter data.	05
Module-4	
Pivot Tables, Pivot Charts, and What-If Analysis	06
Module-5	
Macros, Worksheet Protection, and Hyperlinks	04
Module-6	
MS PowerPoint Overview and Slide Management and Advanced PowerPoint Features	04

REFERENCE BOOKS

1. Excel 2019 Bible" by Michael Alexander, Richard Kusleika, and John Walkenbach
2. Presentation Zen: Simple Ideas on Presentation Design and Delivery" by Garr Reynolds

Savitribai Phule Pune University

(Formerly University of Pune)

Value Education Course (VEC)

Syllabus for F.Y.B.Sc. Students

(As Per National Education Policy-2020)

For Colleges Affiliated to Savitribai Phule Pune University

To be implemented from Academic Year 2024-2025

VEC-101-T: Environment Education-I

Course type: VEC (Theory) **No. of Credits: 2**

Semester : I

Course Outcomes

After the completion of this course, student will be able to-

CO-1: describe how human activities impact the environment.

CO-2: explain principles of sustainable development and resource management.

CO-3: analyze local, regional, and global environmental issues and their effects.

CO-4: evaluate different strategies for conserving biodiversity and ecosystems.

CO-5: apply relevant environmental policies and ethical considerations to real-world scenarios.

CO-6: design and implement action plans for community-based environmental projects**.**

Course Content

Chapter 1: Humans and the Environment [06 hours]

The man-environment interaction: Humans as hunter-gatherers; Mastery of fire; Origin of agriculture; Emergence of city-states; Great ancient civilizations and the environment; Middle Ages and Renaissance; Industrial revolution and its impact on the environment; Population growth and natural resource exploitation; Global environmental change.

The emergence of environmentalism: Anthropocentric and eco-centric perspectives (Major thinkers); The Club of Rome- Limits to Growth; UN Conference on Human Environment 1972; World Commission on Environment and Development and the concept of sustainable development; Rio Summit and subsequent international efforts.

Suggested Readings

1. Fisher, Michael H. (2018) An Environmental History of India- From Earliest Times to the Twenty-First Century, Cambridge University Press.
2. Headrick, Daniel R. (2020) Humans versus Nature- A Global Environmental History, Oxford University Press.
3. Hughes, J. Donald (2009) An Environmental History of the World- Humankind's Changing Role in the Community of Life, 2nd Edition. Routledge.
4. Perman, R., Ma, Y., McGilvray, J., and Common, M. (2003) Natural Resource and Environmental Economics. Pearson Education.
5. Simmons, I. G. (2008). Global Environmental History: 10,000 BC to AD 2000. Edinburgh University Press

Chapter 2: Natural Resources and Sustainable Development [08 hours]

Overview of natural resources: Definition of resource; Classification of natural resources- biotic and abiotic, renewable and non-renewable.

Biotic resources: Major type of biotic resources- forests, grasslands, wetlands, wildlife and aquatic (fresh water and marine); Microbes as a resource; Status and challenges.

Water resources: Types of water resources- fresh water and marine resources; Availability and use of water resources; Environmental impact of over-exploitation, issues and challenges; Water scarcity and stress; Conflicts over water.

Soil and mineral resources: Important minerals; Mineral exploitation; Environmental problems due to extraction of minerals and use; Soil as a resource and its degradation.

Energy resources: Sources of energy and their classification, renewable and non-renewable sources of energy; Conventional energy sources- coal, oil, natural gas, nuclear energy; Non-conventional energy sources- solar, wind, tidal, hydro, wave, ocean thermal, geothermal, biomass, hydrogen and fuel cells; Implications of energy use on the environment.

Introduction to sustainable development: Sustainable Development Goals (SDGs)- targets and indicators, challenges and strategies for SDGs.

Suggested Readings

1. Chiras, D. D and Reganold, J. P. (2010). Natural Resource Conservation: Management for a Sustainable Future.10th edition, Upper Saddle River, N. J. Benjamin/Cummins/Pearson.
2. John W. Twidell and Anthony D. (2015). Renewable Energy Sources, 3rd Edition, Weir Publisher (ELBS)
3. William P.Cunningham and Mary A. (2015) Cunningham Environmental Science: A Global Concern, Publisher (Mc-Graw Hill, USA)
4. Gilbert M. Masters and W. P. (2008). An Introduction to Environmental Engineering and Science, Ela Publisher (Pearson)
5. Singh, J.S., Singh, S.P. & Gupta, S.R. 2006. Ecology, Environment and Resource Conservation. Anamaya Publications https://sdgs.un.org/goals

Chapter 3: Environmental Issues: Local, Regional and Global [08 hours]

Environmental issues and scales: Concepts of micro-, meso-, synoptic and planetary scales; Temporal and spatial extents of local, regional, and global phenomena.

Pollution: Impact of sectoral processes on Environment, Types of Pollution- air, noise, water, soil, municipal solid waste, hazardous waste; Transboundary air pollution; Acid rain; Smog.

Land use and Land cover change: land degradation, deforestation, desertification, urbanization. Biodiversity loss: past and current trends, impact.

Global change: Ozone layer depletion; Climate change.

Suggested Readings

1. Harper, Charles L. (2017) Environment and Society, Human Perspectives on Environmental Issues 6th Edition. Routledge.
2. Harris, Frances (2012) Global Environmental Issues, 2nd Edition. Wiley- Blackwell.
3. William P. Cunningham and Mary A. (2015). Cunningham Environmental Science: A global concern, Publisher (Mc-Graw Hill, USA)
4. Manahan, S.E. (2022). Environmental Chemistry (11th ed.). CRC Press. https://doi.org/10.1201/9781003096238
5. Rajagopalan, R. (2011). Environmental Studies: From Crisis to Cure. India: Oxford University Press.

Chapter 4: Conservation of Biodiversity and Ecosystems [08 hours]

Biodiversity and its distribution: Biodiversity as a natural resource; Levels and types of biodiversity; Biodiversity in India and the world; Biodiversity hotspots; Species and ecosystem threat categories.

Ecosystems and ecosystem services: Major ecosystem types in India and their basic characteristics- forests, wetlands, grasslands, agriculture, coastal and marine; Ecosystem services- classification and their significance.

Threats to biodiversity and ecosystems: Land use and land cover change; Commercial exploitation of species; Invasive species; Fire, disasters and climate change.

Major conservation policies: in-situ and ex-situ conservation approaches; Major protected areas; National and International Instruments for biodiversity conservation; the role of traditional knowledge, community-based conservation; Gender and conservation.

Suggested Readings

1. Bawa, K.S., Oomen, M.A. and Primack, R. (2011) Conservation Biology: A Primer for South Asia. Universities Press.
2. Sinha, N. (2020) Wild and Wilful. Harper Collins, India.
3. Varghese, Anita, Oommen, Meera Anna, Paul, Mridula Mary, Nath, Snehlata (Editors) (2022) Conservation through Sustainable Use: Lessons from India. Routledge.
4. Bhagwat, Shonil (Editor) (2018) Conservation and Development in India: Reimagining Wilderness, Earthscan Conservation and Development, Routledge.
5. Krishnamurthy, K.V. (2003) Textbook of Biodiversity, Science Publishers, Plymouth, UK

VEC-151-T: Environment Education-II

Course type: VEC (Theory) **No. of Credits: 2**

Semester : II

Course Outcomes

After the completion of this course, student will be able to-

CO-1: identify various types of environmental pollution and their impacts on health.

CO-2: explain the basic concepts of climate change, including its causes and effects.

CO-3: analyze different strategies for adapting to and mitigating the effects of climate change.

CO-4: evaluate various environmental management practices and their effectiveness.

CO-5: apply the principles of key environmental treaties and legislation to case studies.

CO-6: create action plans that address specific environmental issues based on current policies and management practices.

Course Content

Chapter 1: Environmental Pollution and Health [08 hours]

Understanding pollution: Production processes and generation of wastes; Assimilative capacity of the environment; Definition of pollution; Point sources and non-point sources of pollution.

Air pollution: Sources of air pollution; Primary and secondary pollutants; Criteria pollutants- carbon monoxide, lead, nitrogen oxides, ground-level ozone, particulate matter and sulphur dioxide; Other important air pollutants- Volatile Organic compounds (VOCs), Peroxyacetyl Nitrate (PAN), Polycyclic aromatic hydrocarbons (PAHs) and Persistent organic pollutants (POPs); Indoor air pollution; Adverse health impacts of air pollutants; National Ambient Air Quality Standards.

Water pollution: Sources of water pollution; River, lake and marine pollution, groundwater pollution; water quality Water quality parameters and standards; adverse health impacts of water pollution on human and aquatic life.

Soil pollution and solid waste: Soil pollutants and their sources; Solid and hazardous waste; Impact on human health.

Noise pollution: Definition of noise; Unit of measurement of noise pollution; Sources of noise pollution; Noise standards; adverse impacts of noise on human health.

Thermal and Radioactive pollution: Sources and impact on human health and ecosystems.

Suggested Readings

1. Jackson, A. R., & Jackson, J. M. (2000). Environmental Science: The Natural Environment and Human Impact. Pearson Education.

2. Masters, G. M., & Ela, W. P. (2008). Introduction to environmental engineering and science (No.60457). Englewood Cliffs, NJ: Prentice Hall.
3. Miller, G. T., & Spoolman, S. (2015) Environmental Science. Cengage Learning.
4. Central Pollution Control Board Web page for various pollution standards. https://cpcb.nic.in/ standards/
5. Ahluwalia, V. K. (2015). Environmental Pollution, and Health. The Energy and Resources Institute (TERI).

Chapter 2: Climate Change: Impacts, Adaptation and Mitigation [06 hours]

Understanding climate change: Natural variations in climate; Structure of atmosphere; Anthropogenic climate change from greenhouse gas emissions– past, present and future; Projections of global climate change with special reference to temperature, rainfall, climate variability and extreme events; Importance of 1.5 °C and 2.0 °C limits to global warming; Climate change projections for the Indian sub-continent.

Impacts, vulnerability and adaptation to climate change: Observed impacts of climate change on ocean and land systems; Sea level rise, changes in marine and coastal ecosystems; Impacts on forests and natural ecosystems; Impacts on animal species, agriculture, health, urban infrastructure; the concept of vulnerability and its assessment; Adaptation vs. resilience; Climate-resilient development; Indigenous knowledge for adaptation to climate change.

Mitigation of climate change: Synergies between adaptation and mitigation measures; Green House Gas (GHG) reduction vs. sink enhancement; Concept of carbon intensity, energy intensity and carbon neutrality; National and international policy instruments for mitigation, decarbonizing pathways and net zero targets for the future; Energy efficiency measures; Renewable energy sources; Carbon capture and storage, National climate action plan and Intended Nationally Determined Contributions (INDCs); Climate justice.

Suggested Readings

1. Pittock, Barrie (2009) Climate Change: The Science, Impacts and Solutions. 2nd Edition. Routledge.
2. www.ipcc.org; https://www.ipcc.ch/report/sixth-assessment-report-cycle/.
3. Adenle A., Azadi H., Arbiol J. (2015). Global assessment of technological innovation for climate change adaptation and mitigation in developing world, Journal of Environmental Management, 161 (15): 261-275.
4. Barnett, J. & S. O'Neill (2010). Maladaptation. Global Environmental Change—Human and Policy Dimensions 20: 211–213.
5. Berrang-Ford, L., J.D. Ford & J. Paterson (2011). Are we adapting to climate change ? Global Environmental Change—Human and Policy Dimensions 21: 25-33.

Chapter 3: Environmental Management [06 hours]

Introduction to environmental laws and regulation: Constitutional provisions- Article 48A, Article 51A (g) and other derived environmental rights; Introduction to environmental legislations on the forest, wildlife and pollution control.

Environmental management system: ISO 14001 Life cycle analysis; Cost-benefit analysis Environmental audit and impact assessment; Environmental risk assessment, Pollution control and management; Waste Management- Concept of 3R (Reduce, Recycle and Reuse) and sustainability; Ecolabeling /Ecomark scheme

Suggested Readings

1. Jørgensen, Sven Marques, Erik João Carlos and Nielsen, Søren Nors (2016) Integrated Environmental Management, A transdisciplinary Approach. CRC Press.
2. Theodore, M. K. and Theodore, Louis (2021) Introduction to Environmental Management, 2nd Edition. CRC Press.
3. Barrow, C. J. (1999). Environmental management: Principles and practice. Routledge.
4. Tiefenbacher, J (ed.) (2022), Environmental Management - Pollution, Habitat, Ecology, and Sustainability, Intech Open, London. 10.5772/
5. Richard A. Marcantonio, Marc Lame (2022). Environmental Management: Concepts and Practical Skills. Cambridge University Press.

Chapter 4: Environmental Treaties and Legislation [10 hours]

An overview of instruments of international cooperation; bilateral and multilateral agreements; conventions and protocols; adoption, signature, ratification and entry into force; binding and non-binding measures; Conference of the Parties (COP)

Major International Environmental Agreements: Convention on Biological Diversity (CBD); Cartagena Protocol on Biosafety; Nagoya Protocol on Access and Benefit-sharing; Convention on International Trade in Endangered Species of Wild Flora and Fauna (CITES); Ramsar Convention on Wetlands of International Importance; United Nations Convention to Combat Desertification (UNCCD); Vienna Convention for the Protection of the Ozone Layer; Montreal Protocol on Substances that Deplete the Ozone Layer and the Kigali Amendment; Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal; Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade; Stockholm Convention on Persistent Organic Pollutants; Minamata Convention on Mercury; United Nations Framework Convention on Climate Change (UNFCCC); Kyoto Protocol; Paris Agreement; India's status as a party to major conventions

Major Indian Environmental Legislations: The Wild Life (Protection) Act, 1972; The Water (Prevention and Control of Pollution) Act, 1974; The Forest (Conservation) Act, 1980; The Air

(Prevention and Control of Pollution) Act, 1981; The Environment (Protection) Act, 1986; The Biological Diversity Act, 2002; The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006; Noise Pollution (Regulation and Control) Rules, 2000; Industry-specific environmental standards; Waste management rules; Ramsar sites; Biosphere reserves; Protected Areas; Ecologically Sensitive Areas; Coastal Regulation Zone; Status phase-out of production and consumption of Ozone Depleting Substances by India; National Green Tribunal; Some landmark Supreme Court judgements

Major International organisations and initiatives: United Nations Environment Programme (UNEP), International Union for Conservation of Nature (IUCN),World Commission on Environment and Development (WCED), United Nations Educational, Scientific and Cultural Organization (UNESCO), Intergovernmental Panel on Climate Change (IPCC), and Man and the Biosphere (MAB) programme.

Suggested Readings

1. UNEP (2007) Multilateral Environmental Agreement Negotiator's Handbook, University of Joensuu, ISBN 978-952-458-992-5
2. Ministry of Environment, Forest and Climate Change (2019) A Handbook on International Environment Conventions & Programmes. https://moef.gov.in/wp-content/uploads/2020/02/ convention-V-16-CURVE-web.pdf
3. Kanchi Kohli and Manju Menon (2021) Development of Environment Laws in India, Cambridge University Press.
4. India Code – Digital repository of all Central and State Acts: https://www.indiacode.nic.in/
5. Bohra, Saroj, Judicial Intervention and Evolution of Environmental Principles and Doctrines (January 7, 2019). Available at SSRN: https://ssrn.com/abstract=3311406 or http://dx.doi. org/10.2139/ssrn.3311406

Note: Case Studies and Field Work is compulsory

The students are expected to be engaged in some of the following or similar identified activities:

1. Discussion on one national and one international case study related to the environment and sustainable development.
2. Field visits to identify local/regional environmental issues, make observations including data collection and prepare a brief report.
3. Documentation of campus biodiversity.
4. Campus environmental management activities such as solid waste disposal, water management, and sewage treatment.

University of Pune

Board of Studies in Mathematics

S. Y. B. Sc. (Computer Science)

Syllabus of Mathematics

Introduction:

Savitribai Phule Pune University, Pune has decided to change the syllabi of various faculties from June, 2020. Taking into consideration the rapid changes in science and technology and new approaches in different areas of mathematics and related subjects Board of studies in Mathematics with concern of teachers of Mathematics from different colleges affiliated to Savitribai Phule Pune University, Pune has prepared the syllabus of S.Y.B.Sc. Computer Science Mathematics. To develop the syllabus the U.G.C. Model curriculum is followed.

Aims:

i) Give the students a sufficient knowledge of fundamental principles ,methods and a clear perception of innumerous power of mathematical ideas and tools and know how to use them by modeling ,solving and interpreting.

ii) Reflecting the broad nature of the subject and developing mathematical tools for continuing further study in various fields of science.

iii) Enhancing students overall development and to equip them with mathematical modeling abilities, problem solving skills , creative talent and power of communication necessary for various kinds of employment .

iv) Enabling students to develop a positive attitude towards mathematics as an interesting and valuable subject of study.

Objectives:

(i) A student should be able to recall basic facts about mathematics and should be able to display knowledge of conventions such as notations, terminology and recognize basic geometrical figures and graphical displays, state important facts resulting from their studies.

(ii) A student should get a relational understanding of mathematical concepts and concerned structures, and should be able to follow the patterns involved, mathematical reasoning.

(iii) A student should get adequate exposure to global and local concerns that explore them many aspects of Mathematical Sciences.

(iv) A student be able to apply their skills and knowledge, that is, translate information presented verbally into mathematical form, select and use appropriate mathematical formulae or techniques in order to process the information and draw the relevant conclusion.

(v) A student should be made aware of history of mathematics and hence of its past, present and future role as part of our culture.

*** Medium of Instruction: English**

*** Eligibility:** F.Y.B.Sc. Computer Science, as per University rules.

Structure of the course:

Semester - I			Semester -II	
Paper I	**MTC-231**	**Groups and Coding Theory**	**MTC-241**	**Computational Geometry**
Paper II	**MTC-232**	**Numerical Techniques**	**MTC-242**	**Operations Research**
Paper III	**MTC-233**	**Mathematics Practical: Python Programming Language-I**	**MTC-243**	**Mathematics Practical: Python Programming Language-II**

* **All three above courses are compulsory.**

*** External Students: Not allowed.**

*** Variation / Revaluation: Allowed for Paper- I and Paper-II.**

*** Qualifications for Teacher: M.Sc. Mathematics (with NET /SET as per existing rules)**

Equivalence of Previous syllabus along with new syllabus:

	Semester-III		Semester-IV	
	New Course	**Old Course**	**New Course**	**Old Course**
Paper I	**MTC-231: Groups and Coding Theory**	**MTC-211 : Applied Algebra**	**MTC-241: Computational Geometry**	**MTC-221: Computational Geometry**
Paper II	**MTC-232: Numerical Techniques**	**MTC-212: Numerical Analysis**	**MTC-242: Operations Research**	**MTC-222: Operations Research**

Paper III	**MTC-233: Mathematics Practical: Python Programming Language-I**	**MTC-213 : Mathematics Practical**	**MTC-243: Mathematics Practical: Python Programming Language-II**	**MTC–223: Mathematics Practical**

Semester III

MTC-231 : Groups and Coding Theory

Unit 1. Integers **[05 Lectures]**

1.1 Division Algorithm (without Proof)
1.2 G.C.D. using division algorithm and expressing it as linear combination
1.3 Euclid's lemma
1.4 Equivalence relation (revision), Congruence relation on set of integers, Equivalence class partition

Unit 2. Groups **[03 Lectures]**

2.1 Binary Operation
2.2 Group: Definition and Examples
2.3 Elementary Properties of Groups

Unit 3. Finite Groups and Subgroups **[10 Lectures]**

3.1 Order of a group, order of an element
3.2 Examples (Zn, +) and (U(n), *)
3.3 Subgroup definition, Finite subgroup test, subgroups of Zn
3.4 Generator, cyclic group, finding generators of Zn(Corollary 3,4 without proof)
3.5 Permutation group, definition, composition of two permutations, representation as product of disjoint cycles, inverse and order of a permutation, even/ odd permutation
3.6 Cosets: Definition, Examples and Properties, Lagrange Theorem(without Proof)

Unit 4. Groups and Coding Theory **[18 Lectures]**

4.1 Coding of Binary Information and Error detection
4.2 Decoding and Error Correction
4.3 Public Key Cryptography

Text Books:-

1. **Contemporary Abstract Algebra By J. A, Gallian (Seventh Edition)**
   Unit 1:Chapter 0, Unit 2: Chapter 2, Unit 3: Chapter 3 ,4, 5 and 7
2. **Discrete Mathematical Stuctures By Bernard Kolman, Robert C. Busby and Sharon Ross (6th Edition) Pearson Education Publication**
   Unit 4: Chapter 11

MTC-232 : Numerical Techniques

Unit 1: Algebraic and Transcendental Equation [04 Lectures]

1.1 Introduction to Errors

1.2 False Position Method

1.3 Newton-Raphson Method

Unit 2: Calculus of Finite Differences and Interpolation [16 Lectures]

2.1 Differences

2.2. Forward Differences

2.3 Backward Differences

2.4 Central Differences

2.5 Other Differences (δ, μ operators)

2.6 Properties of Operators

2.7 Relation between Operators

2.8 Newton's Gregory Formula for Forward Interpolation

2.9 Newton's Gregory Formula for Backward Interpolation

2.10 Lagrange's Interpolation Formula

2.11 Divided Difference

2.12 Newton's Divided Difference Formula

Unit 3: Numerical Integration [08 Lectures]

3.1 General Quadrature Formula

3.2 Trapezoidal Rule

3.3 Simpson's one-Third Rule

3.4 Simpson's Three-Eight Rule

Unit 4: Numerical Solution of Ordinary Differential Equation [08 Lectures]

4.1 Euler's Method

4.2 Euler's Modified Method

4.3 Runge-Kutta Methods

Text Book:-

1. A textbook of Computer Based Numerical and Statistical Techniques, by A. K. Jaiswal and Anju Khandelwal. New Age International Publishers.

Unit 1: Chapter 2: Sec. 2.1, 2.5, 2.7
Unit 2: Chapter 3: Sec. 3.1, 3.2, 3.4, 3.5, Chapter 4: Sec. 4.1, 4.2, 4.3, Chapter 5: Sec. 5.1, 5.2, 5.4, 5.5
Unit 3: Chapter 6: Sec. 6.1, 6.3, 6.4, 6.5, 6.6, 6.7
Unit 4: Chapter 7: Sec. 7.1, 7.4, 7.5, 7.6

Reference Books:-

1. S.S. Sastry; Introductory Methods of Numerical Analysis, 3rd edition, Prentice Hall of India, 1999.
2. H.C. Saxena; Finite differences and Numerical Analysis, S. Chand and Company.
3. K.E. Atkinson; An Introduction to Numerical Analysis, Wiley Publications.
4. Balguruswamy; Numerical Analysis.

MTC-233: Mathematics Practical: Python Programming Language-I

Unit 1: Introduction to Python

1.1 Installation of Python
1.2 Values and types: int, float and str,
1.3 Variables: assignment statements, printing variable values, types of variables.
1.4 Operators, operands and precedence:+, -, /, *, **, % PEMDAS(Rules of precedence)
1.5 String operations: + : Concatenation, * : Repetition
1.6 Boolean operator:
    1.6.1 Comparison operators: ==, !=, >, =, <=
    1.6.2 Logical operators: and, or, not
1.7 Mathematical functions from math, cmath modules.
1.8 Keyboard input: input() statement

Unit 2: String, list, tuple

2.1 Strings:
    2.1.1 Length (Len function)
    2.1.2 String traversal: Using while statement, Using for statement
    2.1.3 String slice
    2.1.4 Comparison operators (>, <, ==)
2.2 Lists:
    2.2.1 List operations
    2.2.2 Use of range function
    2.2.3 Accessing list elements
    2.2.4 List membership and for loop
    2.2.5 List operations
    2.2.6 Updating list: addition, removal or updating of elements of a list

2.3 Tuples:
- 2.3.1 Defining a tuple,
- 2.3.2 Index operator,
- 2.3.3 Slice operator,
- 2.3.4 Tuple assignment,
- 2.3.5 Tuple as a return value

Unit 3: Iterations and Conditional statements

3.1 Conditional and alternative statements, Chained and Nested Conditionals: if, if-else, if-elif-else, nested if, nested if-else

3.2 Looping statements such as while, for etc, Tables using while.

3.3 Functions:
- 3.3.1 Calling functions: type, id
- 3.3.2 Type conversion: int, float, str
- 3.3.3 Composition of functions
- 3.3.4 User defined functions, Parameters and arguments

Unit 4: Linear Algebra

4.1 Matrix construct, eye(n), zeros(n,m) matrices

4.2 Addition, Subtraction, Multiplication of matrices, powers and invers of a matrix.

4.3 Accessing Rows and Columns, Deleting and Inserting Rows and Columns

4.4 Determinant, reduced row echelon form, nullspace, columnspace, Rank

4.5 Solving systems of linear equations (Gauss Elimination Method, Gauss Jordan Method, LU- decomposition Method)

4.6 Eigenvalues, Eigenvectors, and Diagonalization

Unit 5: Numerical methods in Python

5.1 Roots of Equations

5.2 Newton-Raphson Method

5.3 False Position (Regula Falsi) Mehtod

5.4 Numerical Integration:
- 5.1.1 Trapezoidal Rule,
- 5.1.2 Simpson's 1/3rd Rule,
- 5.1.3 Simpson's 3/8th Rule

Text Books:-

1. **Downey, A. et al., How to think like a Computer Scientist: Learning with Python, John Wiley, 2015.**
   Sections: 1, 2, 3
2. **Robert Johansson, Introduction to Scientific Computing in Python**
   Section: 4

Reference Books:-

1. **Lambert K. A., Fundamentals of Python - First Programs, Cengage Learning India, 2015.**
2. **Guzdial, M. J., Introduction to Computing and Programming in Python, Pearson**

India.
3. **Perkovic, L., Introduction to Computing Using Python, 2/e, John Wiley, 2015.**
4. **Zelle, J., Python Programming: An Introduction to Computer Science, Franklin, Beedle & Associates Inc.**
5. **Sandro Tosi, Matplotlib for Python Developers, Packt Publishing Ltd.(2009)**

Practicals:
Practical 1: Introduction to Python, Python Data Types-I (Unit 1)
Practical 2: Python Data Types- II (Unit 2)
Practical 3: Control statements in Python-I (Unit 3- 3.1, 3.2)
Practical 4: Control statements in Python-II (Unit 3- 3.3)
Practical 5: Application : Matrices (Unit 4 – 4.1-4.3)
Practical 6: Application : Determinants, system of Linear Equations (Unit 4- 4.4, 4.5)
Practical 7: Application : System of equations (Unit 4- 4.5)
Practical 8: Application : Eigenvalues, Eigenvectors (Unit 4 – 4.6)
Practical 9: Application : Eigenvalues, Eigenvectors (Unit 4 – 4.6)
Practical 10: Application : Roots of equations (Unit 5 – 5.1)
Practical 11: Application : Numerical integration (Unit 5 – 5.2, 5.3)
Practical 12: Application : Numerical integration (Unit 5 – 5.4)

Semester - IV

MTC-241: Computational Geometry

Unit 1. Two dimensional transformations: [12 Lectures]

1.1 Introduction.
1.2 Representation of points.
1.3 Transformations and matrices.
1.4 Transformation of points.
1.5 Transformation of straight lines
1.6 Midpoint Transformation
1.7 Transformation of parallel lines
1.8 Transformation of intersecting lines
1.5 Transformation: rotations, reflections, scaling, shearing.
1.6 Combined transformations.
1.7 Transformation of a unit square.
1.8 Solid body transformations.
1.9 Translations and homogeneous coordinates.
1.10 Rotation about an arbitrary point.
1.11 Reflection through an arbitrary line.

Unit 2. Three dimensional transformations: [08 Lectures]

2.1 Introduction.
2.2 Three dimensional – Scaling, shearing, rotation, reflection, translation.
2.3 Multiple transformations.
2.4 Rotation about – an axis parallel to coordinate axes, an arbitrary line
2.5 Reflection through – coordinate planes, planes parallel to coordinate planes , an arbitrary plane

Unit 3. Projection **[08 Lectures]**

3.1 Orthographic projections.
3.2 Axonometric projections.
3.3 Oblique projections
3.4 Single point perspective projection

Unit 4. Plane and space Curves: **[08 Lectures]**

4.1 Introduction.
4.2 Curve representation.
4.3 Parametric curves.
4.4 Parametric representation of a circle and generation of circle.
4.5 Bezier Curves – Introduction, definition, properties (without proof), Curve fitting (up to n = 3), equation of the curve in matrix form (upto n = 3)

Textbook:

1. D. F. Rogers, J. A. Adams, Mathematical elements for Computer graphics, Mc Graw Hill Intnl Edition.

Unit 1: Chapter 2: Sec. 2-1 to 2.17
Unit 2: Chapter 3: Sec. 3.1 to 3.10,
Unit 3: Chapter 3: Sec. 3.12 to 3.14
Unit 4: Chapter 4: Sec. 4.1, 4.2, 4.5, Chapter 5: Sec. 5.1, 5.8

Reference books:

1. Computer Graphics with OpenGL, Donald Hearn, M. Pauline Baker, Warren Carithers, Pearson (4th Edition)
2. Schaum Series, Computer Graphics.

MTC-242: Operations Research

Unit 1: Linear Programming Problem I **[12 Lectures]**

1.1 Introduction Definition and Examples
1.2 Problem solving using Graphical method
1.3 Theory of Linear Programming, Slack and surplus variables, Standard form of LPP, Some important definitions, Assumptions in LPP, Limitations of Linear programming, Applications of Linear programming, Advantages of Linear programming Techniques
1.4 Simplex method, Big- M-method

Unit 2: Linear Programming Problem II **[08 Lectures]**

2.1 Special cases of LPP : Alternative solution, Unbounded solution, Infeasible solution
2.2 Duality in Linear Programming, Primal to dual conversion, Examples

Unit 3: Assignment Models **[06 Lectures]**

3.1 Assignmment Model -Introduction
3.2 Hungerian method for Assignment problem

Unit 4: Transportation Models **[10 Lectures]**

4.1 Introduction, Tabular representation

4.2 Methods of IBFS (North-West rule, Matrix-minima, Vogel's Approximation), Algorithms

4.3 The Optimality Test of Transportation Model (MODI method only)

Text Book:-

Operation Research (12 th Edition), by S.D.Sharma.

Unit 1: Chapter 1: Sec. 1.1, 1.3-1, 1.3-2, 1.5, 1.6, 1.8, 1.9, 1.10, 1.11, 1.12, Chapter 3: Sec. 3.1, 3.2, 3.3, 3. 4, 3.5-4,

Unit 2: Chapter 3: Sec. 3.8-1,3.8-2, Chapter 5: Sec. 5.1-1, 5.2-1,5.3,5.7-1, 5.7-2

Unit 3: Chapter 9: Sec. 9.1, 9.2, 9.4-1, 9.4-2, 9.5, 9.6, 9.7-1, 9.7-2

Unit 4: Chapter 10: 10.1, 10.2, 10.5, 10.8-1,10.9, 10.10

Reference Books:-

1. Operations Research by H. A. Taha
2. Operations Research by R. Panneerselvam, Prentice Hall of India.
3. Principles of Operations Research by H. M. Wagner, Prentice Hall of India.
4. Operations Research by Gupta and Hira.
5. Operation Research by J.K. Sharma

MTC-243: Mathematics Practical: Python Programming Language-II

Unit 1: 2D, 3D Graphs

1.1 Installation of numpy, matplotlib packages

1.2 Graphs plotting of functions such as ... etc.

1.3 Different formats of graphs.

1.3 Three-dimensional Points and Lines

1.4 Three-dimensional Contour Plots

1.5 Wireframes and Surface Plots

1.6 Graphs plotting of functions such as... etc.

Unit 2: Computational Geometry

1.1 Points: The distance between two points, Lists of Points - the PointList class, Integer point lists, Ordered Point sets, Extreme Points of a PointList, Random sets of Points not in general position

2.2 Points: Displaying Points and other geometrical objects, Lines, rays, and line segments, The geometry of line segments, Displaying lines, rays and line segments

2.3 Polygon : Representing polygons in Python, Triangles, Signed area of a triangle, Triangles and the relationships of points to lines, is Collinear, is Left, is Left On, is Right, is Right On, Between

2.4 Two dimensional rotation and reflection

2.5 Three dimensional rotation and reflection

2.6 Generation of Bezier curve with given control points

Unit 3: Study of Operational Research in Python

3.1 Linear Programming in Python
3.2 Introduction to Simplex Method in Python

Practicals:

Practical 1: Graph Plotting (Unit 1 – 1.1 – 1.3)
Practical 2: Graph Plotting (Unit 1 – 1.4 – 1.7)
Practical 3: Application to Computational Geometry (Unit 2 – 2.1)
Practical 4: Application to Computational Geometry (Unit 2 – 2.2)
Practical 5: Application to Computational Geometry (Unit 2 – 2.3)
Practical 6: Study of Graphical aspects of Two dimensional transformation matrix using matplotlib
Practical 7: Study of Graphical aspects of Three dimensional transformation matrix using matplotlib
Practical 8: Study of Graphical aspects of Three dimensional transformation matrix using matplotlib
Practical 9: Study of effect of concatenation of Two dimensional and Three dimensional transformations
Practical 10: Generation of Bezier curve using given control points
Practical 11: Study of Operational Research in Python (Unit 3.1)
Practical 12: Study of Operational Research in Python (Unit 3.2)

Text Books:-

1. **Jaan Kiusalaas, Numerical Methods in Engineering with Python, Cambridge University Press, (2005)**
   Sections: 3
2. **Robert Johansson, Introduction to Scientific Computing in Python**
   Section: 1
3. **Jason Brownlee, Basics of Linear Algebra for Machine Learning, Discover the Mathematical Language of Data in Python**
   Sections: 2

Reference Books:-

1. Lambert K. A., Fundamentals of Python - First Programs, Cengage Learning India, 2015.
2. Guzdial, M. J., Introduction to Computing and Programming in Python, Pearson India.
3. Perkovic, L., Introduction to Computing Using Python, 2/e, John Wiley, 2015.
4. Zelle, J., Python Programming: An Introduction to Computer Science, Franklin, Beedle and Associates Inc.
5. Jim Arlow, Interactive Computational Geometry in Python

Note:

(i) In paper -I , paper-II and paper-III, each course is of 50 marks (35 marks theory and 15 marks internal examination).

(ii) Paper III: Mathematics Practical - MTC-233 and MTC-243 is practical course and

is of 50 marks. Practicals shall be perforemed on computer.

Examination:

A) Pattern of examination: Paper- I, Paper-II and paper-III: Semesterwise

B) Pattern of question papers: For Paper -I and Paper-II

Q 1. Attempt any 05 out of 07 questions each of 01 marks. [05 Marks]
Q 2. Attempt any 02 out of 04 questions each of 05 marks. [10 Marks]
Q 3. Attempt any 02 out of 04 questions each of 05 marks. [10 Marks]
Q 4. Attempt any 02 out of 04 questions each of 10 marks. [10 Marks]

C) Instructions Regarding Practical:

Paper-III:Mathematics Practical:

(i) Mathematics Practical, external examiner shall be appointed by Savitribai Phule Pune University, Pune.
(ii) The minimum duration of parctical examination is 3 hours.
(iii) The semester examination is of 35 marks 15 marks are from internal evaluation (Journal, attendence and viva-voce or internal test etc.)
(iv) The slips for the questions on programming and problem solving using python shall be prepared and provided and these can be used at least for 3 years.

D) Standard of passing:

For Paper- I, Paper-II and Papaer -III: 14 Marks out of 35 and 06 marks out of 15 marks and total should be 20 marks for each course.

COMPUTER NETWORKS-I (Effective for the academic year 2024-2025) SEMESTER-IV			
Subject Code	CS 242	**IE Marks**	15
Number of Lecture Hours/Week	03	**UE Marks**	35
Total Number of Lecture Hours	36	**Credits**	2

Course Learning Objectives: This course will enable students to

- Define key networking concepts such as data communication, protocols, standards, and topologies.
- Explain the purpose and applications of various network topologies in real-world scenarios.
- Demonstrate the ability to set up a basic network with appropriate protocols and standards.
- Compare different network topologies and evaluate their advantages and limitations in specific use cases.

Module-1	**Lecture Hours**
Introduction to Networks and Network Models 4 lectures 1.1 Data communication, components, data representation 1.2 Networks, network criteria, network types - LAN, WAN, Switching, The Internet, Accessing the Internet 1.3 Network Software- Protocol hierarchies, Design Issues of the layer, Connection Oriented and Connectionless Services, 1.4 Reference models - OSI Reference Models, TCP/IP Reference model, Connection devices in different layers, Comparison of OSI and TCP/IP Reference Models.	04
Module-2	
Lower Layers 2.1 Communication at the physical layer, data rate limits - Noiseless channel (Nyquist bit rate), noisy channel (Shannon capacity), Performance - bandwidth, throughput, latency, bandwidth-delay product, jitter 2.2 Design issues of Data Link Layer, Services - Framing, flow control, error control, congestion control, Link layer addressing 2.3 Framing Methods - Character Count, Flag bytes with Byte Stuffing, Flags bits with Bit Stuffing, Physical Layer Coding Violations 2.4 The Channel allocation problem, Static and dynamic allocation, Media Access Methods - Taxonomy of multiple-access protocols 2.5 Switching and TCP/IP layers, Types - circuit switching, packet switching and message switching 2.6 Wired LANs - Standard Ethernet characteristics, Addressing, Access method, implementation, Fast and Gigabit Ethernet 2.7 Wireless LANs - Architectural comparison, Characteristics, Access control, IEEE 802.11 CBCS: architecture, Physical layer, MAC sublayer, Bluetooth architecture, Layers	10
Module-3	
Network Layer 3.1 Network layer services - Packetizing, Routing and forwarding, other services 3.2 Open and closed loop congestion control 3.3 IPv4 addressing- Address space, classful addressing, Subnetting, Supernetting, classless addressing, Network address resolution (NAT) 3.4 Forwarding of IP packets- based on destination address, based on label 3.5 Network Layer Protocols- Internet Protocol (IP), IPv4 datagram format, Fragmentation, options 3.6 Mobile IP-addressing, agents, Three phases 3.7 Next Generation IP- IPv6 address representation, address space, address types, IPv6 protocol, packet format, extension header, Difference between IPv4 and IPv6 3.8 Routing - General idea, Algorithms - Distance vector routing, link state routing, path vector routing	12

Module-4	
Transport Layer 4.1 Transport layer Services- Process-to-process communication, Addressing, Encapsulation and decapsulation, Multiplexing and demultiplexing, Flow control, Pushing or pulling, Flow control, Buffers, Sequence numbers, Acknowledgements, sliding window, congestion control 4.2 Connectionless and Connection-oriented service, Port numbers 4.3 Transport layer protocols- User datagram protocol, user datagram, UDP services 4.4 Transmission Control Protocol - TCP Services, TCP Features, TCP Segment format, three-way handshake for connection establishment and termination, State transition diagram, windows in TCP.	10
REFERENCE BOOKS	
1. Computer Networks-Andrew S. Tanenbaum, 5th Edition, Pearson Education 2. Data Communication and Networking- Behrouz Fourouzan, 5th Edition, McGraw Hill	

University of Pune

S.Y.B.Sc.
Environmental Science
Revised Syllabus 2020-21
Course Design

Semester	Paper	Course code	Course Title	Credits	Marks Distribution			
					Internal	University	Subtotal	Total
I	I	EVS – 231	Ecology & Ecosystem	02	15	35	50	150
	II	EVS – 232	Natural Resource Conservation and Management	02	15	35	50	
	III	EVS – 233	Practical Course Based on EVS - 231 & EVS – 232	02	15	35	50	
II	I	EVS-241	Biological Diversity & its Conservation	02	15	35	50	150
	II	EVS-242	Environmental Pollution Control Technology	02	15	35	50	
	III	EVS-243	Practical Course Based on EVS - 241 & EVS - 242	02	15	35	50	
				Total = 12				300

Semester – I, Paper – I, EVS – 231

Ecology & Ecosystem (Total Lectures- 40)

Unit No.	Name of the Unit	Content	Lectures
1.	**Ecology**	• **Introduction & Interdisciplinary nature of Ecology.** • **Levels of Organisation –** a) Biological / Ecological Spectrum. b) Ecological Hierarchy by Barett et al. • **Ecological Classification based on –** a) Taxonomic Affinity (From Kingdom to Species Level Ecology). b) Habitat Types (Terrestrial & Aquatic Ecology). c) Levels of Organisation (Autecology & Synecology – Population, Community, Biome & Ecosystem Ecology)	**08**
2.	**Ecosystem Structure & Function – Energy Flow**	• **Concept** of the Ecosystem. • **Macro & Micro-ecosystems …etc.** • **Ecosystem Structure** – Abiotic & Biotic Components. • **Ecosystem Function : Functional attributes-** a) Food Chain – Grazing & Detritus. b) Food Web & Ecosystem Stability c) Ecological Energetics – i) Energy Input. ii) Energy Flow – Single Channel & Y shaped models. d) Productivity of Ecosystem – i) Primary Production – GPP & NPP. ii) Secondary Production. iii) Standing Crop (Biomass). e) Ecological Pyramids – of Number, Biomass & Energy with examples	**08**
3.	**Ecosystem Function : Nutrient Cycling**	• **Concept of –** a) Macro & Micro-nutrients, deficiency syndrome, Functions b) Nutrient Cycling-- Biogeochemical Cycles – • **Gaseous Cycles** – Hydrological, Carbon & Nitrogen Cycles. • **Sedimentary Cycles** – Phosphorus & Sulphur Cycles. • **Human Impact** on Biogeochemical Cycles. • **Ecosystem Nutrient Cycling Model – Intra-system Cycling & Extra-system Transfers.** a) Nutrient Inputs. b) Biotic Accumulation of Nutrient. c) Nutrient Outputs.	**08**

4.	**Population Ecology**	• **Introduction & Basic Concepts.** • **Population Characteristics** – a) Size & Density. b) Dispersion – Random, Aggregate & Uniform. c) Natality (Potential & Realised). d) Fecundity e) Mortality (Potential & Realised). f) Survivorship Curves. g) Age & Sex Structure. h) Life Table and Viability analysis • **The Concept of Carrying Capacity.** • **Population Growth –** a) Growth Curves – Exponential & Logistic. b) Population Fluctuation. c) Biotic Potential & Environmental Resistance.	**08**
5.	**Community Ecology and Succession**	• **Characteristics of Community** - Species Diversity, Growth form & Structure, Dominance, Succession, Trophic Structure, Ecological Niche, Ecotone & Edge Effect. • **Characters used in Community Structure-** a) Analytical Characters – i) Quantitative. ii) Qualitative. b) Synthetic Characters. • **Inter-specific & Intra-specific Relationships.** • **Causes** of Succession. • **Basic Types** – Primary, Secondary, Autogenic, Allogenic …etc. • **Mechanism** of Succession – a) Nudation. b) Invasion. c) Competition, Co-action & Reaction. d) Stabilisation (Climax). • **Models** of succession – a) Hydrosere. b) Lithosere.	**08**

Semester – I , Paper – II, EVS – 232

Natural Resources & their Management (Total Lectures- 40)

Unit No.	Name of the Unit	Content	Lectures
1.	**Resources**	• **Meaning and Definition** • **Classification of Resources** 1. On the basis of Origin: Biotic & Abiotic 2. On the basis of recovery rate: Renewable and Non Renewable 3. Natural and Artificial Resources 4. Material and Energy Resources • **Importance and Scope** of Resources • **Nature** of Resources : Regenerative and Assimilative Capacity of Earth • **Man's Interaction with Natural Resources**: 1. An important resource 2. A waste sink 3. Cultural Significance • **Problems** associated with Natural Resources	**08**
2.	**Forest and Mineral Resources**	A. **Forest Resources** : • Function of Forest : Protective, Productive, Regulative and Accessory • Importance of Forest : Ecological and Economical • Human Interaction with Forest : Overexploitation, Deforestation (Causes and Effects) • Forest Management in India—JFM, EDP, Protected Areas • Case studies on Timber extraction, Dam construction and its effect on tribal people B. **Mineral Resources** • Origin of Mineral Resources with examples • Need of Mineral Resources • Overexploitation of Mineral Resources • Effects of Mining on Ecosystem with case studies. • Conservation of Mineral resources and its importance	**08**
3.	**Food Resources**	• **World Food Problems**: a) Increasing World Food Demand. b) Nutritional deficiency in food. c) Food Distribution. • **The Green Revolution in India**- Concept, Its Impacts in India. • Introduction of Hybrid Varieties-HYV and Genetically Modified Crops. • Effects of Modern Agriculture technologies • Genetically Modified Crops & Regulations in India • **Fertilizer-Pesticide Problems** –NPK Fertilizers	**08**

4.	**Water Resources**	• **Use and over-utilization** of surface and ground water, • Under-ground water pollution • Water Crisis--- the reasons • Conflicts over water ---World and India • **Conservation & Management** a) Traditional Methods. b) Rain-water Harvesting & Ground Water Recharge. c) Watershed Management– the concept. d) Flood and flood plain management	**08**
5.	**Land Resource**	• **Traditional & Modern Agricultural Systems** • **Major causes of soil degradation**: Soil erosion, Pollution , Use of fertilisers, pesticides, heavy metals, Plastic pollution • **Consequences of soil erosion** a) Decline of soil fertility b) Water logging c) Salinity d) Shifting / jhum cultivation • **Soil conservation methods** • **Sustainable Agriculture Methods**	**08**

Semester – II, Paper – I, EVS -241,

Biological Diversity & its Conservation (Total Lectures- 40)

Unit No.	Name of the Unit	Content	Lectures
1.	**Biological Diversity – Ecosystem Diversity**	**Biological Diversity---** • The Concept, Definition • Levels – Ecosystem, Species & Genetic. • Methods of assessment of Biological diversity **Ecosystem Diversity** • Classification of Ecosystem – a) Udvardy's Classification. b) Bailey's Classification. c) Olsen's Classification. d) Holdridge's Classification. • **Major Ecosystem types of India** with their physical & biological characteristics. • **Major Ecosystem types of the World** with their physical & biological characteristics. • **Importance of Ecosystem in maintaining Ecological balance**	**08**
2.	**Species Diversity**	a) **Species Diversity** at Local , National and International Level b) Special features and Latest estimates for major groups of Plants, Animals & Microbes. • **Measuring Species Diversity** – Species Richness, Species Abundance and Species Evenness. • **Factors** affecting global distribution of Species Richness – Latitudinal, Altitudinal, Rainfall gradients, temperature …etc. • **Endemism** – a) The Concept. b) Types with Examples. c) Endemism in India. • **Centers of Diversity** – a) The Concept. b) Centers of Diversity : Analyses at Global Level – • **Concept of hotspot** i) Myer's Hot-spots. ii) Mega-diversity Centers / Countries. **c) Western Ghat as a Hot-spot.** d) **India as a Mega-diversity Country**.	**08**
3.	**Genetic Diversity**	• **Meaning & Introduction** to Genetic Variations in Species. • **Nature & Origin** of Genetic Variations. • **Factors** affecting Genetic Diversity. • **Darwin's theory of Evolution and Lamarck's theory of Natural Selection** • **Measurement of Genetic Diversity** – a) Based on DNA & Chromosomes. b) Molecular Marker Techniques. • **Transgenic Organisms**. • **Diversity in Domesticated Species** –	**08**

		a) Variations since the first domestication to the present. • Land Races, Advanced Cultivars, Wild Relatives of Cultivated Plants & Feral Plants.	
4.	**Significance & Threat to Biodiversity**	**(Significances)** • Ecological Significances – Contribution of Biodiversity to various Eco- Services. • Non Ecological Significances – Nutritional, Medicinal, Aesthetic, Cultural, Commercial Values …etc. • Optional Values, Use of microorganism in remediation of pollution **(Threats)** • Threats with suitable Examples – a) Large Scale Dev. Projects – Habitat Destruction & Fragmentation. b) Changing Agri. & Forestry Practices. c) Invasion by Introduced Species. d) Over-exploitation. e) Environment Pollution. f) Global Climate Change. g) Loss of Traditional Knowledge. h) Nature of Legal & Mgmt. System – Human Wildlife Conflict. i) Genetically Modified Organisms ...etc.	**08**
5.	**Biodiversity Conservation**	• **Conservation Methods** – In-situ & Ex-situ methods with Example. • **National Conservation Efforts –** a) The laws – Environment Protection Act, Forest Act, Wildlife Act, Biodiversity Act 2002 b) Involving People's Participation – NBSAP, PBR c) Involving Community Participation – JFM, EDP d) People's Movement – Silent Valley Movement, Beej Bachao Andolan • **International Conservation Efforts –** a) IUCN – The World Conservation Union. b) CBD. c) CITES. • **Traditional Methods of Conservation** – Sacred Groves / Ponds / Species, Periodic restrictions on resource harvesting …etc. • **Need & Awareness**.	**08**

Semester – II, Paper – II, EVS – 242

Environmental Pollution Control Technology

(Total Lectures- 40)

Unit No.	Name of Unit	Content	Lectures
1	**Air Quality Parameters and Monitoring**	**Air Quality Monitoring** • **National standards** for ambient air quality by WHO • **Site and Parameter selection**, • **Air Sampling Techniques** • **Monitoring** of important **ambient air components** such as Particulate matter (PM) of 10 micron or less in size and 2.5 micron and less in size, Oxides of Sulfur, Nitrogen, Carbon monoxide • **Methods of analysis of SOx , NOx** • **Monitoring tools/instruments** used for the same and its work principle, Stack gases monitoring technique • **Plume behaviour**	**08**
2	**Water Quality Monitoring**	• **Purpose/objectives** of monitoring • **Water Quality Monitoring Protocol** • **Collection of sample** (types of sample, chain of custody, sampling method, number of samples, sample containers, sample volume, etc.) • **Sample preservation**, handling & storage guidelines/criteria • **Water quality monitoring** on field test parameters, off-field test parameters • **Waste Water Treatment**: a) Primary Treatment – Screening, Grit removal, Sedimentation b) Secondary Treatment - • Aerobic Method- i) Activated Sludge Process. ii) Trickling Filter. iii) Rotating Contractor iv) Oxidation Pond • Anaerobic Method. d) Tertiary Treatment – Disinfection (Chlorination). e) Biogas—one stage and second stage digester, Principle	**08**
3	**Soil Quality Monitoring**	• **Objectives** of soil monitoring/testing • **Sampling** and sample units; sample number, frequency and timing; Sampling methodology a. Site selection b. Infield sampling technique c. Describing the soil profile d. Site description e. Setting a transect instruments / Equipment used	**08**

		• Guidelines for handling and storage of samples • Physiochemical and Biological parameters **Biological Method to control soil pollution**--- a) To reduce dependency on chemicals – Use of Bio fertilizers & Bio pesticides, Conservational Tillage, Mixed Cropping, Crop rotation, Biological Pest Mgmt., Organic Farming b) Bio / Phyto-remediation of contaminated sites. **Soil carbon Flux**	
4	**Forest Monitoring**	• **Classification of forests** • **Measurement of individual trees:** a. Measurement of diameter and girth of trees b. Measurement of heights of trees c. Measurement of form of trees d. Measurement of volume of felled trees e. Measurement of volume of standing trees f. Determination of age of trees • **Forest inventory** • **Kinds of sampling**, sampling units, sampling intensity	**08**
5.	**Noise Quality Parameters**	**Noise and Vibration Monitoring** • Measuring techniques for noise & vibration • Noise monitoring methods • The Basic Noise Unit; Lmax, SEL, Leq(h), Ldn, 24-Hour Exposure from All Events • **Noise Control Techniques-** Sound Insulation, Sound Absorption, Vibration Damping and Isolation • **Noise Control at Source—** a) Selection & Maintenance of machines. b) Control over vibrations. c) Installation of barriers / enclosures. d) Using protective equipment e) Noise proof walls	**08**

Semester I, Paper-III

Practical Course Based on EVS -231 & EVS- 232.

Sr. No.	Description	Practical Type	Practical Sessions
1.	Measurement of Primary Productivity of grassland by Harvest Method.	Field + Laboratory.	**01**
2.	Estimation of Total Chlorophyll from plants in Clean & Polluted Environment.	Laboratory.	**01**
3.	Study of grassland vegetation by List Count Quadrat Method to determine the Frequency, Density & Abundance.	Field + Laboratory	**01**
4.	Determination of Frequency & Abundance of species across terrestrial – aquatic transitional zone, by Line Transect Method.	Field + Laboratory	**01**
5.	Determination of Density of species across terrestrial – aquatic transitional zone by Belt Transect Method.	Field + Laboratory	**01**
6.	Field visit to study Watershed Mgmt. Techniques.	Visit.	**01**
7.	Visit to Nature Interpretation / Information Centre.	Visit.	**01**
8.	Visit to National Park / Wildlife Sanctuary to study Wildlife & various Inter-specific & Intra-specific Relations.	Visit.	**≥ 01 Day**
9.	Continuation of the use of Social Media for e-networking & dissemination of ideas on Environmental Issues Pertaining to the Course.	---	**≥ 02**
10.	Identification of advanced cultivars in the Local market	Visit.	**01**
11.	Field visit to study Rain water Harvesting technique	Visit.	**01**
12.	Determination of minimum area and number of quadrates for vegetation	Field + Laboratory	**01**
13.	Determination of Shannon Diversity Index of a vegetation (Data sheet)	Laboratory.	**01**
14.	Determination of Simpson Diversity Index of a vegetation (Data sheet)	Laboratory.	**01**
15.	Visit to Soil Survey Department.	Visit.	**01**
16.	Study of Soil sampling techniques	Visit.	**01**

Semester II, Paper-III

Practical Course Based on EVS -241 & EVS- 242.

Sr. No.	Description	Practical Type	Practical Sessions
1.	Sampling of Air by High Volume Sampler	Field + Laboratory	**01**
2.	Determination of Optimum Dose of Alum (Coagulant) required for water.	Laboratory.	**01**
3.	Determination of Turbidity of water. (Turbidimeter / Nephelometer)	Laboratory.	**01**
4.	Determination of Residual Chlorine from treated water.	Laboratory.	**01**
5.	Determination of Dissolved Oxygen in water by Winkler's method	Laboratory.	**01**
6.	Determination of Nitrate from water (Calorimeter)	Laboratory.	**01**
7.	Determination of Phosphate from water. (Colorimeter)	Laboratory.	**01**
8.	Determination of Soluble Salts from Soil.	Laboratory.	**01**
9.	Determination of Available Nitrogen from soil.	Laboratory.	**01**
10.	Study of Water Sampling and Preservation techniques	Field + Laboratory	**01**
11.	Measurement of sounds by DB meter / SLM in silent, industrial, residential and commercial zones and Analysis	Field + Laboratory	**01**
12.	Estimation of AGB, BGB and Carbon from sampling of trees	Field + Laboratory	**01**
13.	Determination of Lime required for Acidic soil	Laboratory	**01**
14.	Estimation of Productivity of Lake using DO method	Field + Laboratory	**02**
15.	Estimation of NOx content of given samples	Laboratory.	**01**
16.	Estimation of SOx content of given samples	Laboratory.	**01**

Reference Books

- Understanding Environment; Chokkar K. B., Pandya M. & Raghunathan M.; Centre for Environment Education; Sage Publication, New Delhi.
- An Advanced Textbook on Biodiversity – Principles & Practice; Krishnamurthy K.V.; Oxford & IBH Publishing Co. Pvt. Ltd.; New Delhi.
- Ecology – Principles & Applications; Chapman J. L. & Reiss M. J.; Cambridge University Press.
- Fundamentals of Ecology; Odum P.E.; Natraj Publishers; Dehradun; 3 Edt..
- Ecology, Environment & Resource Conservation; Singh J.S., Singh S.P. & Gupta S.R.; Annamaya Publishers; New Delhi.
- Ecology & Environment; Sharma P.D.; Rastogi Publication; Meerut; 11 Rev. Edt..
- Environment Science; Tyler M.G.; Wadsworth Publishing Co.; 1997.
- Perspective in Environmental Studies; Kaushik & Kaushik; New Age International Pvt. Ltd. Publishers.
- Environmental Science; Santra S.C.; New Central Book Agency (P) Ltd.; 2 Edt..
- Environmental Chemistry, Dey A. K.; New Age International Publishers; 6 Edt..
- Air Pollution; Rao M.N. & Rao H.V.N.; Tata McGraw Hill; New Delhi; 1989.
- Environmental Pollution Control & Environmental Engineering; Rao C. S.; Tata McGraw Hill; New Delhi; 1994.
- Pollution Management; Agarwal S.K.
- Environmental Science; Daniel Chiras.
- Waste Water Engineering, Treatment, Disposal & Reuse; Metcalf & Eddy.
- Manual for Field Ecology; Mishra R.
- Handbook of Methods in Environmental Studies Vol-I &II; Mailti S.K.; ABD Publishers; Jaipur.
- Physico-Chemical Examination of Water, Sewage & Industrial Effluents; Manivasakam N.; Pragati Prakashan; Meerut; 1984.
- Chemical & Biological Methods for Water Pollution Studies; Trivedi R.K. & Goel P.K.; Environmental Publications; Karad; 1986.
- Instrumental Methods of Analysis; Willard; cbpspd; 7 Edt..
- Pollution Management; Agarwal S.K.
- Waste Water Engineering, Treatment, Disposal & Reuse; Metcalf & Eddy
- Advanced Air and Noise Pollution Control – L.K Wang & N.C Pereira
- Textbook of Noise Pollution & Its Control – S.C. Bhatia
- Waste Water Engineering, Treatment, Disposal & Reuse; Metcalf & Eddy
- Environmental Pollution Control & Environmental Engineering; Rao C. S.; Tata McGraw Hill; New Delhi; 1994.
- Air Pollution; Rao M.N. & Rao H.V.N.; Tata McGraw Hill; New Delhi; 1989

SAVITRIBAI PHULE PUNE UNIVERSITY

(Formerly University of Pune)

S.Y. B. Sc. (Computer Science), Electronics

Choice Based Credit System Syllabus

To be implemented from

Academic Year 2020–2021

(Under the faculty of Science and Technology)

Savitribai Phule Pune University

(Formerly University of Pune)

SYLLABUS OF

S. Y. B. Sc. (Computer Science), Electronics

Choice Based Credit System

To be implemented from A.Y. 2020-21

Structure of S. Y. B. Sc.(Computer Science) Electronics

Semester	Paper Code	Paper	Paper title	No. of Credit	Lectures/Week	Evaluation		
						CA	UE	Total
III	ELC-231	I	Microcontroller Architecture & Programming	2	3 (each lecture of 50 minutes)	15	35	50
	ELC-232	II	Digital Communication and Networking	2	3 (each lecture of 50 minutes)	15	35	50
	ELC-233	III	Practical Course I	2	1 pract / week (each practical of 04 hours & 20 minutes)	15	35	50
IV	ELC-241	I	Embedded System Design	2	3 (each lecture of 50 minutes)	15	35	50
	ELC-242	II	Wireless Communication and Internet of Things	2	3 (each lecture of 50 minutes)	15	35	50
	ELC-243	III	Practical Course II	2	1 pract / week (each practical of 04 hours & 20 minutes)	15	35	50

S.Y.B.Sc.(Computer Science), Electronics- Semester III

Paper-I: Microcontroller Architecture & Programming (ELC 231)

Objectives:

1. To study the basics of 8051microcontroller
2. To study the Programming of8051microcontroller
3. To study the interfacing techniques of 8051microcontroller
4. To design different application circuits using 8051microcontroller

Course Outcomes : On completion of the course, student will be able

1. To write programs for 8051 microcontroller
2. To interface I/O peripherals to 8051 microcontroller
3. To design small microcontroller based projects

COURSE CONTENTS

UNIT- 1: Basics of Microcontroller & Intel 8051 architecture [08]

Introduction to microcontrollers, difference in controller and processor.
Architecture of 8051, Internal block diagram, Internal RAM organization, SFRS, pin functions of 8051, I/O port structure & Operation, External Memory Interface.

UNIT-2: Programming model of 8051 [10]

Instruction classification, Instruction set, Addressing Modes: Immediate, register, direct, indirect and relative, assembler directives (ORG, END), features with examples, I/O Bit & Byte programming using assembly language for LED and seven segment display (SSD) interfacing.
Introduction to 8051 programming in C.

UNIT- 3: Timer /Counter, Interrupts [10]

Timer / counter: TMOD, TCON, SCON, SBUF, PCON Registers, Timer modes, programming for time delay using mode 1 and mode 2.
Interrupts: Introduction to interrupt, Interrupt types and their vector addresses, Interrupt enable register and interrupt priority register (IE, IP)

UNIT- 4: Interfacing, Serial Communication [08]

Programming of serial port without interrupt, Serial Communication: Synchronous and asynchronous serial communication, Use of timer to select baud rate for serial communication. Interfacing : ADC, DAC, LCD, stepper motor.

Recommended books:

1. 8051 microcontroller and Embedded system using assembly and C : Mazidi and McKinley, Pearson publications
2. The 8051 microcontroller – Architecture, programming and applications: K.Uma Rao and Andhe Pallavi, Pearson publications.

S.Y.B.Sc. Computer Science), Electronics, Semester III

Paper-II, Digital Communication and Networking, ELC- 232

Objectives:

1. To introduce to all aspects of data communication system
2. To introduce various digital modulation schemes
3. To identify the need of data coding and error detection/correction mechanism.
4. To study bandwidth utilization techniques : multiplexing and Spectrum spreading
5. To know data link layer protocol: Media Access Control
6. To study OSI and TCP/IP models of Networking.

Course Outcomes : On completion of the course, student will be able

1. Define and explain terminologies of data communication
2. Understand the impact and limitations of various digital modulation techniques
3. To acknowledge the need of spread spectrum schemes.
4. Identify functions of data link layer and network layer while accessing communication link
5. To choose appropriate and advanced techniques to build the computer network

.

COURSE CONTENTS

UNIT 1: Introduction to Electronic Communication (9)

Introduction to Communication: Elements of Communication system, types of noise sources, Electromagnetic spectrum, signal and channel bandwidth,

Types of communication: simplex, half duplex, full duplex, baseband and broadband,

Serial communication: asynchronous and synchronous,

Information Theory: Information entropy, rate of information (data rate, baud rate), channel capacity, Nyquist theorem, Signal to noise ratio, Noise Figure, Shannon theorem,

Error handling codes: Necessity, Hamming code, CRC

UNIT 2: Modulation and Demodulation (5)

Introduction to modulation and demodulation: Concept and need of modulation and demodulation,

Digital Modulation techniques: Pulse Code Modulation (PCM), FSK, QPSK, QAM.

UNIT 3: Multiplexing, Spectrum Spreading and Media Access Control (12)

Multiplexing techniques: Frequency division multiplexing, wavelength division multiplexing, Time division multiplexing

Spread Spectrum techniques: Frequency hopping Spread Spectrum, Direct Sequence Spread Spectrum

Media Access Control (MAC):

Random Access Protocol: ALOHA, CSMA, CSMA/CD, CSMA/CA,
Controlled Access Protocols: Reservation, Polling, Token passing,
Channelization Protocols: FDMA, TDMA, CDMA.

UNIT 4: Computer Networking (10)

Introduction to computer networks

Types of networks : LAN, MAN, WAN, Wireless networks, Switching, Internet,

Network topology : point to point, Star, Ring, Bus, Mesh, Tree, Daisy Chain, Hybrid

Network devices : Repeater, Switch, Networking cables, Router, Bridge, Hub, Brouter, Gateway.

Wired LANs:-

Ethernet: Ethernet protocol, standard Ethernet, 100 MBPS Ethernet, Gigabit Ethernet, 10 Gigabit Ethernet,

Computer network model: OSI and TCP/IP.

Recommended books:

1.Communication Electronics: Principles and Applications, Frenzel, Tata Mc Graw Hill publication, 5th edition.

2. Data Communication and Networking, Forouzan, Mc Graw Hill publication, 5th edition

3. Computer Networks, Tanenbaum, pHI publication, 5th edition

S.Y.B.Sc.(Computer Science), Electronics, Semester III
Paper III, Practical Course (ELC-233)

Objectives:

1. To get hands on training of Embedded C
2. To study experimentally interfacing of microcontroller
3. To design, build and test modulator and demodulators of digital communication
4. To build and test experimentally various techniques of wired communication
5. To develop practical skills of network setup

Course Outcomes : On completion of the course, student will be able

1. To design and build his/her own microcontroller based projects.
2. To acquire skills of Embedded C programming
3. To know multiplexing and modulation techniques useful in developing wireless application
4. Do build and test own network and do settings.

Guidelines for Practical:

- Practical batch size : 12
- Minimum no of Practical to be performed : 10
- At least five practical from each Group
- Electronics lab should have set up for embedded programming (Computers and microcontroller target and interfacing boards)

COURSE CONTENTS

Group A: (Any 5)

1. Arithmetic, logical & code conversion problems using assembly/C programming
2. Interfacing of thumbwheel & seven segment display to 8051 microcontroller
3. Traffic light controller using 8051 microcontroller
4. Interfacing LCD to 8051Microcontroller
5. Waveform generation using DAC Interface to 8051Microcontroller

6. Event counter using opto-coupler, seven segment LED/LCD display interface to 8051Microcontroller
7. Speed Control of stepper motor using 8051 microcontroller

Group B: (Any 5)

1. Study of 3 or 4 Bit Pulse Code Modulation technique
2. Study of Frequency Shift Keying
3. Study of Time Division Multiplexing
4. Study of Frequency Division Multiplexing
5. Study of Code Division Multiple Access System
6. Study of Error detection and correction by using Hamming Code technique
7. Study of Computer network components : Cables, Connectors, Routers, Switches, Ethernet and related interfacing cards
8. To study Configuration of IP and MAC address and to study Local Area Network setup

S.Y.B. Sc. (Computer Science), Electronics, Semester IV

Paper I : Embedded System Design (ELC-241)

Objectives:

1. To understand the concept of Embedded systems.
2. To study the design flow and available tools for an Embedded system.
3. To understand the implementation of embedded system using firmware and hardware components.
4. To acquire programming skills for the development of Embedded system design.
5. To develop practical skills for designing embedded system Applications.

Course Outcomes : On completion of the course, student will be able

1. To understand the difference between general computing and the Embedded systems.
2. To know the fundamentals of embedded systems.
3. Understand the use of Single board Computer (Such as Raspberry Pi) for an embedded system application.
4. Familiar with the programming environment to develop embedded systems and their interfaces with peripheral devices.
5. To develop familiarity with tools used to develop in an embedded environment.

COURSE CONTENTS

Unit 1:Introduction to Embedded systems using single board computers (SBC) (08)

Single boards computer block diagram, types, Comparison of SBC models, Specifications, I/O devices (Storage, display, keyboard and mouse), Network access devices

Unit 2: Architecture of System on Chip (SOC) (08)

Architecture of SoC, Basic version Broad Coprocessor, Pin Description of Raspberry Pi, Architectural features: CPU Overview, CPU Pipeline stages, CPU Cache Organization, Branch Prediction & Folding (Concept), GPU Overview

Unit 3:Programming using Python (10)

Overview of Rasberian OS (Operating System), Installation, different types of Operating Systems

Basic Python Programming (Script programming):Variable & data types, Flow Control structures, Conditional statements (If…Then…else),
Functions: I/O function (GPIO, Digital),Time functions, Library functions
Basic Arithmetic Programs: Addition, Subtraction, Multiplication, Division

Unit 4 : Interfacing of devices using Python Programming (10)

Basic interfacing: LED, Switch, LCD
Internal Advanced: Bluetooth, Wifi, Ethernet,
External advanced: Camera, Serial Communication GSM, Ultrasonic Sensor, PIR, Finger Print reader.

Recommended Books:

1. Rasberry Pi CookBook: Software & Hardware problems and Solutions By Simon Monk(O'Reilly Media Inc.)
2. Raspberry Pi Hardware Reference by Warren Gay (Apress)
3. Rasberry Pi User Guide By Eben Upton, Greath Halfacree (John Wiley & Sons, Inc.)
4. Learning Python with Rasberry Pi, by Alex Bradbury, Ben Everard, John Wiley & Sons, Inc
5. Learn Raspberry Pi programming with Python By Wolfram Donat (Apress)

S.Y.B.Sc.(Computer Science), Electronics, Semester IV
Paper II: Wireless Communication and Internet of Things (ELC242)

Objectives:

1. To learn and understand applications of wireless communication system
2. To learn and understand cellular system
3. To learn and understand architecture of short range Wireless Technologies
4. To learn and understand basics of Internet of Things
5. To study applications of IoT

Course Outcomes: Students will be able to

1. Know working of wireless technologies such as Mobile communication, GSM, GPRS
2. Become familiar with 3G and 4G Cellular Network Technologies for Data Connections.
3. Understand working principles of short range communication application
4. Get introduce to upcoming technology of Internet of Things
5. Explore themselves and develop new IoT based applications

COURSE CONTENTS

Unit1: Wireless Communication: Cellular Telephony (12)

Overview of wireless communication,

Introduction of cellular telephony system: Frequency reuse, handoff strategies, Co-channel and adjacent channel interference, block diagram of mobile handset

Overview of Cellular Telephony generations: 1G to 5G,3G (W-CDMA, UMTS), 4G(LTE)

GSM: architecture, frame structure, mobility management,

GPRS : architecture, application

Unit 2 : Short Range Wireless Technologies and Location Tracking (12)

Short range Technologies :

Bluetooth: Bluetooth architecture, Bluetooth protocol stack, Bluetooth frame structure
Zigbee: Architecture, topologies, applications, Z wave: Protocol architecture, applications
RFID: working of RFID system, types of RFID tags, RFID frequencies, applications

Location Tracking: GPS system**:** components of GPS system (space segment, control segment, user segment), GPS receiver, Applications

Unit 3: IoT Architecture (08)

Introduction to IOT : Evolution of IOT, M2M and/or IOT, Seven layer architecture of IoT, Role of cloud in IoT, cloud topologies, Cloud access, Protocols in IoT, Cross connectivity across IoT system components:

- Device to Gateway-short range Wireless: cellphone as gateway, dedicated wireless Access points
- Gateway to cloud: Long range connectivity, (wired, cellular, Satellite, WAN)
- Direct Device to Cloud connectivity ,

Networking technologies: Low power local area networking (LPLAN), Low power wide area networking (LPWAN) technologies, comparison of LoRa, sigfox NB-IoT, Cat –M.

Unit 4: IoT Applications (04)

Application domains,

Challenges in IoT : Power consumption, Physical security, durability, Secure Connectivity, Secure Data Storage, Data volume, Scalability

Case studies:

Case Study 1: Smart Irrigation system for Agricultural field
Case Study 2:Home Automation
Case Study 3: Smart Cities

Recommended books:

1. Wireless Communications Principles and Practice, Rappaport, Pearson publication
2. Mobile Communications, Jochen Schiller, Pearson publication
3. Internet of Things : Principles and Paradigms, Rajkumar Buyya and Dastjerdi, MK publishers
4. Internet of Things, Mayur Ramgir, Pearson publication

S.Y.B.Sc.(Computer Science), Electronics, Semester IV
Paper III, Practical Course (ELC-243)

Objectives:

1. To use basic concepts for building various applications of embedded electronics.
2. To build experimental setup and test the circuits.
3. To develop skills of analyzing test results of given experiments.
4. Developing Trained Personals for educating and training for upcoming graduates in wireless communication.
5. Implement basic IoT applications on embedded platform

Course Outcomes : On completion of the course, students will be able

1. To design and develop own smart applications using Rasberry-Pi
2. To write Python program for simple applications
3. To build own IoT based system

Guidelines :

- Practical batch size : 12
- Minimum no of Practical to be performed : 10
- Eight compulsory experiments: At least four practical from each Group
- One activity equivalent to 2 experiments by the student.
  a. Continuation of F. Y. activity.
  b. Electronics project Based on the Theory Courses learnt
  c. Documentation type experiments
  d. Presentation/Seminar on Electronics /advanced topic/research topics.

Prerequisite: Rasberry Pi boards, Arduino / LoRa boards

COURSE CONTENTS

Group A (any 4)

1. Programming of Raspberry Pi to control LEDs attached to the GPIO pins
2. Programming of Raspberry Pi to get feedback from a switch connected to the GPIO pins

3. Programming of Raspberry Pi to detect temperature using temperature sensor

4. Programming of Raspberry Pi to detect light intensity using photocell sensor
5. Programming of Raspberry Pi for Motion detection
6. Programming of Raspberry Pi for image detection

Group B (any 4)

1. Study of GSM system (Message transmission & Reception).
2. To study working of SIM card in GSM handset
3. Study of GPRS system
4. Study of Zig-bee for one application
5. Study of RFID system
6. Introduction to Python programming.
7. To study Arduino based LED switching using mobile
8. Temperature and humidity sensing using Arduino
9. LoRa Interfacing.



Savitribai Phule Pune University

(Formerly University of Pune)

Three Year B.Sc. Degree Program in Computer Science

(Faculty of Science & Technology)

S.Y.B.Sc. (Computer Science)

Choice Based Credit System Syllabus
To be implemented from Academic Year
2020-2021

S. Y. B. Sc.(Computer Science)

Semester III

(Total credits=22)

Course type	Paper Code	Paper title	Credits	Evaluation		
				CA	UE	TOTAL
CC-VIII	CS 231	Data Structures and Algorithms – I	2	15	35	50
	CS 232	Software Engineering	2	15	35	50
	CS 233	Practical course on CS 231 and CS 232	2	15	35	50
CC-IX		Mathematics - I	2	15	35	50
		Mathematics - II	2	15	35	50
		Practical course in Mathematics	2	15	35	50
CC-X		Electronics - I	2	15	35	50
		Electronics - II	2	15	35	50
		Practical course in Electronics	2	15	35	50
AECC-I		Environment Science – I	2			
AECC-II		Language Communication – I	2			

Semester IV

(Total credits=22)

Course type	Paper Code	Paper title	Credits	Evaluation		
				CA	UE	TOTAL
CC-XI	CS 241	Data Structures and Algorithms – II	2	15	35	50
	CS 242	Computer Networks - I	2	15	35	50
	CS 243	Practical course on CS 241 and CS 242	2	15	35	50
CC-XII		Mathematics - I	2	15	35	50
		Mathematics - II	2	15	35	50
		Practical course in Mathematics	2	15	35	50
CC-XIII		Electronics - I	2	15	35	50
		Electronics - II	2	15	35	50
		Practical course in Electronics	2	15	35	50
AECC-I		Environment Science – II	2			
AECC-II		Language Communication –II	2			

- Each theory Lecture time for S.Y. B.Sc Computer Science is of 50 min (3 lectures/ week for 2 credit course)
- Each practical session time for S.Y. B.Sc Computer Science is of 4 hrs 20 minutes (260 min)
- Practical batch size =12

Savitribai Phule Pune University
S.Y.B.Sc. (Computer Science)
Computer Science Paper - I
Course Code: CS 231
Title : Data Structures and Algorithms – I

Teaching Scheme	No. of Credits	Examination Scheme
3 Lectures / week (50 mins duration)	2	IE : 15 marks UE: 35 marks

Prerequisites:
Basic knowledge of algorithms and problem solving
Knowledge of C Programming Language

Course Objectives
1. To learn the systematic way of solving problem
2. To understand the different methods of organizing large amount of data
3. To efficiently implement the different data structures
4. To efficiently implement solutions for specific problems
5. To apply linear data structures.

Course Outcomes: On completion of the course, student will be able to
1. To use well-organized data structures in solving various problems.
2. To differentiate the usage of various structures in problem solution.
3. Implementing algorithms to solve problems using appropriate data structures.

Course Contents

Chapter 1	**Introduction to Data Structures and Algorithm Analysis**	**4 lectures**

1.1 Introduction
- 1.1.1 Need of Data Structure
- 1.1.2 Definitions - Data and information, Data type, Data object, ADT, Data Structure
- 1.1.3 Types of Data Structures

1.2 Algorithm analysis
1.2.1 Space and time complexity, Graphical understanding of the relation between different functions of n, examples of linear loop, logarithmic,quadratic loop etc.
1.2.2 Best, Worst, Average case analysis, Asymptotic notations (Big O, Omega Ω, Theta θ), Problems on time complexity calculation.

Chapter 2	**Array as a Data Structure**	**10 lectures**

2.1 ADT of array, Operations
2.2Array applications - Searching
2.2.1 Sequential search, variations - Sentinel search, Probability search, ordered list search
- 2.2.2 Binary Search
- 2.2.3 Comparison of searching methods

2.3 Sorting Terminology- Internal, External, Stable, In-place Sorting
- 2.3.1 Comparison Based Sorting - Lower bound on comparison based sorting, Methods- Bubble Sort, Insertion Sort, Selection Sort, Algorithm design strategies - Divide and Conquer strategy, Merge Sort, Quick Sort, complexity analysis of sorting methods.

2.3.2 Non Comparison Based Sorting: Counting Sort, Radix Sort, complexity analysis.
2.3.3 Comparison of sorting methods

Chapter 3	Linked List	10 lectures

3.1 List as a Data Structure, differences with array.
3.2 Dynamic implementation of Linked List, internal and external pointers
3.3 Types of Linked List – Singly, Doubly, Circular
3.4 Operations on Linked List - create, traverse, insert, delete, search, sort, reverse, concatenate, merge, time complexity of operations.
3.5 Applications of Linked List – polynomial representation, Addition of two polynomials
3.6 Generalized linked list – concept, representation, multiple-variable polynomial representation using generalized list.

Chapter 4	Stack	6 lectures

4.1 Introduction
4.2 Operations – init(), push(), pop(), isEmpty(), isFull(), peek(), time complexity of operations.
4.3 Implementation- Static and Dynamic with comparison
4.4 Applications of stack
4.4.1 Function call and recursion, String reversal, palindrome checking
4.4.2 Expression types - infix, prefix and postfix, expression conversion and evaluation (implementation of infix to postfix, evaluation of postfix)
4.4.3Backtracking strategy - 4 queens problem (implementation using stack)

Chapter 5	Queue	6 lectures

5.1 Introduction
5.2 Operations - init(), enqueue(), dequeue(), isEmpty(), isFull(), peek(),time complexity of operations, differences with stack.
5.3 Implementation - Static and Dynamic with comparison
5.4 Types of Queue - Linear Queue, Circular Queue, Priority Queue, Double Ended Queue (with implementation)
5.5 Applications – CPU Scheduling in multiprogramming environment, Round robin algorithm

Reference Books:

1. Classic Data Structures-D. Samanta, Prentice Hall India Pvt. Ltd.
2. Fundamentals of Data Structures in C- Ellis Horowitz, SartajSahni,Susan Anderson-Freed, 2nd Edition, Universities Press.
3. Data Structures using C and C++-YedidyahLangsam, Moshe J. Augenstein, Aaron M. Tenenbaum, Pearson Education
4. Data Structures: A Pseudo code approach with C, Richard Gilberg ,Behrouz A. Forouzan, Cengage Learning.
5. Introduction to Data Structures in C-Ashok Kamthane, Pearson Education
6. Algorithms and Data Structures, Niklaus Wirth, Pearson Education

Savitribai Phule Pune University
S.Y.B.Sc. (Computer Science)
Computer Science Paper -II
Course Code: CS 232
Title : Software Engineering

Teaching Scheme	No. of Credits	Examination Scheme
3 lectures / week (50 mins duration)	**2**	IE : 15 marks UE: 35 marks

Prerequisites
ER Modeling

Course Objectives

1. To get knowledge and understanding of software engineering discipline.
2. To learn analysis and design principles for software project development.

Course Outcomes
On completion of the course, student will be able to-

1. Compare and chose a process model for a software project development.
2. Identify requirements analyze and prepare models.
3. Prepare the SRS, Design document, Project plan of a given software system.

Course Contents

Chapter 1	**Title : Introduction To Software Engineering and Process Models**	**8 lectures**

1.1 Definition of Software
1.2 Nature of Software Engineering
1.3 Changing nature of software
1.4 Software Process
- 1.4.1 The Process Framework
- 1.4.2 Umbrella Activities
- 1.4.3 Process Adaptation

1.5 Generic Process Model
1.6 Prescriptive Process Models
- 1.6.1 The Waterfall Model
- 1.6.2 Incremental Process Models
- 1.6.3 Evolutionary Process Models
- 1.6.4 Concurrent Models
- 1.6.5 The Unified Process

Chapter 2	**Title : Agile Development**	**5lectures**

2.1 What is Agility?
2.2 Agile Process
- 2.2.1 Agility Principles
- 2.2.2 The Politics Of Agile Development
- 2.2.3 Human Factors

2.3 Extreme Programming(XP)
- 2.3.1XP Values
- 2.3.2XP Process
- 2.3.3 Industrial XP

2.4 Adaptive Software Development(ASD)
2.5 Scrum
2.6 Dynamic System Development Model (DSDM)
2.7 Agile Unified Process (AUP)

Chapter 3	Title : Requirements Analysis	7 lectures

3.1 Requirement Elicitation,
3.2 Software requirement specification (SRS)
3.2.1 Developing Use Cases (UML)
3.3 Building the Analysis Model
3.3.1 Elements of the Analysis Model
3.3.2 Analysis Patterns
3.3.3 Agile Requirements Engineering
3.4 Negotiating Requirements
3.5 Validating Requirements

Chapter 4	Title : Requirements Modeling	10 lectures

4.1 Introduction to UML
4.2Structural Modeling
4.2.1 Use case model
4.2.2Class model
4.3Behavioral Modeling
4.3.1 Sequence model
4.3.2 Activity model
4.3.3 Communication or Collaboration model
4.4 Architectural Modeling
4.4.1 Component model
4.4.2 Artifact model
4.4.3 Deployment model

Chapter 5	Title :Design Concepts	6lectures

5.1 Design Process
5.1.1 Software Quality Guidelines and Attributes
5.1.2 Evolution of Software Design
5.2 Design Concepts
5.2.1 Abstraction
5.2.2 Architecture
5.2.3 Patterns
5.2.4 Separation of Concerns
5.2.5 Modularity
5.2.6 Information Hiding
5.2.7 Functional Independence
5.2.8 Refinement
5.2.9 Aspects
5.2.10 Refactoring
5.2.11 Object Oriented Design Concepts
5.2.12 Design Classes
5.2.13 Dependency Inversion
5.2.14 Design for Test
5.3 The Design Model
5.3.1 Data Design Elements
5.3.2 Architectural Design Elements

5.3.3 Interface Design Elements
5.3.4 Component-Level Diagram
5.4.5 Deployment-Level Diagram

Reference Books:

1. Software Engineering : A Practitioner's Approach - Roger S. Pressman, McGraw hill(Eighth Edition) ISBN-13: 978-0-07-802212-8, ISBN-10: 0-07-802212-6
2. A Concise Introduction to Software Engineering - Pankaj Jalote, Springer ISBN: 978-1-84800-301-9
3. The Unified Modeling Language Reference Manual - James Rambaugh, Ivar Jacobson, Grady Booch ISBN 0-201-30998-X

Savitribai Phule Pune University
S.Y.B.Sc. (Computer Science)
Computer Science Paper - III
Course Code: CS 233
Title : Practical course on CS 231 (Data Structures and Algorithms I) and CS 232 (Software Engineering)

Teaching Scheme	No. of Credits	Examination Scheme
4 hrs 20 mins / week Batch Size : 12	2	IE : 15 marks UE: 35 marks

Operating Environment:
For Data Structures:

- **Operating system:** Linux
- **Editor:** Any linux based editor like vi, gedit etc.
- **Compiler** : cc or gcc

Lab Book:
The lab book is to be used as a hands-on resource, reference and record of assignment submission and completion by the student. The lab book contains the set of assignments which the student must complete as a part of this course.

Programming Assignments:
Programs should be done individually by the student intheir respective login. The codes should be uploaded on either the local server, Moodle, Github or any open source LMS. Print-outs of the programs and output may be taken but not mandatory for assessment.

Assessment:
Continuous assessment of laboratory work is to be done based on overall performance and lab assignments performance of student. Each lab assignment assessment will be assigned grade/marks based on parameters with appropriate weightage. Suggested parameters for overall assessment as well as each lab assignment assessment include-timely completion, performance, innovation, efficient codes and good programming practices.

- **Internal Evaluation :**
  - 10 marks will be given based on a mini project of Software Engineering.
  - 5 marks will be allocated for Assignment completion and practical attendance.
- **University Evaluation :**
  - The Practical slip will be of 35 Marks which will be based on Data structures.

Course Contents:

Suggested Assignments for Data Structures – I

Assignment1: Searching Algorithms
Implementation of searching algorithms to search an element using: Linear Search, Sentinel Search, Binary Search (with time complexity)

Assignment 2: Sorting Algorithms - I
Implementation of sorting algorithms: Bubble Sort, Insertion Sort, Selection Sort

Assignment 3: Sorting Algorithms - II
Implementation of sorting algorithms: Quick Sort, Merge Sort , Counting Sort

Assignment 4: Singly Linked List

1. Dynamic implementation of Singly Linked List to perform following operations: Create, Insert, Delete, Display, Search, Reverse
2. Create a list in the sorted order.

Assignment 5: Doubly Linked List

1. Dynamic implementation of Doubly circular Linked List to perform following operations: Create, Insert, Delete, Display, Search

Assignment 6: Linked List Applications

1. Merge two sorted lists.

Addition of two polynomials in a single variable.

Assignment 7: Stack

1. Static and Dynamic implementation of Stack to perform following operations: Init, Push, Pop, Peek, Isempty, Isfull

Assignment 8: Applications of Stack

1. Implementation of an algorithm that reverses string of characters using stack and checks whether a string is a palindrome.
2. Infix to Postfix conversion.
3. Evaluation of postfix expression.

Assignment 9: Linear Queue

1. Static and Dynamic implementation of linear Queue to perform following operations: Init, enqueue, dequeue Peek, IsEmpty, IsFull.

Assignment 10: Circular and Priority Queue

1. Implementation of circular queue
2. Implementation of priority queue

Suggested Assignments for Software Engineering mini Project	**3**

1. Prepare detailed statement of problem for the selected mini project
2. Identify suitable process model for the same.
3. Develop Software Requirement Specification for the project.
4. Identify scenarios and develop UML Use case
5. Other artifacts: Class Diagram, activity diagram, sequence diagram, component diagram and any other diagrams as applicable to the project.

Sample project titles: (These are just samples, students are suggested to take up different case studies)

1. Online mobile recharge system
2. Credit calculation system
3. Image sharing and editing system
4. Internal examination system
5. e-learning management system

Savitribai Phule Pune University
S.Y.B.Sc. (Computer Science)
Computer Science Paper - I
Course Code: CS 241
Title : DATA STRUCTURES AND ALGORITHMS-II

Teaching Scheme	No. of Credits	Examination Scheme
3 Lectures / week (50 mins. duration)	02	IE : 15 marks UE: 35 marks

Prerequisites :

- Knowledge of C Programming Language
- Basic knowledge of algorithms
- Basic knowledge of linear data structures

Course Objectives

- To learn the systematic way of solving problems
- To design algorithms
- To understand the different methods of organizing large amount of data
- To efficiently implement the non-linear data structures

Course Outcomes: On completion of this course students will be able to

- Implementation of different data structures efficiently
- Usage of well-organized data structures to handle large amount of data
- Usage of appropriate data structures for problem solving

Course Contents

Chapter 1	**Tree**	**10 lectures**

1.1 Concept and Terminologies
1.2 Types of Binary trees - Binary tree, skewed tree, strictly binary tree, full binary tree, complete binary tree, expression tree, binary search tree, Heap
1.3 Representation – Static and Dynamic
1.4 Implementation and Operations on Binary Search Tree - Create, Insert, Delete, Search, Tree traversals– preorder, inorder, postorder (recursive implementation), Level-order traversal using queue, Counting leaf, non-leaf and total nodes, Copy, Mirror.
1.5 Applications of trees
1.5.1 Heap sort, implementation
1.5.2 Introduction to Greedy strategy, Huffman encoding (implementation using priority queue)

Chapter 2	**Efficient Search Trees**	**8 lectures**

2.1 Terminology: Balanced trees - AVL Trees, Red Black tree, splay tree, Lexical search tree -Trie
2.2 AVL Tree- concept and rotations
2.3 Red Black trees - concept, insertion and deletion.
2.4 Multi-way search tree - B and B+ tree - Insertion, Deletion

Chapter 3	**Graph**	**12 lectures**

3.1 Concept and terminologies
3.2 Graph Representation –Adjacency matrix, Adjacency list, Inverse Adjacency list, Adjacency multilist
3.3 Graph Traversals – Breadth First Search and Depth First Search (with implementation)
3.4 Applications of graph

3.4.1 Topological sorting
3.4.2 Use of Greedy Strategy in Minimal Spanning Trees (Prims and Kruskals algorithm)
3.4.3 Single source shortest path - Dijkstra's algorithm
3.4.4 Dynamic programming strategy, All pairs shortest path - Floyd Warshall algorithm
3.4.5 Use of graphs in social networks

Chapter 4	Hash Table	6 lectures

4.1 Concept of hashing
4.2 Terminologies – Hash table,Hash function, Bucket, Hash address, collision, synonym, overflow etc.
4.3 Properties of good hash function
4.4 Hash functions : division function, MID square , folding methods
4.5 Collision resolution techniques
4.5.1 Open Addressing - Linear probing, quadratic probing, rehashing
4.5.2 Chaining - Coalesced , separate chaining

Reference Books:

1. Fundamentals of Data Structures in C- Ellis Horowitz, SartajSahni,Susan Anderson-Freed, 2nd Edition, Universities Press.
2. Data Structures using C and C++-YedidyahLangsam, Moshe J. Augenstein, Aaron M. Tenenbaum, Pearson Education
3. Data Structures: A Pseudo code approach with C, Richard Gilberg ,Behrouz A. Forouzan, Cengage Learning.
4. Introduction to Data Structures in C-Ashok Kamthane, Pearson Education
5. Algorithms and Data Structures, Niklaus Wirth, Pearson Education
6. Introduction to Algorithms—Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein--MIT Press
7. Fundamentals of Computer Algorithms-- Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, Universities Press
8. The Algorithm Design Manual - Steven S Skiena, Springer

Savitribai Phule Pune University
S.Y.B.Sc. (Computer Science)
Computer Science Paper - I Semester II
Course Code: CS 242 Title : Computer Networks-I

Teaching Scheme	No. of Credits	Examination Scheme
3 lectures / week (50 mins. duration)	**02**	IE : 15 marks UE: 35 marks

Prerequisites
Principles of Digital Electronics
Communication Principles

Course Objectives
To prepare students with basic networking concepts: data communication, protocolsand standards, various topologies and applications of network.

Course Outcomes

1. Have a good understanding of the OSI and TCP/IP Reference Models and in particular have a good knowledge of Layers.
2. Understand the working of various protocols.
3. Analyze the requirements for a given organizational structure and select the most appropriate networking architecture and technologies

Course Contents

Chapter 1	**Introduction to Networks and Network Models**	**4 lectures**

1.1 Data communication, components, data representation
1.2 Networks, network criteria, network types - LAN, WAN, Switching, The Internet, Accessing the Internet
1.3 Network Software- Protocol hierarchies, Design Issues of the layer, Connection Oriented and Connectionless Services,
1.4 Reference models - OSI Reference Models, TCP/IP Reference model, Connection devices in different layers, Comparison of OSI and TCP/IP Reference Models.

Chapter 2	**Lower Layers**	**10 lectures**

2.1 Communication at the physical layer, data rate limits - Noiseless channel (Nyquist bit rate), noisy channel (Shannon capacity), Performance - bandwidth, throughput, latency, bandwidth-delay product, jitter
2.2 Design issues of Data Link Layer, Services - Framing, flow control, error control, congestion control, Link layer addressing
2.3 Framing Methods - Character Count, Flag bytes with Byte Stuffing, Flags bits with Bit Stuffing, Physical Layer Coding Violations
2.4 The Channel allocation problem, Static and dynamic allocation, Media Access Methods - Taxonomy of multiple-access protocols
2.5 Switching and TCP/IP layers, Types - circuit switching, packet switching and message switching
2.6 Wired LANs - Standard Ethernet characteristics, Addressing, Access method, implementation, Fast and Gigabit Ethernet
2.7 Wireless LANs - Architectural comparison, Characteristics, Access control, IEEE 802.11

architecture, Physical layer, MAC sublayer, Bluetooth architecture, Layers

Chapter 3	Network Layer	12 lectures

3.1 Network layer services - Packetizing, Routing and forwarding, other services
3.2 Open and closed loop congestion control
3.3 IPv4 addressing- Address space, classful addressing, Subnetting, Supernetting, classless addressing, Network address resolution (NAT)
3.4 Forwarding of IP packets- based on destination address, based on label
3.5 Network Layer Protocols- Internet Protocol (IP), IPv4 datagram format, Fragmentation, options
3.6 Mobile IP-addressing, agents, Three phases
3.7 Next Generation IP- IPv6 address representation, address space, address types, IPv6 protocol, packet format, extension header, Difference between IPv4 and IPv6
3.8 Routing - General idea, Algorithms - Distance vector routing, link state routing, path-vector routing

Chapter 4	Transport Layer	10 Lectures

4.1 Transport layer Services- Process-to-process communication, Addressing, Encapsulation and decapsulation, Multiplexing and demultiplexing, Flow control, Pushing or pulling, Flow control, Buffers, Sequence numbers, Acknowledgements, sliding window, congestion control
4.2 Connectionless and Connection-oriented service, Port numbers
4.3 Transport layer protocols- User datagram protocol, user datagram, UDP services
4.4 Transmission Control Protocol - TCP Services, TCP Features, TCP Segment format, three-way handshake for connection establishment and termination, State transition diagram, windows in TCP.

Reference Books:

1. Computer Networks-Andrew S. Tanenbaum, 5th Edition, Pearson Education
2. Data Communication and Networking- BehrouzFourouzan, 5th Edition, McGraw Hill Pvt. Ltd.

Savitribai Phule Pune University
S.Y.B.Sc. (Computer Science)
Computer Science Paper - III
Course Code: CS 243
Title : Practical course on CS 241(Data Structures and Algorithms II) and CS 242 (Computer Networks I)

Teaching Scheme	No. of Credits	Examination Scheme
4 hrs 20 mins / week Batch size : 12	2	IE : 15 marks UE: 35 marks

Lab Book:

The lab book is to be used as a hands-on resource, reference and record of assignment submission and completion by the student. The lab book contains the set of assignments which the student must complete as a part of this course.

Programming Assignments:

Programs should be done individually by the student in the respective login. The codes should be uploaded on either the local server, Moodle, Github or any open source LMS. Print-outs of the programs and output may be taken but not mandatory for assessment.

Assessment:

Continuous assessment of laboratory work is to be done based on overall performance and lab assignments performance of student. Each lab assignment assessment will be assigned grade/marks based on parameters with appropriate weightage. Suggested parameters for overall assessment as well as each lab assignment assessment include-timely completion, performance, innovation, efficient codes and good programming practices.

- **Internal Evaluation :**
  - 10 marks will be given based on Networking assignments.
  - 5 marks will be allocated for Assignment completion and practical attendance
- **University Evaluation :**
  - The Practical slip will be of 35 Marks which will be based on Advanced Data structures.

Operating Environment:

For Data Structures:

- **Operating system:** Linux
- **Editor:** Any linux based editor like vi, gedit etc.
- **Compiler** : cc or gcc

Course Contents :-

Assignment 1 Binary Search Tree and Traversals

1. Implement Binary Search Tree (BST) to perform following operations on BST– Create, Recursive Traversals - Inorder, Preorder, Postorder
2. Perform following operations: insert, delete

Assignment 2 Binary Search Tree Operations

1. Implement Binary Search Tree (BST) to perform following operations on BST–copy and mirror image of BST, counting leaf, non-leaf and total nodes.
2. Level-order traversal of binary search tree using queue.

Assignment 3 Applications of Binary Tree

1. Sort set of elements using Heap sort
2. Encode a set of characters using Huffman encoding

Assignment 4 Graph implementation

1. Implement Graph as adjacency matrix and adjacency list
2. Calculate indegree and outdegree of vertices
3. Graph traversals: BFS and DFS.

Assignment 5 Graph Applications - I

1. Implementation of Topological sorting
2. Implementation of Prims/Kruskals Minimum spanning tree algorithm

Assignment 6 Graph Applications - II

1. Implementation of Dijkstra's shortest path algorithm for finding Shortest Path from a given source vertex using adjacency cost matrix.
2. Implementation of Floyd Warshall algorithm for all pairs shortest path.

Assignment 7 Hash Table

1. Implementation of static hash table with Linear Probing.
2. Implementation of static hash table with chaining.

Assignment 8 Hash Table-2

1. Implementation of linked hash table with chaining.

Assignment 9 Networking Assignment

Assignment 10 Networking Assignment

SYBSC & SYBSC (Computer Science)

ENGLISH

(Ability Enhancement Compulsory Course-AECC)

(Choice Based Credit System-35:15-Pattern)
(w. e. f- 2020- 2021)

(02-Credit Course)

Text: ***Horizons: English in Multivalent Contexts*** (Board of Editors- Orient BlackSwan)

Preamble:

This is an ABILITY ENHANCEMENT COURSE. Considering the needs of students and the requirements of professional sectors, the syllabus of this paper is designed to enhance linguistic and professional skills of the students. In the age of technology, it's high time for the students to acquire and exercise the skill and sub-skills of using English in multivalent contexts.

The paper aims at a balanced up-gradation of the students, focussing on their ability enhancement. Hence, to avoid a lopsided professional development, the humane values are also taken care of by accommodating literature section in the syllabus. The other units cater to the needs of enhancing speaking ability, writing ability, the ability to face an interview, the ability of using soft skills effectively while planning one's work and working on the plans. By and large, the present syllabus is an attempt to galvanise the existing competencies of the students and enhancing their abilities for a better performance and better results.

Objectives:

1. To introduce the use of English in multimedia
2. To acquaint the students with the language skills in multivalent contexts
3. To acquaint and enlighten students regarding the speaking skill in various contexts
4. To acquaint and familiarize the students with advanced writing skills in different contexts
5. To acquaint and familiarize the students with soft skills
6. To minimize the gap between the existing communicative skills of the students and the skills they require at professional level
7. To develop competence among the students to appreciate and analyze short stories and poetry

Semester-III

Text: ***Horizons: English in Multivalent Contexts*** (Board of Editors- Orient BlackSwan)

Content-

UNIT-I- LITERATURE **10 Clock Hours**

1. **Short Story:**
   i) 'A Shadow': R. K. Narayan
2. **Poetry:**
   i) La Belle Dame sans Merci: John Keats
   ii) Where the Mind is without Fear: Rabindranath Tagore

UNIT-II-CONVERSATIONAL SKILL **10 Clock Hours**

(Sample Dialogues, Useful Expressions and Exercises)

1. Introducing Yourself and Others
2. Asking, Giving and Refusing Permission
3. Describing Daily Routine
4. Complaining and Apologizing
5.

UNIT-III-INTERVIEW TECHNIQUES **10 Clock Hours**

1. Job Application Letter
2. Resume Writing
3. GDPI
4. Presentations

Semester-IV

Text: ***Horizons: English in Multivalent Context (***Board of Editors- Orient BlackSwan)

UNIT-I-LITERATURE **10 Clock Hours**

1. **Short Story:**
   i) My Lost Dollar: Stephen Leacock
2. **Poetry:**
   i) The Bird Sanctuary: Sarojini Naidu
   ii) Stopping by Woods on a Snowy Evening: Robert Frost

UNIT-II-WRITING SKILLS **10 Clock Hours**

(Sample Passages, Useful Techniques and Exercises)

1. Notices
2. Agenda
3. Minutes
4. Content Writing

UNIT-III-SOFT SKILLS AND PERSONALITY DEVELOPMENT

(Sample Situations, Useful Techniques and Exercises) 10 Clock Hours

1. An Introduction to Soft Skills

2. SWOC Analysis

3. Goal Setting

4. Project Management

.....

BIBLIOGRAPHY:

1. Adair, John. Effective Communication, London: Pan Macmillan Ltd. 2003.

2. Amos, Julie-Ann. Handling Tough Job Interviews. Mumbai: Jaico Publishing, 2004.

3. Baron, N.S., (2008).Always On: Language in an Online and Mobile World. Oxford University

Press. Oxford.

4. Borg, James.(2010). Body Language: 7 Easy Lessons to Master the Silent Language. FT Press.

5. Collins, Patrick. Speak with Power and Confidence. New York: Sterling, 2009.

6. Kroehnert, Gary. Basic Presentation Skills. Sidney: McGraw Hill, 2010.

7. Linda B., Iris V. (2001). Intercultural Communication in the Global Workplace. 2nd Edition. Tata McGraw

8. Mitra, B. (2011). Personality Development & Soft Skills.1st edition. Oxfor.

9. Moore, Ninja-Jo, et al. Nonverbal Communication: Studies and Applications. New York: Oxford University Press, 2010.

10. Nelson, Paul E. & Judy C. Pearson, Confidence in Public Speaking.

11. Prasad, H. M. How to Prepare for Group Discussion and Interview. New Delhi: Tata McGraw-Hill Publishing Company Limited, 2001.

12. Pease, Allan. Body Language. Delhi: Sudha Publications, 1998.

13. Raman, Meenakshi & Sangeeta Sharma. Technical Communication: Principles and Practice. Second Edition. New Delhi: Oxford University Press, 2011.

14. Rutherford, Andrea J. Basic Communication Skills for Technology: Second Edition. Delhi: Pearson Education, 2007.

15. Sharma, R. C. & Krishna Mohan. Business Correspondence and Report Writing: Third Edition. New Delhi: Tata McGraw-Hill Publishing company Limited, 2007.

WEB LINKS:

http://networketiquette.net/

https://public.wsu.edu/~brians/errors/

http://users3.ev1.net/~pamthompson/body_language.htm

http://www.albion.com/netiquette/corerules.html

http://www.bbc.co.uk/worldservice/learningenglish/radio/specials/1535_questionanswer/page15.shtml

http://www.colostate.edu/Depts/Speech/rccs/theory44.html

http://www.dailywritingtips.com/

EVALUATION PATTERN

Considering the choice-based credit system (CBCS) and the semester pattern, both Semesters-III & IV will have a uniform evaluation pattern of **50 marks** each. There will be an **'Internal Examination'** for **15 marks** and **Semester-end Examination** for **35 marks**.

The Internal Examination for **15 marks** will be conducted in two parts.

1) Practical Examination for 05 marks:

(The choices like Group Discussion, Mock Interviews, Seminar, Project Presentation, Role Play, Home Assignment, Library Work, Lecture Notes etc. can be considered.)

2) A Mid-semester Written Test for 10 marks:

The Mid-semester Written Test will be based on the book prescribed for the syllabus. The test aims at assessing students' writing competence in general. Hence, descriptive and essay type questions can be considered while setting the question paper.

Semester-end Examination:

The Semester-end Examination will have a uniform question paper pattern for both semesters. The pattern of the question paper is given below.

SEMESTER-END EXAMINATION

Question Paper Patterns (III & IV)

Marks: 35 **Time: 2 Clock Hours**

.....

Instructions:

1. All questions are compulsory.

2. Figures to the right indicate full marks.

.....

Q.1) Long-answer question on Unit-1

(Any 1 out of 2) (15)

Q.2) Short notes on Unit-2

(Any 2 out of 3) (10)

Q.3) Short notes on Unit-3

(Any 2 out of 3) (10)

****



Savitribai Phule Pune University

(Formerly University of Pune)

Three Year B.Sc. Degree Program in Computer Science

(Faculty of Science & Technology)

T.Y.B.Sc. (Computer Science)

Choice Based Credit System Syllabus

To be implemented from Academic Year 2021 – 2022

Course Structure T.Y.B.Sc. (Computer Science)

Semester V (Total credits=22)

Course type	Paper Code	Paper title	Credits		Evaluation		
			T	P	CA	UA	TOTAL
DSEC - I	CS-351	Operating Systems – I	2		15	35	50
	CS-352	Computer Networks – II	2		15	35	50
	CS-357	Practical course based on CS 351		2	15	35	50
DSEC - II	CS-353	Web Technologies – I	2		15	35	50
	CS-354	Foundations of Data Science	2		15	35	50
	CS-358	Practical course based on CS 353 and CS 354		2	15	35	50
DSEC - III	CS-355	Object Oriented Programming using Java - I	2		15	35	50
	CS-356	Theoretical Computer Science	2		15	35	50
	CS-359	Practical Course based on CS 355		2	15	35	50
SECC - I	CS-3510	Python Programming	2	0	15	35	50
SECC - II	CS-3511	Blockchain Technology	2	0	15	35	50

Semester VI (Total credits=22)

Course type	Paper Code	Paper title	Credits		Evaluation		
			T	P	CA	UA	TOTAL
DSEC - I	CS-361	Operating Systems – II	2		15	35	50
	CS-362	Software Testing	2		15	35	50
	CS-367	Practical course based on CS 361		2	15	35	50
DSEC - II	CS-363	Web Technologies – II	2		15	35	50
	CS-364	Data Analytics	2		15	35	50
	CS-368	Practical course based on CS 363 and CS 364		2	15	35	50
DSEC - III	CS-365	Object Oriented Programming using Java - II	2		15	35	50
	CS-366	Compiler Construction	2		15	35	50
	CS-369	Practical Course based on CS 365		2	15	35	50
SECC - III	CS-3610	Software Testing Tools	2	0	15	35	50
SECC - IV	CS-3611	Project	2	0	15	35	50

SavitribaiPhule Pune University
T.Y.B.Sc. (Computer Science) - Sem – V
Course Type: DSEC – I Course Code : CS - 351
Course Title : Operating Systems – I

Teaching Scheme:	No. of Credits:	Examination Scheme:
03 Lect / week	2	IE : 15 marks UE: 35 marks

Prerequisites
Data structures like stack, queue, linked list, tree, graph, hashing, file structures, any structured programming language

Course Objectives:
1. To understand the concept of operation system and its principle
2. To study the various functions and services provided by operating system
3. To understand the notion of process and threads

Course Outcomes: After completion of this course students will be able to understand the concept of
1. Processes and Thread Scheduling by operating system
2. Synchronization in process and threads by operating system
3. Memory management by operating system using with the help of various schemes

Course Contents

Chapter 1	**Introduction to Operating Systems**	**6 lectures**

- Operating Systems Overview- system Overview and Functions of operating systems
- What does an OS do?
- Operating system Operations
- Operating system structure
- Protection and security
- Computing Environments- Traditional, mobile , distributed, Client/server, peer to peer computing
- Open source operating System
- Booting
- Operating System services,
- System calls Types of System calls and their working.

Chapter 2	**Processes and Threads**	**6 lectures**

- Process Concept – The processes, Process states, Process control block.
- Process Scheduling – Scheduling queues, Schedulers, context switch
- Operations on Process – Process creation with program using fork(), Process termination
- Thread Scheduling- Threads, benefits, Multithreading Models, Thread Libraries

Chapter 3	**Process Scheduling**	**7 lectures**

- Basic Concept – CPU-I/O burst cycle, Scheduling Criteria ,CPU scheduler, Preemptive scheduling, Dispatcher
- Scheduling Algorithms – FCFS, SJF, Priority scheduling, Round-robin scheduling, Multiple queue scheduling, Multilevel feedback queue scheduling

Chapter 4	**Synchronization**	**5 lectures**

- Background
- Critical Section Problem
- Semaphores: Usage, Implementation

- Classic Problems of Synchronization – The bounded buffer problem, The reader writer problem, The dining philosopher problem

Chapter 5	**Memory Management**	**12 lectures**

- Background – Basic hardware, Address binding, Logical versus physical address space, Dynamic loading, Dynamic linking and shared libraries
- Swapping
- Contiguous Memory Allocation – Memory mapping and protection, Memory allocation, Fragmentation
- Paging – Basic Method, Hardware support, Protection, Shared Pages
- Segmentation – Basic concept, Hardware
- Virtual Memory Management – Background, Demand paging, Performance of demand paging, Page replacement – FIFO, Optimal, LRU, MFU

Reference Books:

1. Operating System Concepts, Avi Silberschatz, Peter Galvin, Greg Gagne, Student Edition, Wiley Asia
2. Operating Systems: Internals and Design Principles, William Stallings, Prentice Hall of India.
3. Advanced Concepts in Operating Systems, M Singhal and NG Shivaratri, Tata McGraw Hill Inc, 2001
4. The 'C' Odyssey, UNIX-the open boundless C, Meeta Gandhi,Tilak Shetty,Rajiv Shah, BPB publication

SavitribaiPhule Pune University
T.Y.B.Sc. (Computer Science) Sem - V
Course Code: DSEC - I Course Code : CS - 352
Course Title :Computer Networks - II

Teaching Scheme	No. of Credits	Examination Scheme
03 Lect/ week	2	IE :15 marks UE: 35 marks

Prerequisites:
Prerequisites: Basic knowledge of Networking and ISO/OSI model

Course Objectives

- To understand different protocols of application layer.
- To understand concepts of multimedia.
- Explore the different methods used for Network/INTERNET security.

Course Outcomes

On completion of the course, student will be able to–

- Student will understand the different protocols of Application layer.
- Develop understanding of technical aspect of Multimedia Systems
- Develop various Multimedia Systems applicable in real time.
- Identify information security goals.
- Understand, compare and apply cryptographic techniques for data security.

Course Contents

Chapter 1	**Application Layer**	**10 Lect**

Domain Name System

- Name space-Flat name space, Hierarchical name space
- Domain Name Space -Label ,Domain name, FQDN,PQDN
- Distribution of Domain Name Space-Hierarchy of name servers, zone, Root server, Primary and secondary servers.
- DNS in the Internet: Generic domains, Country domains,inverse domain
- Resolution-Resolver,mapping names to address,mapping addresses to names,recursive resolution,iterative resolution,caching

Electronic Mail-

- Architecture-First scenario, second scenario, Third scenario, Fourth scenario
- User agent-services of user agent, types of UA Format of e-mail
- MIME-MIME header
- Message transfer agent-SMTP
- Message Access Agent: POP and IMAP

File Transfer

FTP-Communication over data control connection,File type,data structure,Transmission mode,anonymous FTP

Chapter 2	**Multimedia**	**08 Lect**

Digitizing audio and video, Audio and Video compression

Streaming Stored audio/video

- First approach
- Second approach
- Third approach
- Fourth approach

Streaming live audio/video

Real time interactive audio/video- Characteristics, Time relationship, timestamp, Playback buffer, ordering multicasting, translation

RTP-Packet format

RTCP-Message types

Voice over IP-SIP,SIP sessionH.323-

Architecture, Protocols

Chapter 3	**Cryptography and Network Security**	**09 Lect**

Terminology: Cryptography, plain text and cipher text, cipher key, categories of cryptography-Symmetric key, asymmetric key

Encryption model

Symmetric key cryptography

- Traditional ciphers – substitution cipher, shift cipher, Transposition cipher
- Simple Modern ciphers-XOR, Rotation cipher, s-box,p-box
- Modern round ciphers-DES
- Mode of operation-ECB,CBC,CFB,OFB

Asymmetric key cryptography-RSA

Security Services

- Message confidentiality-With Symmetric key cryptography, with asymmetric key cryptography
- Message integrity-Document and fingerprint, message and message digest
- Message authentication-MAC,HMAC
- Digital signature
- Entity Authentication-Passwords, Fixed passwords challenge-response

Chapter 4	**Security in the Internet**	**09 Lect**

IPSecurity(IPSec)

- Two modes
- Two security protocols
- Services provided by IPSec
- Security association
- Internet key exchange
- Virtual private network

SSL/TLS

- SSL services
- Security parameters
- Sessions and connections
- Four protocols
- Transport layer security

PGP

- Security parameters
- Services
- PGP algorithms
- Key rings
- PGP certificates

Firewalls

- Packet filter firewall
- Proxy firewall

Reference Books:

1. Data communications and networking by Behrouz Forouzan 4th/5th edition, McGraw Hill Pvt Ltd.
2. Computer Networks by Andrew S Tanenbaum, 4th/5th edition, Pearson Education
3. Cryptography and Network Security: Principles and Practice, William Stallings, 7th edition, Pearson Education
4. Network Security Essentials: Applications and Standards (For VTU), William Stallings, 3rd edition, Pearson Education

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem - V
Course Type:DSEC – II Course Code: CS - 353
Course Title : Web Technologies - I

Teaching Scheme	No. of Credits	Examination Scheme
03 Lect/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

HTML basics for form designing

Course Objectives

- To Design dynamic and interactive Web pages.
- To Learn Core-PHP, Server Side Scripting Language
- To Learn PHP-Database handling

Course Outcomes

On completion of the course, student will be able to–

- Understand how to develop dynamic and interactive Web Page

Course Contents

Chapter 1	**Introduction to HTML, HTTP and PHP**	**10 Lects**

Overview of HTML and Basic Tags , Creating Forms ,Tables, HTML5 Semantics.
CSS basic concept ,Three ways to use CSS, Box Model, Navigation Bar .
Introduction to Web server and Web browser .
HTTP basics .
PHP Basics: Use of PHP, Lexical structure, Language basics .

Chapter 2	**Function and String**	**8 Lects**

Defining and calling a function
Default parameters
Variable parameters, Missing parameters
Variable function, Anonymous function
Types of strings in PHP
Printing functions
Encoding and escaping
Comparing strings
Manipulating and searching strings
Regular expressions

Chapter 3	**Arrays**	**6 Lectures**

Indexed Vs Associative arrays
Identifying elements of an array
Storing data in arrays
Multidimensional arrays
3.4Extracting multiple values
Converting between arrays and variables
Traversing arrays
Sorting
Action on entire array

Chapter 4	**Files and database handling**	**10 Lectures**

Working with files and directories
Opening and Closing, Getting information about file, Read/write to file,
Splitting name and path from file, Rename and delete files
Reading and writing characters in file
Reading entire file

Random access to file data
Getting information on file
Ownership and permissions
Using PHP to access a database
Relational databases and SQL
PEAR DB basics
Advanced database techniques

Chapter 5	**Handling email with php**	**2 Lectures**

Email background
Internet mail protocol
Structure of an email message
Sending email and validation of Email_id with php

Reference Books:

1. HTML & CSS: The Complete Reference, Fifth Edition Author: Thomas A. Powell First published: 01 Jan 2010.
2. Programming PHP By Rasmus Lerdorf and Kevin Tatroe, O'Reilly publication
3. Beginning PHP 5 , Wrox publication
4. PHP web sevices, Wrox publication
5. Mastering PHP , BPB Publication
6. PHP cookbook, O'Reilly publication
7. PHP for Beginners, SPD publication
8. Programming the World Wide Web , Robert W Sebesta(3rd Edition)
9. HTML 5 Black Book : Covers Css3, Javascript, XML, XHTML, Ajax, PHP And Jquery by Kogent Learning Solutions Inc, Published November 2011 by Dreamtech Press
10. Spurlock Jake, Bootstrap: Responsive Web development. O'Reilly Media, Inc

Ref. Links

11. www.php.net.in
12. www.W3schools.com
13. www.wrox.com
14. https://coreui.io/docs/layout/grid/#grid-options
15. https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/bootstrap-grid-system.php

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) – Sem - V
Course Type:DSEC – II Course Code: CS - 354
Paper Title : Foundations of Data Science

Teaching Scheme	No. of Credits	Examination Scheme
03 lectures / week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Problem solving using computers
- Basic mathematics and statistics
- Knowledge of Databases

Course Objectives

- Provide students with knowledge and skills for data-intensive problem solving and scientific discovery
- Be prepared with a varied range of expertise in different aspects of data science such as data collection, visualization, processing and modeling of large data sets.
- Acquire good understanding of both the theory and application of applied statistics and computer science based existing data science models to analyze huge data sets originating from diversified application areas.
- Be better trained professionals to cater the growing demand for data scientists in industry.

Course Outcomes

On completion of the course, student will be able to–

- Perform Exploratory Data Analysis
- Obtain, clean/process, and transform data.
- Detect and diagnose common data issues, such as missing values, special values, outliers, inconsistencies, and localization.
- Demonstrate proficiency with statistical analysis of data.
- Present results using data visualization techniques.
- Prepare data for use with a variety of statistical methods and models and recognize how the quality of the data and the means of data collection may affect conclusions.

Course Contents

Chapter 1	**Introduction to Data Science**	**6 lectures**

Introduction to data science, The 3 V's: Volume, Velocity, Variety
Why learn Data Science?
Applications of Data Science
The Data Science Lifecycle
Data Scientist's Toolbox
Types of Data

Structured, semi-structured, Unstructured Data, Problems with unstructured data
Data sources
Open Data, Social Media Data, Multimodal Data, standard datasets
Data Formats
Integers, Floats, Text Data, Text Files, Dense Numerical Arrays, Compressed or Archived Data, CSV Files, JSON Files, XML Files, HTML Files , Tar Files, GZip Files, Zip Files, Image Files: Rasterized, Vectorized, and/or Compressed

Chapter 2	Statistical Data Analysis	10 lectures

2.1.Role of statistics in data science

2.2.Descriptive statistics

Measuring the Frequency

Measuring the Central Tendency: Mean, Median, and Mode

Measuring the Dispersion: Range, Standard deviation, Variance, Interquartile Range

2.3.Inferential statistics

Hypothesis testing, Multiple hypothesis testing, Parameter Estimation methods,

2.4.Measuring Data Similarity and Dissimilarity

Data Matrix versus Dissimilarity Matrix, Proximity Measures for Nominal Attributes, Proximity Measures for Binary Attributes, Dissimilarity of Numeric Data: Euclidean, Manhattan, and Minkowski distances, Proximity Measures for Ordinal Attributes

2.5.Concept of Outlier, types of outliers, outlier detection methods

Chapter 3	Data Preprocessing	10 lectures

Data Objects and Attribute Types: What Is an Attribute?, Nominal , Binary, Ordinal Attributes, Numeric Attributes, Discrete versus Continuous Attributes

Data Quality: Why Preprocess the Data?

3.3.Data munging/wrangling operations

Cleaning Data - Missing Values, Noisy Data (Duplicate Entries, Multiple Entries for a Single Entity, Missing Entries, NULLs, Huge Outliers, Out-of-Date Data, Artificial Entries, Irregular Spacings, Formatting Issues - Irregular between Different Tables/Columns, Extra Whitespace, Irregular Capitalization, Inconsistent Delimiters, Irregular NULL Format, Invalid Characters, Incompatible Datetimes)

Data Transformation – Rescaling, Normalizing, Binarizing, Standardizing,Label and One Hot Encoding

Data reduction

Data discretization

Chapter 4	Data Visualization	10 lectures

Introduction to Exploratory Data Analysis

Data visualization and visual encoding

Data visualization libraries

Basic data visualization tools

Histograms, Bar charts/graphs, Scatter plots, Line charts, Area plots, Pie charts, Donut charts

Specialized data visualization tools

Boxplots, Bubble plots, Heat map, Dendrogram, Venn diagram, Treemap, 3D scatter plots

Advanced data visualization tools- Wordclouds

Visualization of geospatial data

Data Visualization types

Reference Books:

1) Data Science Fundamentals and Practical Approaches, Gypsy Nandi, Rupam Sharma, BPB Publications, 2020.
2) The Data Science Handbook, Field Cady, John Wiley & Sons, Inc, 2017
3) Data Mining Concepts and Techniques, Third Edition, Jiawei Han, Micheline

Kamber, Jian Pei, Morgan Kaufmann, 2012.
4) A Hands-On Introduction to Data Science, Chirag Shah, University of Washington Cambridge University Press

SavitribaiPhule Pune University
T.Y.B.Sc. (Computer Science) Sem – V
Course Type:DSEC – III Course Code: CS - 355
Course Title: Object Oriented Programming using Java - I

Teaching Scheme	No. of Credits	Examination Scheme
03 Lect / week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Knowledge of C Programming language

Course Objectives

- To learn Object Oriented Programming language
- To study various java programming concept like Interface, File and Exception Handling etc.
- To design User Interface using Swing and AWT

Course Outcomes

On completion of the course, student will be able to–

- Understand the concept of classes, object, packages and Collections.
- To develop GUI based application.

Course Contents

Chapter 1	**An Introduction to Java**	**6 Lect**

Object Oriented Programming Concepts
A short history of Java
Features OR Buzzwords of Java
Java Environment
Simple Java Program
Java Tools – jdb, javap, javadoc
Types of Comments
Data Types
Final Variable
Declaring 1D, 2D Array
Accepting Input
(Command Line Arguments, BufferedReader, Scanner)

Chapter 2	**Objects and Classes**	**7 Lect**

Defining your own classes
Access Specifiers
(public, protected, private, default)
Array of Objects
Constructors, Overloading Constructors and Use of 'this' keyword
static block, static fields And methods
Predefined Classes

- Object Class, Methods (equals(), toString(),hashcode(), getClass())
- String Class And StringBuffer Class,Formatting String data using format() method

Creating , Accessing And Using Packages
Wrapper Classes

Chapter 3	Inheritance and Interface	8 Lect

Inheritance Basics (extends Keyword) and Types of Inheritance
Superclass, Subclass and use of Super Keyword
Method Overriding and runtime polymorphism
Use of final keyword related to method and class
Use of abstract class and abstract methods
Defining and Implementing Interfaces
Runtime polymorphism using interface
Concept of Marker and Functional Interfaces

Chapter 4	Exception and File Handling	5 Lect

Dealing with errors , Exception class, Checked And Unchecked Exception
Catching Exceptions, Multiple Catch Block, Nested try block
Creating User Defined Exception
Introduction to Files And Streams
Input-OutputStream : FileInput/OutputStream, BufferedInput/OutputStream, DataInput/OutputStream
Reader-Writer : FileReader/Writer, BufferedReader/Writer, InputStreamReader, OutputStreamWriter

Chapter 5	User Interface with AWT and Swing	10 Lect

What is AWT? What is Swing? Difference between AWT and Swing
The MVC Architecture And Swing
Layouts And Layout Managers
Containers And Components – JFrame, JButton, JLabel, JText, JTextArea, JCheckBox And JRadioButton, JList, JComboBox, JMenu And related Classes
Dialogs (Message, Confirmation, Input), JFileChooser, JColorChooser
Event Handling: Event Sources, Listeners
Adapters And Anonymous Inner Class

Reference Books:

R1. Complete reference Java by Herbert Schildt(5th edition)
R2. Java 2 programming black books, Steven Horlzner
R3. Programming with Java , A primer ,Forth edition , By E. Balagurusamy
R4. Core Java Volume-I-Fundamentals, Eighth Edition, Cay S. Horstmann, Gary Cornell, Prentice Hall, Sun Microsystems Pres

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) Sem - V
Course Type: DSEC - III Course Code: CS - 356
Paper Title: Theoretical Computer Science

Teaching Scheme	No. of Credits	Examination Scheme
3 Lect/ week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Mathematical Preliminaries Sets (Subset, Set Operations), Relations (Properties of Relations, Closure of Relations) and Functions
- Discrete Mathematics- Graphs, Trees, Logic and Proof Techniques

Course Objectives

- To understand the Finite Automata, Pushdown Automata and Turing Machine.
- To understand the Regular Language, Context Free Language, Context Sensitive Language and Unrestricted Language.
- To understand the relation between Automaton and Language

Course Outcomes

On completion of the course, student will be able to–

- Understand the use of automata during language design.
- Relate various automata and Languages.

Course Contents

Chapter 1	**Finite Automaton**	**10 Lect**

Introduction: Symbol, Alphabet, String, Prefix & Suffix of Strings, Formal Language, Operations on Languages.
Deterministic finite Automaton – Definition, DFA as language recognizer, DFA as pattern recognizer.
Nondeterministic finite automaton – Definition and Examples.
NFA To DFA (Myhill Nerode Method)
NFA with ε- transitions Definition and Examples.
NFA with ε-Transitions to DFA & Examples
Finite automaton with output – Mealy and Moore machine, Definition and Examples.
Minimization of DFA, Algorithm & Problem using Table Method.

Chapter 2	**Regular Expressions and Languages**	**6 Lect**

Regular Expressions (RE): Definition & Example
Regular Expressions Identities.
Regular language-Definition and Examples.
Conversion of RE to FA-Examples.
Pumping lemma for regular languages and applications.
Closure Properties of regular Languages

Chapter 3	**Context-Free Grammars and Languages**	**10 Lect**
Grammar - Definition and Examples. Derivation-Reduction - Definition and Examples. Chomsky Hierarchy. CFG: Definition & Examples. LMD, RMD, Parse Tree Ambiguous Grammar: Concept & Examples. Simplification of CFG: Removing Useless Symbols, Unit Production, ϵ-production and Nullable Symbol. Normal Forms: Greibach Normal Form (GNF) and Chomsky Normal Form (CNF) Regular Grammar: Definition. Left linear and Right Linear Grammar-Definition and Example. Equivalence of FA & Regular Grammar Construction of regular grammar equivalent to a given DFA. Construction of a FA from the given right linear grammar		
Chapter 4	**Push Down Automata**	**5 Lect**
Definition of PDA and examples. Construction of PDA using empty stack and final State method: Examples using stack method. Definition DPDA & NPDA, their correlation and Examples of NPDA CFG (in GNF) to PDA: Method and examples		
Chapter 5	Turing Machine	**5 Lect**
The Turing Machine Model, Definition and Design of TM Problems on language recognizers. Language accepted by TM. Types of Turing Machines (Multitrack TM, Two-way TM, Multitape TM, Non-deterministic TM) Introduction to LBA (Basic Model) & CSG. (Without Problems)		
Reference Books		
1. Introduction to Automata Theory, Languages and Computation, John E. Hopcraft, Rajeev Motwani, Jeffrey D. Ullman, Third Edition, Pearson Education Publication, 2008 2. Introduction to Automata theory, Languages and computation By John E. Hopcroft and JeffreyUllman – Narosa Publishing House, 1995 3. Theory of Computer Science Automata, Languages and Computation, K.L.P. Mishra, N. Chandrasekaran, Publication- Prentice Hall of India, 2008 4. Introduction to Computer Theory Daniel I. A. Cohen – 2nd edition – John Wiley & Sons, 1996 5. Introduction to Languages and The Theory of Computation John C. Martin The McGraw-Hill, Fourth Edition, 2011		

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem - V
Course Type: DSEC - I Course Code: CS - 357
Course Title : Practical Course based on CS - 351

Teaching Scheme:	No. of Credits:	Examination Scheme:
5 Lect/ week	2	IE : 15 marks UE: 35 marks

Course Objectives:

1. To understand the concept of process scheduling with the help of simulation.
2. To study the concept demand paging concepts in operating system.
3. To understand the working of operating system shell.

Course Outcomes: After completion of this course students will be able to understand the concept of

1. Process synchronization
2. Processes and Thread Scheduling by operating system
3. Memory management by operating system using with the help of various schemes

Guidelines:

1. Operating system platform – Linux
2. Programming language - C

List of Assignments:

- Operations on processes : (2 slot)
  (Create a child process using fork() and commands like exec(),execv() and execvp())
- Simulation of Operating System Shell and its working (commands)(2 slots)
- Simulation of CPU Scheduling Algorithms – FCFS, SJF, Priority and Round Robin(4 slots)
- Simulation of demand paging using memory page replacement algorithms – FIFO, LRU, OPT, MFU(4 slots)

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) Sem – V
Course Type: DSEC - II **Course Code: CS - 358**
Course Title : Practical Course based on CS - 353 and CS - 354

Teaching Scheme:	No. of Credits:	Examination Scheme:
5 Lect/ week Batch Size : 12	2	IE : 15 marks UE: 35 marks

Course Objectives:

- To Design dynamic and interactive Web pages.
- To Learn Core-PHP, Server Side Scripting Language
- To Learn PHP- Database handling
- To apply statistical, data preprocessing and visualization techniques on data sets

Course Outcomes:

- Understand how to develop dynamic and interactive Web Page
- Prepare data for use with a variety of statistical methods and recognize how the quality of the data may affect conclusions.
- Perform exploratory data analysis

Guidelines:

Operating Environment for web technologies: HTML5.0, PHP 5.0 and above , Webserver
Operating Environment for Data Science: Linux + python

List of Assignments on web technologies:

1 : HTML and HTML5.0
2 : CSS, Box Model, Navigation Bar
3 : Bootstrap
4 : Function and String
5 : Arrays
6: Files
7: Databases (PHP-PostgreSQL)

Suggested Assignments for Foundations of Data Science

Assignment 1: The Data Science environment
Getting introduced to Python IDLE, command line, online tools like google colaboratory and essential packages like NumPy, SciPy, pandas, scikit-learn, matplotlib, jupyter, beautiful-soup, etc.

Assignment 2: Loading the dataset
Select a dataset from a list of publicly available datasets at UCI Machine Learning Repository and load it using Pandas. (Import different dataformat files like .CSV,.htm,.json etc. Briefly describe what the dataset is about and size of the dataset (e.g. number of tables, number of instances and attributes, etc.)

Assignment 3: Basic statistical operations
Apply basic statistical operations on a dataset. For example - compute the mean, median, mode, range, quartiles, and variance for one or more attributes.

Assignment 4: Data preprocessing

Apply data preprocessing techniques that are likely required for the dataset.

1) Partition them into appropriate number of bins by equal-frequency as well as equal-width partitioning.

2) Use smoothing by bin means to smooth the data based on the above partitioning,

3) Normalize the attribute based on min-max normalization and z-score normalization.

Comment on which method you would prefer to use for partitioning, smoothing, and normalization for the given attribute.

Assignment 5: Data Visualization with matplotlib

View the data using various 2-D, 3-D plots and charts, setting styles, saving the figures, customizing the legends, multiple subplots,

SavitribaiPhule Pune University
T.Y.B.Sc. (Computer Science) - Sem – V
Course Type: DSEC - III Course Code: CS - 359
Course Title : Practical Course based on CS - 355

Teaching Scheme	No. of Credits	Examination Scheme
5 Lect / week Batch Size : 12	2	IE : 15 marks UE: 35 marks

Course Objectives:

Covers the complete scope of the syllabus.

1. Bringing uniformity in the way course is conducted across different colleges.
2. Continuous assessment of the students.

Course Outcomes:

1. Use an integrated development environment to write, compile, run, and test simple object-oriented Java programs.
2. Read and make elementary modifications to Java programs that solve real-world problems.
3. Validate input in a Java program.

Guidelines:

Operating Environment :

- **Operating system :** Linux
- **Editor :** Anylinux based editor like vi, gedit and Use of IDE – Eclipse etc.
- **Compiler** : javac

Submission :

Each assignment will be assessed on a scale of 0 to 5 as indicated below.

- Not done 0
- Incomplete 1
- Late Complete 2
- Needs improvement 3
- Complete 4
- Well Done 5

Assessment :

Easy : All exercises are compulsory.
Medium : All exercises are compulsory.

List of Assignments :

Assignment 1 : Java Tools and IDE, Simple java programs [Slot – 1]

Introduction to the java environment
Use of java tools like java, javac, jdb and javadoc
Defining simple classes and creating objects.

Assignment 2 : Array of Objects and Packages [Slot – 2]

Defining a class.
Creating an array of objects.
Creating a package.

Assignment 3 : Inheritance and Interfaces [Slot – 2]

To implement inheritance in java.
To define abstract classes.
To define and use interfaces and Functional Interface.

Assignment 4 : Exception And File Handling [Slot – 2]

Demonstrate Exception Handling Mechanism in Java.
Use of try, catch, throw, throws ,finally blocks
Defining User defined Exception classes.
Creation of files and demonstration of I-O operations

Assignment 5 : GUI Designing, Event Handling [Slot – 5]

To demonstrate GUI creation using Swing Package and Layout managers.
To understand Event handling mechanism in Java.
Using Event classes, Event Listeners and Adapters

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem – V
Course Type: SECC – I Course Code : CS-3510
Course Title: Python Programming

Teaching Scheme	No. of Credits	Examination Scheme:
03 Lect / week	2	IE : 15 marks UE: 35 marks

Course Objectives

1. To introduce programming concepts using python
2. Student should be able to develop Programming logic using python
3. To develop basic concepts and terminology of python programming
4. To test and execute python programs

Course Outcomes

On completion of the course, student will be able to–

- Develop logic for problem solving
- Determine the methods to create and develop **Python programs** by utilizing the data
- structures like lists, dictionaries, tuples and sets.
- To be familiar about the basic constructs of programming such as data, operations, conditions, loops, functions etc.
- To write python programs and develop a small application project

Course Contents

Chapter 1	**An Introduction to Python**	**3 Lect**

Introduction to Python

The Python Programming Language, History, features, Applications, Installing Python, Running Simple Python program

Basics of Python

Standard data types - basic, none, Boolean (true & False), numbers, Variables, Constants,Python identifiers and reserved words, Lines and indentation, multi-line statements and Comments,Input/output with print and input ,functions Declaration, Operations on Data such as assignment, arithmetic, relational, logical and bitwise operations, dry run, Simple Input and output etc.

Chapter 2	**Control Statements**	**4 Lect**

Sequence Control – Precedence of operators, Type conversion

Conditional Statements: if, if-else, nested if-else,

Looping- for, while, nested loops, loop control statements (break, continue, pass)

a. **Strings**: declaration, manipulation, special operations, escape character, string formatting operator, Raw String, Unicode strings, Built-in String methods.

Chapter 3	**Lists, functions, tuples and dictionaries, Sets**	**7 Lect**

Python Lists: Concept, creating and accessing elements, updating & deleting lists, traversing a List, reverse Built-in List Operators, Concatenation, Repetition, In Operator, Built-in List functions and methods.

Functions: Definitions and Uses, Function Calls, Type Conversion Functions, Math Functions, Composition, Adding New Functions, Flow of Execution, Parameters and Arguments, Variables and Parameters, Stack Diagrams, Void Functions, Anonymous functions Importing with from, Return Values, Boolean Functions, More Recursion, Functional programming tools - filter(), map(), and reduce(),recursion, lambda forms.

Tuples and Dictionaries: Tuples, Accessing values in Tuples, Tuple Assignment, Tuples as return values, Variable-length argument tuples, and Basic tuples operations, Concatenation, Repetition, in Operator, Iteration, Built-in tuple functions, indexing, slicing and matrices. Creating a Dictionary, Accessing Values in a dictionary, Updating Dictionary, Deleting Elements from Dictionary, Properties of Dictionary keys, Operations in Dictionary, Built-In Dictionary Functions, Built-in Dictionary Methods.

Sets- Definition, transaction of set(Adding, Union, intersection), working with sets

Chapter 4	**Modules ,Working with files, Exception handling**	**4 Lect**

Modules: Importing module, Creating & exploring modules, Math module, Random module, Time module

Packages: Importing package, creating package,examples

Working with files: Creating files and Operations on files (open, close, read, write), File object attributes, file positions, Listing Files in a Directory, Testing File Types, Removing files and directories, copying and renaming files, splitting pathnames, creating and moving directories

Regular Expression- Concept of regular expression, various types of regular expressions, using match function.

Exception Handling: Built-in Exceptions, Handling Exceptions, Exception with Arguments, User-defined Exceptions.

Demonstration	**Programming Assignments:**	**18 Lect**

Out of 36 lectures, 18 are assigned for demonstration. Teacher should give demonstration of various programs mentioned below in the classroom or in the laboratory as per their convenience.

Programming assignments should be done individually by the student in their respective login from the list given in Labbook. The codes should be uploaded on either the local server, Moodle, Github or any LMS.

Assignment 1 - Python Basics
Assignment 2 – Arrays, Strings, and Functions
Assignment 3 - List, Tuples, Sets, and Dictionary
Assignment 4 - File Handling and Date-Time
Assignment 5 - Exception handling and Regular expression

Reference Books:

1. An Introduction to Computer Science using Python 3 by Jason Montojo, Jennifer Campbell, Paul Gries, The pragmatic bookshelf-2013
2. James Payne, "Beginning Python: Using Python and Python 3.1,Wrox Publication
3. Introduction to Computer Science Using Python- Charles Dierbach, Wiley Publication Learning with Python ", Green Tea Press, 2002
4. Introduction to Problem Solving with Python by E balguruswamy,TMH publication-2016
5. Beginning Programming with Python for Dummies Paperback – 2015 by John Paul Mueller
6. Object-oriented Programming in Python, Michael H. Goldwasser, David Letscher, Pearson Prentice Hall-2008

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem – V
Course Type: SECC – II Course Code : CS-3511
Course Title: Blockchain Technology

Teaching Scheme	No. of Credits	Examination Scheme:
03 Lect / week	2	IE : 15 marks UE: 35 marks

Prerequisite:
Understanding of Object Oriented Programming Concepts
Knowledge of Python

Course Objectives

1. Understand what and why of blockchain technology.
2. Explore major components of blockchain.
3. Learn about Bitcoin, Cryptocurrency and Ethereum.
4. To learn blockchain programming using Python, Flask Web Framework, and HTTP client Postman.

Course Outcomes

On completion of the course, student will be able to–
1. Learn the fundamentals of Blockchain Technology.
2. Learn Blockchain programming
3. Basic knowledge of Smart Contracts and how they function.

Course Contents

Chapter 1	**Introduction to Blockchain**	**7 Lect**

- Foundational Computing Concepts (Client-Server systems vs Peer to Peer Systems)
- Evolution of Blockchain
- Blockchain Vs Database
- Essentials of Blockchain (Blockchain generations, types of blockchain, benefits and challenges of blockchain usage)
- Types of Networks
- Layered Architecture of Blockchain Ecosystem
- Components of blockchain
- Cryptography (private and public keys, Hashing &
- Digital Signature)
- Consensus Mechanisms
- Cryptocurrency, Digital Currency Bitcoin and Ethereum
- Smart Contracts
- Blockchain use cases

Chapter 2	**How Blockchain Works?**	**5 Lect**

- Understanding SHA256 Hash
- Immutable Ledger
- Distributed P2P Network
- How Mining Works? (The NONCE and Cryptographic Puzzle)
- Byzantine Fault Tolerance
- Consensus Protocols: Proof of Work, Proof of State, Défense Against Attackers, Competing Chains
- Blockchain Demo

Chapter 3	**Smart Contracts**	6 Lect

- Ethereum Network
- What is a Smart Contract?
- Ethereum Virtual Machine, Ether, Gas
- DApps
- Decentralized Autonomous Organizations (DAO)
- Hard and Soft Forks
- Initial Coin Offerings
- Demo of Smart Contracts

Demonstration	**Programming Assignments:**	**18 Lect**

Out of 36 lectures, 18 are assigned for demonstration. Teacher should give demonstration of various programs mentioned below in the classroom or in the laboratory as per their convenience.

Assignment 1 –Demonstration of Blockchain
https://andersbrownworth.com/blockchain

Assignment 2 – Installation of Ganache, Flask and Postman

Assignment 3 –Write a Simple Python program to create a Block class that contains index, timestamp, and previous hash. Connect the blocks to create a Blockchain.

Assignment 4 –Demo of Remix-Ethereum IDE https://remix.ethereum.org and Test Networks

Assignment5–1. Write a Simple Smart Contract for Bank with withdraw and deposit functionality.

Assignment 6 – 2. Write a Smart Contract for storing and retrieving information of Degree Certificates.

Reference Books:

Textbook:

1. **Beginning Blockchain : A Beginner's Guide to Building Blockchain Solutions By Bikramaditya Singhal, Gautam Dhameja, Priyansu Sekhar Panda, Apress Media**

Reference Books:

2. Mastering Blockchain by Imran Bashir, Third Edition, Packt Publication
3. Waterhole, The Science of the Blockchain
4. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System
5. Mastering Ethereum: Building Smart Contracts and DAPPS, by Andreas Antonopoulos, Dr. Gavid Wood, Oreilly Publication

Reference Web Links

1. https://www.investopedia.com/terms/b/blockchain.asp

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem – VI
Course Type: DSEC - IV Course Code: CS - 361
Course Title : Operating Systems-II

Teaching Scheme: 03 Lect / week	No. of Credits: 2	Examination Scheme: IE : 15 marks UE: 35 marks

Prerequisites
Concepts of Operating System, Processes and Threads Scheduling, Synchronization

Course Objectives:
1. To understand the issue of Deadlocks in Process management.
2. To understand the concept of File system management & disk scheduling
3. To study the concept of distributed and mobile operating systems

Course Outcomes: After completion of this course students will be able to understand the concept of
1. Management of deadlocks and File System by operating system
2. Scheduling storage or disk for processes
3. Distributed Operating System and its architecture and the extended features in mobile OS.

Course Contents

Chapter 1	**Process Deadlocks**	**7 lectures**

- System model
- Deadlock Characterization – Necessary conditions, Resource allocation graph
- Deadlock Methods- Prevention and Deadlock Avoidance - Safe state, Resource allocation graph algorithm, Banker's Algorithm
- Deadlock Detection
- Recovery from Deadlock – Process termination, Resource preemption

Chapter 2	**File system Management**	**6 lectures**

- File concept , File attributes, File operations
- Access Methods – Sequential, Direct, Other access methods
- Directory overview, Single level directory, Two level directory, Tree structure directory, Acyclic graph directory, General graph directory
- Allocation Methods – Contiguous allocation, Linked allocation, Indexed allocation
- Free Space Management – Bit vector, Linked list, Grouping, Counting, Space maps

Chapter 3	**Disk scheduling**	**4 lectures**

- Overview, Disk Structure
- Disk Scheduling, FCFS Scheduling, SSTF Scheduling, Scan Scheduling-Scan Scheduling, Look Scheduling , Disk Management

Chapter 4	**Introduction to Distributed operating systems & Architecture**	**11 lectures**

- What is a distributed system, Design goals
- Types of distributed systems
- Architectural styles : Layered architectures , Object-based architectures, Resource-centered architectures
- System architecture – Centralized organization, Decentralized organizations, peer-to-peer systems, Hybrid architectures.

- Example architectures : Network file system(NFS), Web-based distributed systems

Chapter 5	**Mobile Operating Systems**	**7 lectures**

- Introduction
- Features
- Special Constraints and Requirements of Mobile Operating System
- Special Service Requirements
- ARM & Intel architectures – Power management
- Mobile OS architectures – Underlying OS, kernel structure & native level programming, Runtime issues, Approaches to power management
- Commercial Mobile Operating Systems - Windows Mobile, iPhone OS (iOS), Android
- A Comparative Study of Mobile Operating Systems (Palm OS, Android, Symbian OS, Blackberry OS, Apple iOS)

Reference Books:

1) Advanced Concepts in Operating Systems, M Singhal and NG Shivaratri, Tata McGraw Hill Inc, 2001 (Text Book)

2) Operating System Concepts, Avi Silberschatz, Peter Galvin, Greg Gagne, Student Edition, Wiley Asia

3) Operating Systems: Internals and Design Principles, William Stallings, Prentice Hall of India.

4) Distributed Operating Systems Concepts and Design, Pradeep K. Sinha, PHI

5) Prasant Kumar Pattnaik, Rajib Mall, "Fundamentals of Mobile Computing", PHI Learning Pvt.Ltd, New Delhi – 2012.

6) A. Tannenbum, Herbert Bos, "Modern Operating systems", Pearson Publication, 4th Edition

7) A. Tannenbum, Maarten van Steen, "Distributed systems", 3rd Edition

8) Source wikipedia, Mobile operating systems, General books, LLC,2010

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem – VI
Course Type: DSEC - IV Course Code: CS - 362
Course Title : Software Testing

Teaching Scheme:	No. of Credits:	Examination Scheme:
3 Lect / week	2	IE : 15 marks UE: 35 marks

Prerequisites:

- Basic knowledge of algorithms, problem solving, expected inputs/outputs
- Knowledge of C and java Programming Language, compilation, debugging.

Course Objectives:

- To provide the knowledge of software testing techniques
- To understand how testing methods can be used as an effective tools in quality assurance of software.
- To provide skills to design test case plan for testing software.
- To provide knowledge of latest testing methods

Course Outcomes:

- To understand various software testing methods and strategies.
- To understand a variety of software metrics, and identify defects and managing those defects for improvement in quality for given software.
- To design test cases and test plans, review reports of testing for qualitative software.
- 4. To understand latest testing methods used in the software industries.

Course Contents

Chapter 1	**Introduction to Software Testing**	**5 lectures**

Basics of Software Testing – faults, errors and failures
Testing objectives
Principles of testing
Testing and debugging
Testing metrics and measurements
Verification and Validation
Testing Life Cycle

Chapter 2	**Software Testing Strategies & Techniques**	**10 lectures**

Testability - Characteristics lead to testable software.
Test characteristics
Test Case Design for Desktop, Mobile, Web application using Excel
White Box Testing - Basis path testing, Control Structure Testing.
Black Box Testing- Boundary Value Analysis, Equivalence partitioning.
Differences between BBT & WBT

Chapter 3	**Levels of Testing**	**10 lectures**

A Strategic Approach to Software Testing
Test strategies for conventional Software
Unit testing
Integration testing – Top-Down, Bottom-up integration
System Testing – Acceptance, performance, regression, Load/Stress testing, Security testing, Internationalization testing.
Alpha, Beta Testing

Usability and accessibility testing Configuration, compatibility testing		
Chapter 4	**Testing Web Applications**	**6 lectures**
Dimension of Quality, Error within a WebApp Environment Testing Strategy for WebApp Test Planning The Testing Process –an overview		
Chapter 5	**Agile Testing**	**5 lectures**
Agile Testing, Difference between Traditional and Agile testing, Agile principles and values, Agile Testing Quadrants, Automated Tests.		

Reference Books:

1. Software Engineering – A Practitioners Approach, Roger S. Pressman, 7th Edition, Tata McGraw Hill, 20
2. Effective Methods of Software Testing, William E Perry, 3rd Edition, Wiley Publishing Inc
3. Managing the Testing Process: Practical Tools and Techniques for Managing Hardware and Software Testing, Rex Black, Microsoft Press, 1999
4. Agile Testing: A Practical Guide for Testers and Agile Teams, Lisa Crispin and Janet Gregory, 1st Edition, Addison-Wesley Professional, 2008
5. Software Testing Principles and Practices By Srinivasan Desikan, Gopalaswamy Ramesh, Pearson

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science)- Sem - VI
Course Type:DSEC – V Course Code: CS - 363
Course Title : Web Technologies - II

Teaching Scheme	No. of Credits	Examination Scheme
3 Lect / week	2	IE : 15 marks UE: 35 marks

Prerequisites
- HTML5,CSS
- Core PHP
- Bootstrap framework utility

Course Objectives
- To Learn different technologies used at client Side Scripting Language
- To Learn XML and XML parsers.
- To One PHP framework for effective design of web application.
- To Learn Java Script to program the behavior of web pages.
- To Learn AJAX to make our application more dynamic.
- Framework has

Course Outcomes
On completion of the course, student will be able to–
- Build dynamic website.
- Using MVC based framework easy to design and handling the errors in dynamic website.

Course Contents

Chapter 1	**Introduction to Web Techniques**	**6 Lect**

Variables
- Server information
- Processing forms
- Setting response headers
- Maintaining state
- PHP error handling

Chapter 2	**XML**	**6 Lect**

What is XML?
- XML document Structure
- PHP and XML
- XML parser
- The document object model
- The simple XML extension
- Changing a value with simple XML

Chapter 3	**Java Script and Jquery**	**10 Lect**

Overview of JavaScript
- Object Orientation and JavaScript Basic Syntax(JS datatypes, JS variables)

Primitives, Operations and Expressions
Screen Output and keyboard input(Verification and Validation)
JS Control statements and JS Functions
JavaScript HTML DOM Events(onmouseup, onmousedown, onclick, onload, onmouseover, onmouseout).

JS Strings and JS String methods

JS popup boxes(alert, confirm, prompt).
Jquery library , Including jquery library in page
Jquery selector , DOM manipulation using jquery

Chapter 4	AJAX	**6 Lect**

Introduction of AJAX
AJAX web application model
AJAX –PHP framework
Performing AJAX validation
Handling XML data using php and AJAX
Connecting database using php and AJAX

Chapter 5	PHP framework CodeIgniter	**8 Lect**

CodeIgniter - Overview, Installing CodeIgnite
Application Architecture
MVC Framework , Basic concept of CodeIgniter, Libraries
Working with databases
Load external JS and CSS page & redirecting from controller , Adding JS and CSS ,
Page redirection.
Loading dynamic data on page & session management, cookies management

Reference Books:

1. **Programming PHP By Rasmus Lerdorf and Kevin Tatroe O'Reilly publication**
2. **Beginning PHP 5, Wrox publication**
3. **AJAX Black Book Kogent solution**
4. **Mastering PHP BPB Publication**
5. **Professional Codeigniter By Thomas Myer ,Wrox Publication,**
6. **Codeihniter 2 CookBook By Rob Foster ,PACKT Publication ,**
7. **JQuery CookBook, O'reilly Publication.**

Ref. Links:

1. **www.php.net.in**
2. **www.W3schools.com**
3. **https://www.tutorialspoint.com/codeigniter/index.htm**
4. **https://api.jquery.com/**
5. **http://codeigniter.com/docs**

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) – Sem VI
Course Type:DSEC – V Course Code: CS - 364
Course Title : Data Analytics

Teaching Scheme	No. of Credits	Examination Scheme
03 lectures / week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Basic of mathematics and statistics
- Basic programming Knowledge of python
- Knowledge of databases

Course Objectives

- Deploy the Data Analytics Lifecycle to address data analytics projects.
- Develop in depth understanding of the key technologies in data analytics.
- Apply appropriate analytic techniques and tools to analyze data, create models, and identify insights that can lead to actionable results.

Course Outcomes

On completion of the course, student will be able to–

- Use appropriate models of analysis, assess the quality of input, and derive insight from results.
- Analyze data, choose relevant models and algorithms for respective applications
- Understand different data mining techniques like classification, prediction, clustering and association rule mining
- Apply modeling and data analysis techniques to the solution of real world business problems

Course Contents

Chapter 1	**Introduction to Data Analytics**	**6 lectures**

Concept of data analytics
Data analysis vs Data analytics
Types of analytics
Diagnostic Analytics, Predictive Analytics , Prescriptive Analytics, Exploratory Analysis, Mechanistic Analysis
Mathematical models - Concept
Model evaluation: metrics for evaluating classifiers - Class imbalance - AUC, ROC (Receiver-Operator Characteristic) curves, Evaluating value prediction models

Chapter 2	**Machine Learning Overview**	**6 Lectures**

Introduction to Machine Learning, deep learning, Artificial intelligence
Applications for machine learning in data science
The modeling process
Engineering features and selecting a model, Training the model, Validating the model, Predicting new observations
Types of machine learning
Supervised learning, Unsupervised learning, Semi-supervised learning, ensemble techniques
Regression models
Linear Regression

Polynomial Regression
Logistic Regression

2.6. Concept of classification, clustering and reinforcement learning.

Chapter 3	Mining Frequent Patterns, Associations, and Correlations	12 lectures

What kind of patterns can be mined

Class/Concept Description: Characterization and Discrimination, Mining Frequent Patterns, Associations, and Correlations, Classification and Regression for Predictive Analysis, Cluster Analysis, Outlier Analysis

Mining frequent patterns - Market Basket Analysis.

Frequent Itemsets, Closed Itemsets, and Association Rules

Frequent Itemset Mining Methods

Apriori Algorithm

Generating Association Rules from Frequent Itemsets

Improving efficiency of apriori algorithm

Frequent pattern growth (FP-growth) algorithm

Chapter 4	Social Media and Text Analytics	12 lectures

Overview of social media analytics

Social Media Analytics Process, Seven layers of social media analytics, accessing social media data

Key social media analytics methods

Social network analysis

Link prediction, Community detection, Influence maximization, Expert finding, Prediction of trust and distrust among individuals

Introduction to Natural Language Processing

Text Analytics : Tokenization, Bag of words, Word weighting : TF-IDF, n-Grams, stop words, Stemming and lemmatization, synonyms and parts of speech tagging

Sentiment Analysis

Document or text summarization

Trend analytics

Challenges to social media analytics

Reference Books:

1) Data Science Fundamentals and Practical Approaches, Gypsy Nandi, Rupam Sharma, BPB Publications, 2020.
2) The Data Science Handbook, Field Cady, John Wiley & Sons, Inc, 2017
3) Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, Morgan Kaufmann, Third Edition, 2012.
4) A Hands-On Introduction to Data Science, Chirag Shah, University of Washington Cambridge University Press
5) The Data Science Design Manual, Steven S. Skiena, Springer, 2017
6) Introducing data science: big data, machine learning, and more, using Python tools, Cielen D., Meysman A. D., & Ali M., Manning Publications Co., 2016

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem – VI
Course Type: DSEC – VI Course Code : CS - 365
Course Title : Object Oriented Programming using Java – II

Teaching Scheme	No. of Credits	Examination Scheme
03 Lect / week	2	IE : 15 marks UE: 35 marks

Prerequisites

- Knowledge of Core Java (CS – 355)

Course Objectives

- To learn database programming using Java
- To study web development concept using Servlet and JSP
- To develop a game application using multithreading
- To learn socket programming concept

Course Outcomes

On completion of the course, student will be able to–

- To access open database through Java programs using Java Data Base Connectivity (JDBC) and develop the application.
- Understand and Create dynamic web pages, using Servlets and JSP.
- Work with basics of framework to develop secure web applications.

Course Contents

Chapter 1	**Collections**	**6 Lect**

Introduction to the Collection framework
List - ArrayList, LinkedList
Set - HashSet, TreeSet,
Map - HashMap and TreeMap
Interfaces such as Comparator, Iterator, ListIterator, Enumeration

Chapter 2	**Multithreading**	**6 Lect**

What are threads?
Life cycle of thread
Creating threads - Thread class , Runnable interface
Thread priorities
Running multiple threads
Synchronization and interthread communication

Chapter 3	**Database Programming**	**6 Lect**

The design of jdbc
Types of drivers
Executing sql statements, query execution
Scrollable and updatable Resultset

Chapter 4	**Servlets and JSP**	**12 Lect**

Introduction to Servlet and Hierarchy of Servlet
Life cycle of servlet
Handing get and post request (HTTP)
Handling data from HTML to servlet
Retrieving data from database to servlet

Session tracking – User Authorization, URL rewriting, Hidden form fields,Cookies and HttpSession

Introduction to JSP, Life cycle of JSP
Implicit Objects
Scripting elements - Declarations, Expressions, Scriplets, Comments
JSP Directives - Page Directive, include directive
Mixing Scriplets and HTML
JSP Actions - jsp:forward , jsp:include, jsp:useBean, jsp:setProperty and jsp:getProperty

Chapter 5	**Spring Framework**	**6 Lect**

Introduction of Spring framework
Spring Modules / Architecture
Spring Applications
Spring MVC
Spring MVC Forms, Validation

Reference Books:

R1. Complete reference Java by Herbert Schildt(5th edition)
R2. Java 2 programming black books, Steven Horlzner
R3. Programming with Java , A primer ,Forth edition , By E. Balagurusamy
R4. Core Java Volume-I-Fundamentals, Eighth Edition, Cay S. Horstmann, Gary Cornell, Prentice Hall, Sun Microsystems Press
R5. Core Java Volume-II-Advanced Features, Eighth Edition, Cay S. Horstmann, Gary Cornell, Prentice Hall, Sun Microsystems Press
R6. Getting started with Spring Framework: covers Spring 5 by J Sharma and Ashish Sarin
R7. Spring 4 for Developing Enterprise Applications: An End-to-End Approach by Henry H. Liu

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem - VI
Course Type: DSEC - VI Course Code: CS - 366
Course Title: Compiler Construction

Teaching Scheme	No. of Credits	Examination Scheme
3 Lect / week	2	IE: 15 marks UE: 35 marks

Prerequisites

- Knowledge of Automata Theory and Languages.

Course Objectives

- To understand design issues of a lexical analyzer and use of LEX tool.
- To understand design issues of a parser and use of YACC tool.
- To understand and design code generation and optimization techniques.

Course Outcomes

On completion of the course, student will be able to–

- Understand the process of scanning and parsing of source code.
- Learn the conversion code written in source language to machine language.
- Understand tools like LEX and YACC.

Course Contents

Chapter 1	**Introduction**	**4 Lect**

Definition of Compiler, Aspects of compilation.
The structure of Compiler.
Phases of Compiler – Lexical Analysis, Syntax Analysis, Semantic Analysis,
Intermediate Code generation, code optimization, code generation.
Error Handling.
Introduction to one pass & Multipass compilers, cross compiler, Bootstrapping.

Chapter 2	**Lexical Analysis (Scanner)**	**4 Lect**

Review of Finite automata as a lexical analyzer,
Applications of Regular Expressions and Finite Automata (lexical analyzer,
searching using RE), Input buffering, Recognition of tokens.
LEX: A Lexical analyzer generator (Simple Lex Program)

Chapter 3	**Syntax Analysis (Parser)**	**14 Lect**

Definition, Types of Parsers
Top-Down Parser –
Top-Down Parsing with Backtracking: Method & Problems
Drawbacks of Top-Down parsing with backtracking, 3.2.3Elimination of Left Recursion (direct & indirect) 3.2.4Need for Left Factoring & examples
Recursive Descent Parsing: Definition
Implementation of Recursive Descent Parser Using Recursive Procedures
3.4 Predictive [LL (1)] Parser (Definition, Model)
3.4.1Implementation of Predictive Parser [LL (1)]
3.4.2 FIRST & FOLLOW

Construction of LL (1) Parsing Table
Parsing of a String using LL (1) Table.
Bottom-Up Parsers
Operator Precedence Parser -Basic Concepts
Operator Precedence Relations form Associativity & Precedence
Operator Precedence Grammar
Algorithm for LEADING & TRAILING (with ex.)
Algorithm for Operator Precedence Parsing (with ex.)
Precedence Functions
Shift Reduce Parser
Reduction, Handle, Handle Pruning
Stack Implementation of Shift Reduce Parser (with examples)
LR Parser: Model, Types [SLR (1), Canonical LR, LALR]-Method & examples.
YACC (from Book 3) –program sections, simple YACC program for expression evaluation

Chapter 4	**Syntax Directed Definition**	**7 Lect**

Syntax Directed Definitions (SDD)
Inherited & Synthesized Attributes
Evaluating an SDD at the nodes of a Parse Tree, Example
Evaluation Orders for SDD's
Dependency Graph
Ordering the Evaluation of Attributes
S-Attributed Definition
L-Attributed Definition
Application of SDT
Construction of syntax trees,
The Structure of a Type
4. 4 Translation Schemes
4.4.1 Definition, Postfix Translation Scheme

Chapter 5	**Code Generation and Optimization**	**7 Lect**

Compilation of expression –
Concepts of operand descriptors and register descriptors with example.
Intermediate code for expressions – postfix notations,
Triples, Quadruples and Expression trees.
Code Optimization – Optimizing transformations – compile time evaluation, elimination of common sub expressions, dead code elimination, frequency reduction, strength reduction.
Three address code
DAG for Three address code
The Value-number method for constructing DAG's.
Definition of basic block, Basic blocks, and flow graphs
Directed acyclic graph (DAG) representation of basic block.
Issues in design of code generator.

Reference Books
1. Compilers: Principles, Techniques, and Tools, Alfred V. Aho, Ravi Sethi, Jeffrey D. Ullman, 2004 2. Principles of Compiler Design By: Alfred V. Aho, Jeffrey D. Ullman, Narosa Publication House, 2002 3. LEX & YACC, 2nd edition, O'reilly Publication, 2012

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem – VI
Course Type: DSEC- IV Course Code: CS - 367
Course Title : Practical Course based on CS - 361

Teaching Scheme:	No. of Credits:	Examination Scheme:
5 Lect/ week	2	IE : 15 marks UE: 35 marks

Course Objectives:

1. To implement Banker's algorithm for Deadlocks in Process management.
2. To simulate File system management
3. To study and implement various algorithms of disk scheduling

Course Outcomes: After completion of this course students will be able to understand the concept of

1. Management of deadlocks by operating system
2. File System management
3. Disk space management and scheduling for processes

Guidelines:

1. Operating system platform – Linux
2. Programming language - C

List of Assignments:

- Simulation of Banker's algorithm of deadlock avoidance in processes of operating system (3 slots)
- Simulation of File Allocation methods and free space management in storage - Contiguous allocation, Linked allocation, Indexed allocation (4 slots)
- Simulation of Disk Scheduling algorithms – FCFS, SSTF, Scan, Look (2 slots)
- Assignment based on distributed and mobile OS (3 slots)

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem – VI
Course Type: DSEC - V **Course Code: CS - 368**
Course Title : Practical Course based on CS - 363 and CS - 364

Teaching Scheme	No. of Credits:	Examination Scheme:
5 Lect/ week Batch Size : 12	2	IE : 15 marks UE: 35 marks

Course Objectives:

- To Learn different technologies used at client Side Scripting Language
- To Learn XML and XML parsers.
- To One PHP framework for effective design of web application.
- To Learn Java Script to program the behavior of web pages.
- To Learn AJAX to make our application more dynamic.

Framework has some utility features that make easy to write API in more efficient way than

Core PHP

Course Outcomes:

- Build dynamic website.
- Using MVC based framework easy to design and handling the errors in dynamic website.

Guidelines:

Operating Environment :Linux, HTML, PHP5.0 and above, Codeigniter, Python

List of Assignments based on Web Technology CS-363:

1 : Self Processing Forms, Sticky Forms, File Upload.
2 : COOKIES and SESSIONS.
3 : XML documents and DOM
4 : JavaScript
5 : Ajax
6 : PHP framework CodeIgniter

List of Assignments for Data Analytics

Assignment 1: Frequent itemset and association rule mining

Load Transactional data set. Do the needful data preprocessing. Display the set of frequent 2-itemsets and 3-itemsets. Repeat the process for different min_sup value.

Assignment 2: Linear and Logistic regression

For Given dataset predict the value of specific attribute.

Assignment 3: Text Analytics

Take text file as input. Create bag of words. Find frequent item sets. Display word cloud

Assignment 4: Sentiment analysis

SavitribaiPhule Pune University
T.Y.B.Sc. (Computer Science) - Sem – VI
Course Type: DSEC - VI Course Code: CS - 369
Course Title : Practical Course based on CS - 365

Teaching Scheme	No. of Credits	Examination Scheme
5 Lect/ week Batch Size : 12	2	IE : 15 marks UE: 35 marks

Course Objectives:

1. Covers the complete scope of the syllabus.
2. Bringing uniformity in the way course is conducted across different colleges.
3. Continuous assessment of the students.
4. Advanced Java is designed to develop web based, network centric, Enterprise level applications

Course Outcomes:

1. To Learn database Programming using Java
2. Understand and Create dynamic web pagesusing Servlets and JSP.
3. Work with basics of framework to develop secure web applications

Guidelines:

Operating Environment :

- **Operating system:** Linux
- **Editor:**Anylinux based editor like vi, gedit and Use of IDE – Eclipse etc.
- **Compiler** :javac
- **Database** :postgresql

Submission :

Each assignment will be assessed on a scale of 0 to 5 as indicated below.

- Not done 0
- Incomplete 1
- Late Complete 2
- Needs improvement 3
- Complete 4
- Well Done 5

Assessment :

Easy : All exercises are compulsory.
Medium : All exercises are compulsory.

List of Assignments:

Assignment 1 : Database Programming [Slot-2]

Study the Collection framework in java.
To Implement various Interfaces and classes through algorithms.
To Demonstrate Cursor Objects (Enumeration, Iterator, ListIterator, Comparator)

Assignment 2 : Multithreading [Slot-2]

To create and use threads in java.
To demonstrate multithreading using Thread Synchronization, Inter-thread .
Communication, Thread Priorities.

Assignment 3 : Database Programming [Slot-2]

To communicate with a database using java.
To execute queries on tables.
To obtain information about the database and tables.

Assignment 4 : Servlets [Slot-2]

To understand server-side programming.
Simple steps to create and execute servlets.
How to pass parameters using doGet and doPost methods.
Handling data from HTML to servlet .
How to connect servlet to a database .
Use of various session tracking methods like Cookies.

Assignment 5 : Java Server Pages [Slot-2]

JSP life-cycle.
Use of JSP implicit objects.
JSP Directives.
Use of Scripting Elements.
To understand actionstags in JSP.
Understanding flow of JSP custom tags.

Assignment 6 : Spring Framework [Slot-2]

To create and understand the steps to develop Spring application.

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem – VI
Course Type: SECC - III Course Code: CS - 3610
Course Title: Software Testing Tools

Teaching Scheme:	No. of Credits:	Examination Scheme:
03 Lect / week	2	IE:15 marks UE: 35 marks

Prerequisites

- Basic knowledge of algorithms, problem solving, expected inputs/outputs
- Knowledge of C and Java Programming Language, compilation, debugging

Course Objectives:

- To provide the knowledge of software testing methods and strategies.
- To understand how testing methods can be used as an effective tool in quality assurance of software.
- To provide skills to design test case plan for testing software.
- 4.To provide knowledge of latest testing tools

Course Outcomes:

- To understand various software testing methods and strategies.
- To understand a variety of software metrics and identify defects and managing those defects for improvement in quality for given software.
- To design test cases and test plans, review reports of testing for qualitative software.
- 4. To understand latest testing tools used in the software industries.

Course Contents

Chapter 1	**Introduction to Test case design**	**4 lectures**
How to identify errors, bugs in the given application. Design entry and exit criteria for test case, design test cases in excel. Describe feature of a testing method used.		
Chapter 2	**Test cases for simple programs**	**4 lectures**
Write simple programs make use of loops and control structures. Write Test Cases for above programs.		
Chapter 3	**Test cases and Test plan**	**4 lectures**
Write Test Plan for given application with resources required. Write Test case for given application. Prepare Test report for test cases executed.		
Chapter 4	**Defect Report**	**3 lectures**
Defect Life Cycle Classification of Defect Write Defect Report		
Chapter 5	**Testing Tools**	**3 lectures**
How to make use of Automation Tools Types of Testing Tools		
Demonstration	**Programming Assignments**	**18 Lect**

Out of 36 lectures, 18 are assigned for demonstration. Teacher should give demonstration of various assignments based on above theory topics in the classroom or in the laboratory as per their convenience. Demonstration of any open source testing tool should be given.

Programming assignments should be done individually by the student in their respective login from the list given in Labbook. The code/ documentation should be uploaded on either the local server, Moodle, Github or any LMS.

Reference Books:

1. Software Engineering – A Practitioners Approach, Roger S. Pressman, 7thEdition, Tata McGraw Hill, 20
2. Effective Methods of Software Testing, William E Perry, 3rd Edition, Wiley Publishing Inc
3. Managing the Testing Process: Practical Tools and Techniques for Managing Hardware and Software Testing, Rex Black, Microsoft Press, 1999
4. Software Testing Principles and Practices by Srinivasan Desikan, Gopalaswamy Ramesh, Pearson.

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) - Sem - VI
Course Type: SECC - IV Course Code: CS - 3611
Course Title : Project

Teaching Scheme	No. of Credits	Examination Scheme
03 Lect/ week/Batch Batch Size : 20	2	IE : 15 marks UE: 35 marks

Project Guidelines:

- Students should work in a team of minimum 3 and maximum 4 students.
- Students can choose a project topic and implement the same using any language/technology covered in the curriculum so far. The operating environment must be linux.
- The student group will work independently throughout the project work including: problem identification, information searching, literature study, design and analysis, implementation, testing, and the final reporting.
- Project guide must conduct project presentations (minimum 2) to monitor the progress of the project groups.
- At the end of the project, the group should prepare a report which should conform to international academic standards. The report should follow the style in academic journals and books, with clear elements such as: abstract, background, aim, design and implementation, testing, conclusion and full references, Tables and figures should be numbered and referenced to in the report.
- The final project presentation with demonstration (UE) will be evaluated by the project guide (appointed by the college) and one external examiner (appointed by the University).

Recommended Documentation contents:

Abstract

Introduction

- motivation
- problem statement
- purpose/objective and goals
- literature survey
- project scope and limitations

System analysis

- Existing systems
- scope and limitations of existing systems
- project perspective, features
- stakeholders
- Requirement analysis - Functional requirements, performance requirements, security requirements etc.

System Design

- Design constraints
- System Model: Using OOSE
- Data Model
- User interfaces

Implementation details

- Software/hardware specifications

Outputs and Reports Testing

- Test Plan, Black Box Testing or Data Validation Test Cases, White Box Testing or Functional Validation Test cases and results

Conclusion a*nd* Recommendations

Future Scope

Bibliography and References

Project Related Assignments

Guidelines:

- The project assignments are a compulsory part of the project course and should be carried out by each project group.
- Project assignments are to be given by the guide for continuous internal evaluation.
- The project assignments are to be allotted to each group separately by the project guide on the basis of the implementation technology. A suggested list of assignments is given below.
  1. Project Time management: plan (schedule table), Gantt chart, Roles and responsibilities, data collection, Implementation
  2. Simple assignments to evaluate choice of technology
  3. Assignments on UI elements in chosen technology
  4. Assignments on User interfaces in the project
  5. Assignments on event handling in chosen technology
  6. Assignments on Data handling in chosen technology
  7. Online and offline connectivity
  8. Report generation
  9. Deployment considerations
  10. Test cases
- Each student within the group must work actively and contribute to the assignments, project work and report writing.

Evaluation guidelines:

IA (15 marks)			**UE (35 marks)**		
First presentation	Second presentation	Assignments	Project Logic/ Presentation	Assignments and Project Documentation	Viva
05	05	05	20	10	05